SKEW SCHUR EQUIVALENCE BY JACOBI–TRUDI NEGATIVE COUNT

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Abstract. We classify connected skew diagrams with at most six negative indices in their full row Jacobi–Trudi board, against arbitrary skew diagrams with the same Schur function. Seven parameter families give all exceptional classes, each with four members; every other source is determined up to rotation. When the negative pattern requires at least seven rows to realize, nine negatives force rigidity. At ten negatives under the same minimum-row condition, exactly two seven-row quartet families are nonrigid, and every higher-row source is rigid. Their complete classes have legal marked-composition presentations. Classical overlap invariants recover negative count and minimum realizing row count. Boundary coefficients identify cofactor components, and endpoint comparisons determine their placements. A saturated first negative column with at most two negatives elsewhere forces rotation rigidity in at least six rows, without a bound on negative count. Six-row quartets and non-dense placement resonances remain outside the classification.

1. Introduction

We classify connected skew diagrams with at most six negative indices in their full row Jacobi–Trudi board. The comparison is with an arbitrary skew diagram having the same skew Schur function, without a bound on area or row count and without excluding common beta entries. When the negative pattern requires at least seven rows to realize, rigidity holds through nine negatives. At ten under the same minimum-row condition, exactly two seven-row quartet families are nonrigid; every such source with more rows is rigid.

For a connected diagram $D = \lambda/\mu$ with n nonempty rows, its *full row board* is

$$(\lambda_i - \mu_j - i + j)_{1\le i,j\le n}.$$

Write $\nu(D)$ for the number of strictly negative entries. A zero index contributes $h_0 = 1$ and is not counted; common beta entries are not canceled before computing ν. Identify diagrams up to translation, write D^* for rotation through 180 degrees, and put

$$[D] = \{H : s_H = s_D,\ H \text{ connected}\}.$$

Call D *rotation-rigid* when $[D] = \{D, D^*\}$, with repetitions removed.

Theorem 1.1 (Classification by negative count)**.** *Let D be a connected skew diagram with $n \ge 1$ nonempty rows, and let H be any skew diagram with $s_H = s_D$. Common beta entries are allowed in both.*

(i) If $\nu(D) \le 2$, then $H \in \{D, D^*\}$.
(ii) If $\nu(D) = 3$, the only exceptions to rotation rigidity have four rows and are the complete four-member classes in (4.1), with $0 \le c < w$ and $b > c + w$.
(iii) If $\nu(D) = 4$, then $H \in \{D, D^*\}$.
(iv) If $\nu(D) = 5$, the only exceptions have five rows and are the two complete four-member families in (5.1), with $1 \le q \le b < g$.

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(v) If $\nu(D) = 6$, the only exceptions have five rows and are the complete four-member classes M1–M4 of (6.3)*, under the parameter inequalities stated there.*

Every exceptional class has two distinct rotation orbits. The seven parameter families are not disjoint: M1 and M2 coincide at their stated boundary. The six-negative exceptions have the staircase negative set

$$\mathcal{T}_n = \{(i, j) : i - j \geq n - 3\}, \qquad n \geq 5. \tag{1.1}$$

Sources with this pattern and at least six rows are rigid, as are all other six-negative sources.

Read the numbers of negative entries in successive rows from bottom to top, omitting zero parts, to obtain the *negative partition* $\rho(D)$. For a nonempty partition ρ, let $n_{\min}(\rho)$ be the least number of nonempty rows in a connected diagram realizing it. We prove

$$n_{\min}(\rho) = 1 + \max_{1 \leq i \leq \ell(\rho)} (\rho_i + i), \qquad n_{\min}(\varnothing) = 1. \tag{1.2}$$

This is a minimum over realizations of a pattern, not over Schur-equal diagrams or the actual height of the source.

Theorem 1.2 (Negative-index rigidity threshold)**.** *Let D be a connected skew diagram satisfying*

$$\nu(D) \leq 9, \qquad n_{\min}(\rho(D)) \geq 7.$$

For every skew diagram H,

$$s_H = s_D \quad \Longrightarrow \quad H \in \{D, D^*\},$$

where diagrams are identified up to translation and D^* denotes rotation through 180 *degrees. Common beta entries are allowed. The bound nine is sharp: the two seven-row families in Proposition 12.1 have ten negatives and minimum realizing row count seven, with nonrotation Schur-equal members.*

Theorem 1.3 (Ten-negative classification)**.** *Let D be connected with*

$$\nu(D) = 10, \qquad n_{\min}(\rho(D)) = 7.$$

Its complete connected Schur class is $\{D, D^*\}$, except for the classes

$$\{F_i(a, b, d), F_i(a, b, d)^*, H_i(a, b, d), H_i(a, b, d)^*\}, \qquad i \in \{A, B\},$$

with the profiles in (12.2) *and* (12.3)*, for integers a, b, d satisfying $a \geq 1, d \geq 0, b > a + d$. Each exceptional class has four distinct members and seven nonempty rows, with proper legal charts in* (12.6)*. Every source in this stratum with at least eight rows is rotation-rigid. Common beta entries are allowed, and diagrams are identified up to translation.*

Together with Corollary 11.1, this classifies the entire ten-negative domain with minimum realizing row count at least seven. The minimum-six case is not included. Negative count and minimum realizing row count are Schur invariants, but the negative partition need not be. Rotation conjugates it. The equal thirteen-cell profiles in Example 2.4 have partitions $(3, 1, 1)$ and $(3, 2)$, which are neither equal nor conjugate. None of these theorems prescribes a mate's negative partition.

Billera, Thomas and van Willigenburg classified ribbon equality by factor reversal [2]. Reiner, Shaw and van Willigenburg established the overlap invariants used here [6]. We use connected-skew-Schur irreducibility in the corrected form of Barekat, Reiner and van Willigenburg [1]. McNamara and van Willigenburg's marked compositions provide a broader source of equalities [5]. The Toeplitz-minor correspondence and row/column reversal used below are classical [3].

The maximal-label coefficient recovers component functions, not their positions or orientations. For five negatives, opening one second-subdiagonal cell gives a sixteen-plus-eight expansion. For the staircase, minimum-gap specialization recovers primitive factors before the full coefficients assemble them. At minimum row count six, the remaining component budget leaves only a singleton beside a two-negative four-row component or two dense components. Endpoint coefficients settle both cases.

The singleton argument also gives Theorem 8.5: at six or more rows, a saturated first negative column and at most two other negative entries force rotation rigidity. This criterion has no bound on the full negative count, but retains its specific column geometry.

At higher minimum row count, the nine-negative budget leaves a dense four-row component and two singletons. A three-variable specialization proves that its endpoint difference has a nonzero lower remainder. At ten negatives, the four-row component can have one negative, and a singleton can accompany an exceptional five-row core. Rank-two and rank-three specializations settle the former; an endpoint-coefficient table determines every exceptional singleton context in the latter. Connected bordering excludes further nonrotation classes at larger heights. The six-row quartets in Proposition 12.2 remain lower inclusions rather than complete class descriptions.

Section 2 gives polynomial readings, negative partitions, and their overlap invariants. Section 3 develops homogenization and bordering. Sections 4–7 classify the small-row exceptions and exclude their higher-row borders. Section 8 treats disconnected cofactors, including the saturated-column criterion and the other minimum-six patterns. Sections 9–11 prove the sharp threshold and finish the six-negative classification. Section 12 gives the sharpness families and boundary examples. Section 13 classifies the ten-negative stratum at minimum row count seven and excludes its higher-row borders. Appendix A contains the four-row calculation; Appendix B gives a further component-determination result, not an input to the parent classification.

2. Full boards and invariant negative count

2.1. **Profiles and polynomial readings.** Work in $\mathbb{Q}[h_1, h_2, \ldots]$, with $h_0 = 1$ and $h_j = 0$ for $j < 0$; see [4]. For one-based row indices put $x_i = \lambda_i - i$, $y_i = \mu_i - i$, and translate both strictly decreasing lists to have $y_n = 0$. Then

$$s_D = \det(h_{x_i - y_j}), \qquad N = x_1, \qquad r_i = x_i - y_i, \qquad z_i = x_{i+1} - y_i.$$

The *profile* $(r; z)$ records row lengths and adjacent overlaps minus one. Its exact connectedness conditions are

$$r_i > 0, \qquad 0 \le z_i < \min(r_i, r_{i+1}). \tag{2.1}$$

It determines the diagram: starting with $y_n = 0$, recover $y_i = y_{i+1} + r_{i+1} - z_i$, then $x_i = y_i + r_i$. Rotation reverses both profile words. A board is *zero-free* if $X \cap Y = \varnothing$, and *dense* if $x_n > y_1$. These are different conditions. Ordinary degree assigns degree one to each h_j, $j > 0$; weighted degree assigns degree j. The square weight of a monomial is the sum of the squares of its generator indices, with multiplicity. Units contribute zero to all three readings.

Lemma 2.1. *For a connected diagram with full lists X, Y, the maximum ordinary degree of s_D is n. Its unique minimum-square-weight monomial is $\prod_i h_{r_i}$, with coefficient one. The largest generator occurring is h_N, and*

$$(-1)^{n-1}[h_N]s_D = \det(h_{x_{i+1} - y_j})_{1 \le i,j < n}. \tag{2.2}$$

The minimum-square monomial of this cofactor reads the multiset of the z_i, padded with zeros to length $n - 1$. The minimum ordinary degree of s_D is $n - |X \cap Y|$.

Proof. The maximum degree, minimum-square monomial, largest generator, and cofactor readings are the polynomial-readings lemma of [7], applied to the full lists. For the minimum degree put $Z = X \cap Y$, $A = X \setminus Z$, and $B = Y \setminus Z$. The exact common-beta path expansion in [7] gives

$$s_D(th_1, th_2, \ldots) = \epsilon t^{n - |Z|} \det(h_{a_i - b_j}) + \text{terms of higher degree in } t,$$

where $\epsilon \in \{1, -1\}$ is the regrouping sign. The residual strict lists have positive sorted diagonal, so its determinant is nonzero by strict rearrangement. The minimum degree is therefore $n - |Z|$. □

The polynomial readings are also consequences of the overlap theory in [6]. We use the corrected irreducibility theorem of Barekat, Reiner and van Willigenburg [1]: the Schur polynomial of a nonempty connected skew diagram is irreducible in the complete-function algebra. A disconnected diagram contributes the product of its connected component polynomials. Their

coefficient-one minimum-square monomials fix the scalar associates. Thus factorization identifies the component polynomials, not their positions or orientations.

2.2. **The invariant and the pattern.**

Lemma 2.2 (Negative-count invariance)**.** *Connected Schur-equal diagrams have the same row count and the same full-board negative count. In particular, every connected mate of a five-row five-negative board is again a five-row five-negative board.*

Proof. For consecutive rows $j, \dots, j+k-1$, put $o_{j,k} = \max(0, X_{j+k-1} - Y_j + k - 1)$, their intersection size. For $k \geq 2$, the index $X_{j+k-1} - Y_j$ is negative exactly when $o_{j,k} \leq k-2$. The diagonal and upper-triangular indices are positive, so

$$\nu(D) = \sum_{k=2}^{n} \#\{j : 1 \leq j \leq n-k+1,\ o_{j,k} \leq k-2\}.$$

The row count is read by Lemma 2.1. The unordered row-overlap partitions are Schur invariants [6]; padding the k-th partition to length $n-k+1$ recovers its zero parts. Hence every summand is invariant. At a common beta the corresponding overlap is $k-1$, so that zero index is correctly excluded. □

Lemma 2.3. *Let D have n nonempty rows and full beta lists X, Y. Put $c_i = \#\{j : x_i - y_j < 0\}$. Its negative sites are closed southwest, and $c_1 \leq \cdots \leq c_n$. Reading these counts from bottom to top and omitting zero parts gives a partition $\rho(D)$, with*

$$|\rho(D)| = \nu(D), \qquad \rho(D^*) = \rho(D)'.$$

If D is connected, then $c_i \leq \max(0, i-2)$.

Proof. If $x_i < y_j$, strict decrease of both lists gives $x_{i'} < y_{j'}$ whenever $i' \geq i$ and $j' \leq j$. Thus each negative row is an initial segment and these segments increase downward. In a connected board the first subdiagonal is nonnegative, which gives the stated bound.

For normalized lists, rotation sends (X, Y) to

$$\big((N - y_n, \dots, N - y_1), (N - x_n, \dots, N - x_1)\big).$$

Its index at (i, j) is the original index at $(n+1-j, n+1-i)$. In coordinates measured upward and rightward from the southwest corner, this map interchanges the two coordinates. It therefore conjugates the negative partition. □

Example 2.4. The thirteen-cell profiles

$$F = (3,4,3,2,1;1,2,1,0), \qquad H = (4,3,3,2,1;2,1,1,0)$$

have $s_F = s_H$ by Theorem 5.1, at $b = q = 1, g = 2$ in the first family of (5.1). Their negative partitions are

$$\rho(F) = (3,1,1), \qquad \rho(H) = (3,2).$$

The former is self-conjugate, so the two partitions are neither equal nor conjugate. Negative partition is consequently not an invariant of a connected Schur class, even after identifying conjugate partitions.

2.3. **Realizability and minimum row count.**

Theorem 2.5. *A nonempty partition ρ is the negative partition of a connected skew diagram with exactly n nonempty rows if and only if*

$$\rho_i \leq n - i - 1 \qquad (1 \leq i \leq \ell(\rho)). \tag{2.3}$$

Consequently (1.2) *holds, and every partition is realizable when the row count is unrestricted.*

Proof. Connectedness gives $x_j \geq y_{j-1}$ for $j \geq 2$. Thus row j has at most $j-2$ negative entries. Reading the rows from below gives (2.3).

Conversely, pad ρ with zeros to length n, put $t_i = \rho_{n+1-i}$ and $M = n+1$, and set

$$X_i = M(n-t_i) - i, \qquad Y_j = M(n-j) \qquad (1 \leq i, j \leq n). \tag{2.4}$$

The t_i are nondecreasing, so the X_i are strictly decreasing; the Y_j are strictly decreasing and $Y_n = 0$. The board index is

$$X_i - Y_j = M(j-t_i) - i.$$

Since $1 \leq i < M$, it is negative exactly when $j \leq t_i$. The prescribed pattern is therefore obtained. Also $t_1 = 0$ and $t_i \leq i-2$ for $i \geq 2$. The diagonal is positive, and

$$X_i - Y_{i-1} = M(i-1-t_i) - i \geq M - i > 0 \qquad (i \geq 2).$$

The two strict lists consequently define nonempty rows of a connected skew diagram. The least admissible n in (2.3) is exactly (1.2). □

For $1 \leq k \leq n$, the overlap of rows $j, \ldots, j+k-1$ is

$$o_k(j) = \max(0, x_{j+k-1} - y_j + k - 1). \tag{2.5}$$

The multiset of these overlaps, sorted and padded with zeros to length $n-k+1$, is a row-overlap partition. These partitions are Schur invariants [6].

Proposition 2.6. *The quantity $n_{\min}(\rho(D))$ is constant on a connected Schur class. If $\rho(D) \neq \varnothing$, let k_0 be the first order $k \geq 2$ for which the padded overlap partition contains an entry at most $k-2$. Then*

$$n_{\min}(\rho(D)) = n + 3 - k_0. \tag{2.6}$$

Proof. Row count and negative count are invariant by Lemma 2.2; in particular the empty-pattern case stays empty. By (2.5), an index in position $(j+k-1, j)$ is negative exactly when $o_k(j) \leq k-2$. If t_i is the number of negative entries in row i, the first such order is

$$k_0 = 1 + \min_{t_i > 0}(i - t_i).$$

Substituting $\rho_j = t_{n+1-j}$ into (1.2) gives (2.6). Both n and the padded overlap partitions are invariant. This argument includes zero indices and does not recover the ordering of the negative row counts within the board. □

3. Boundary coefficients and propagation

3.1. **Faithful passage to zero-free boards.**

Lemma 3.1. *The map*

$$\Gamma(X, Y) = (2X+1, 2Y) \tag{3.1}$$

is injective on translation classes, commutes with rotation, preserves every negative position, and sends every connected board to a connected zero-free board of the same order. For $s_D = \sum_\alpha c_\alpha h_\alpha$,

$$s_{\Gamma D} = \sum_\alpha c_\alpha h_1^{n-\ell(\alpha)} \prod_{j \in \alpha} h_{2j+1}. \tag{3.2}$$

Consequently any rotation-rigidity statement about a specified negative pattern descends from zero-free boards to full boards.

Proof. An index j becomes $2j+1$, which is negative exactly when $j < 0$. The two lists have opposite parity. Connectedness is preserved, and the profile becomes $(2r+1; 2z+1)$. This also proves injectivity and rotation covariance. In each surviving determinant term the omitted unit factors number $n - \ell(\alpha)$, proving (3.2). Equality supplies the common n by Lemma 2.1, so it transports through this formula. No claim about composition-irreducibility is needed. □

By Lemma 2.1, a connected mate of a zero-free source is zero-free: its minimum and maximum ordinary degrees both equal its row count. The maximal cofactor of a connected zero-free board has positive rows. In the component arguments below, all cofactor components refer to this zero-free setting.

Lemma 3.2. *Suppose the maximal cofactor has connected components of row counts $n_1, \dots, n_r$ and internal negative counts $\nu_1, \dots, \nu_r$. Then*

$$\nu(D) = \sum_{i<j} n_i n_j + \sum_i \nu_i. \tag{3.3}$$

Proof. In the ordered cofactor board every block below a diagonal component block is entirely negative, and every block above it is positive. The diagonal blocks contribute their internal negative counts. The deleted first row and last column of the original board have no negative indices. This gives (3.3). □

For fixed $m = \sum_i n_i$ and r, the cross cost is $(m^2 - \sum_i n_i^2)/2$. Moving one row from a smaller nonsingleton component to a largest component increases the sum of squares. Hence the least cross cost occurs at $(m - r + 1, 1, \dots, 1)$ and equals

$$(r-1)m - \frac{r(r-1)}{2}.$$

3.2. **Connected children.** For a normalized diagram $R = (U, V)$ with nonempty rows and span $S = U_1$, define its positive border and its two endpoint extensions by

$$\mathfrak{B}_{u,v} R = ((S + u, U); (V, -v)), \qquad u, v > 0. \tag{3.4}$$

$$R^{\uparrow t} = ((U_1 + t, U_2, \dots); V), \qquad R^{\downarrow t} = (U; (V_1, \dots, V_m - t)), \qquad t > 0.$$

Lists are translated together when normalization is required.

Lemma 3.3. *Let R be connected and zero-free with $m \geq 2$ rows and span S. For $P = \mathfrak{B}_{u,v} R$, put $N = S + u + v$, $q = \min(u, v)$, $M = N - q$. Then $[h_N] s_P = (-1)^m s_R$. After setting $h_N = 0$, the largest remaining label is M. If $u < v$, its coefficient is $(-1)^{m+1} s_{R^{\uparrow q}}$; if $v < u$, it is $(-1)^{m+1} s_{R^{\downarrow q}}$. If $u = v$,*

$$[h_M^2] s_P = \det(h_{U_i - V_j})_{2 \leq i \leq m,\ 1 \leq j < m} \neq 0. \tag{3.5}$$

Thus the full polynomial and the cofactor class determine $\{u, v\}$. If the literal cofactor has been aligned and its opposite endpoint extensions can be equal only when $R = R^*$, equal borders are rotations.

Proof. The largest noncorner entries are $N - u$ at $(2, m+1)$ and $N - v$ at $(1, m)$. At unequal margins the larger is private and gives the stated connected extension. Its span $S + q$ is smaller than M. At equal margins both sites contribute, with positive joint sign; their complementary sorted diagonal is positive. Every coefficient label in (3.5) is below M. These bounds justify extraction after the corner term has been removed. The span determines $u + v$, and M determines their minimum. A nonliteral swap equates the two endpoint extensions. If $R = R^*$, it rotates the parent. □

Lemma 3.4. *If $\rho(D) \neq \varnothing$ and $n > n_{\min}(\rho(D))$, the zero-free maximal cofactor is connected and has negative partition $\rho(D)$. If $n = n_{\min}(\rho(D))$, it is disconnected. Positive endpoint extensions preserve the negative partition.*

Proof. Deleting the first row and last column removes no negatives. Above the minimum,

$$\rho_i + i \leq n_{\min}(\rho) - 1 \leq n - 2,$$

so every first-subdiagonal entry of the maximal cofactor is positive. At the minimum some $\rho_i + i = n - 1$, placing a negative entry on that subdiagonal and separating components. The two endpoint changes affect only the first row or last column, which are positive, so preserve the negative positions. □

Lemma 3.5. *Every connected diagram with at most three nonempty rows is rotation-rigid. Every connected diagram whose full board is nonnegative is rotation-rigid, in arbitrary row count.*

Proof. By Lemma 3.1, it suffices to treat zero-free boards; Lemma 2.1 supplies the mate's row count and the two multisets. One row is immediate. Two rows have polynomial $h_a h_b - h_{a+b-z} h_z$, recovering $\{a, b\}$ and z. In three rows write the profile as $(a, b, c; s, t)$. Its corner cofactor is $h_s h_t - h_b h_{s+t-b}$. If $s + t - b \geq 0$, its top label recovers b, and its diagonal recovers $\{s, t\}$. After orientation only interchange of a, c remains. For $a \neq c$, $s \neq t$, interchange the comparisons and rotate to have $a > c, s > t$. The largest label left in their difference is $a + b - t$, with coefficient

$$h_t h_c - h_{s+t-b} h_{c+b-s} \neq 0;$$

the latter two indices are respectively below t and above c. If $a = c$ nothing changes; if $s = t$ the interchange is rotation.

If $s + t - b < 0$, first take $s < t$. After removing the row monomial and full-run term, the two negative terms are

$$h_{a+b-s} h_s h_c, \qquad h_a h_{b+c-t} h_t.$$

Only the first carries the smallest index s. Its other factors identify c among the row multiset, since $a + b - s > a, b$; the second then identifies a. When $s = t$, uniform shortening reduces to a three-row ribbon, which is rotation-rigid by [2]. Shortening is performed on the homogeneous determinant with its unit variable retained, so it is injective.

The arbitrary-order nonnegative-board assertion is the nonnegative-index rigidity theorem of [7]. That theorem compares against arbitrary connected mates and allows common beta entries. □

3.3. **A singleton beside a connected component.** For a normalized connected zero-free component $Q = (C, D)$, with $C_1 = S, D_m = 0$, and $s, u, v, g > 0$, put $a = s + g$, $z = v + S + g$, and

$$\begin{aligned} X &= (z + s + u, z + s, v + C_1, \ldots, v + C_m), \\ Y &= (z, v + D_1, \ldots, v + D_m, 0). \end{aligned} \tag{3.6}$$

The maximal cofactor is a singleton of length s followed by Q. The separation g is a gap between these cofactor components, not a parent overlap. Write $\Phi_Q(w, t) = s_{\mathfrak{B}_{w,t}Q}$. First-column expansion gives the full identity

$$P_Q = h_{s+u} \Phi_Q(a, v) - h_s \Phi_Q(a + u, v). \tag{3.7}$$

For any normalized zero-free diagram $Q = (C, D)$ with $m \geq 2$ nonempty rows, put $S = C_1$, $D_m = 0$, and $D_Q(t) = s_{Q^{\uparrow t}} - s_{Q^{\downarrow t}}$, for $t > 0$. Let M_{ij} be its unsigned minors. Canceling the common moving corner in the top-row and last-column Laplace expansions gives

$$D_Q(t) = \sum_{j=1}^{m-1} (-1)^{1+j} h_{S - D_j + t} M_{1j} - \sum_{i=2}^{m} (-1)^{i+m} h_{C_i + t} M_{im}. \tag{3.8}$$

Every label in a first-row cofactor is at most C_2, and every label in a last-column cofactor is at most $S - D_{m-1}$.

Lemma 3.6. *Suppose $Q \neq Q^*$ is connected and zero-free, and either is dense of order $m \geq 2$, or has exactly one negative and order three or four. For every $t > 0$,*

$$\max \operatorname{label} D_Q(t) = \kappa + t, \qquad [h_{\kappa+t}] D_Q(t) = c_Q \neq 0, \quad \kappa = \max(C_2, S - D_{m-1}), \tag{3.9}$$

where $0 < \kappa < S$ and c_Q is independent of t. Also $D_Q(t) = 0$ for symmetric Q.

Proof. The cofactor bounds in (3.8) show that only the top level can carry $h_{\kappa+t}$. Off the tie $C_2 = S - D_{m-1}$, there is one nonzero minor. At the tie, the coefficient is the difference of the opposite endpoint extensions of the maximal cofactor, up to sign. For a dense component of order at least three that child is connected and dense; for a four-row one-negative component it is connected with three rows. Their rigidity, proved in Lemma 3.5, makes the difference nonzero

unless the child is symmetric. Equal exterior gaps would then make Q symmetric. In order two the same conclusion follows directly.

For a three-row one-negative component write its strict endpoint chain as

$$0 < D_2 < C_3 < D_1 < C_2 < S,$$

with consecutive gaps $(\alpha, \pi, \theta, \rho, \gamma)$. At the top tie $\gamma = \alpha$, the two minors have difference

$$h_\rho h_{\alpha+\pi} - h_{\rho+\gamma} h_\pi,$$

which vanishes exactly when $\rho = \pi$. Together with $\gamma = \alpha$, that is exactly symmetry. Off the tie only the appropriate single minor is used, not their difference. All cofactor labels are at most κ, completing the claimed extraction. □

Lemma 3.7. *Let Q be connected and zero-free of order at least two. Assume that Q and its positive endpoint extensions are rotation-rigid against arbitrary connected mates, and that non-symmetric Q satisfies the nonzero, shift-independent leading-coefficient assertion of* (3.9)*. Every connected zero-free parent whose maximal cofactor consists of a singleton and Q is rotation-rigid.*

Proof. By Lemma 2.1, an arbitrary mate is zero-free of order $m+2$. Its corner coefficient is $\varepsilon h_s s_Q$, where $\varepsilon = (-1)^{m+1}$. Unique factorization separates the singleton from the degree-m irreducible factor. Component rigidity identifies that factor with Q or Q^*. Rotate the mate to put the singleton first; it has exactly the form (3.6).

Put $N = S + a + u + v$, $q = \min(u, v)$, $M = N - q$. Expansion gives

$$-\varepsilon[h_M]P_Q = \begin{cases} h_{s+q} s_Q, & u < v, \\ h_s s_{Q^{\downarrow q}}, & v < u, \\ h_{s+q} s_Q + h_s s_{Q^{\downarrow q}}, & u = v. \end{cases} \tag{3.10}$$

All coefficient labels are below M. Divisibility by h_s, h_{s+q} distinguishes the three cases: respectively only the second, only the first, or neither divides. The tied sum is nonzero by Schur positivity. The connected degree-m factors cannot have a linear factor. When $v \le u$, the endpoint rigidity of Q fixes its orientation. At a tie N fixes g; for $v < u$, the second terms of (3.7) agree, and unique factorization of the first terms fixes u, then g.

Suppose $u < v$. The polynomial fixes u and $a + v$. The entire first two terms of last-column expansion are the common part

$$J = \varepsilon s_Q (h_s h_N - h_{s+u} h_{N-u}).$$

If two candidates have $v_- < v_+$, subtract J and set $K = \max(N - v_-, S + v_+)$. For the difference "larger bottom margin minus smaller bottom margin," its coefficient at K is

$$\varepsilon \begin{cases} h_s s_{Q_-^{\downarrow v_-}}, & N - v_- > S + v_+, \\ s_T, & N - v_- < S + v_+, \\ h_s s_{Q_-^{\downarrow v_-}} + s_T, & N - v_- = S + v_+. \end{cases} \tag{3.11}$$

Here T deletes the first component row and last column from the larger-margin parent. Its subdiagonal labels are $s, C_2 - D_1, \ldots, C_m - D_{m-1} > 0$, so it is connected. Its span is $N - v_+ < K$; the product's labels are at most $\max(s, S + v_-) < K$; and $K > \max(S, s + u)$. The tied coefficient is a same-sign sum of nonzero Schur-positive functions. Thus v, and then g, are fixed.

Only reflection of a nonsymmetric component at these unchanged margins remains. Write

$$\mathcal{D}(w) = \Phi_Q(w, v) - \Phi_Q(v, w).$$

Rotation identifies the second term with $\Phi_{Q^*}(w, v)$. This difference is zero at $w = v$; otherwise its largest label is $S + \max(w, v)$, with a signed coefficient $D_Q(\min(w, v))$. The full difference is

$$h_{s+u} \mathcal{D}(a) - h_s \mathcal{D}(a + u).$$

A zero summand leaves the other nonzero. For $v < a + u$, the second nonzero summand has strictly larger top label. For $v > a + u$, both tops are $S + v$, and their coefficient, up to sign, is

$$h_{s+u} D_Q(a) - h_s D_Q(a + u). \tag{3.12}$$

The second term has label $\kappa + a + u$, greater than both $\kappa + a$ and $s + u$, since $a = s + g > s$. It cannot cancel. This includes equal endpoint rows of Q, unchanged margins, and a top attained twice in both border differences. □

Corollary 3.8. *Every parent in* (3.6) *is rotation-rigid if Q is dense of order $m \geq 2$, or has exactly one negative and order three or four. Its parameters s, u, v, g are positive integers.*

Proof. Dense components and three-row components are rigid by Lemma 3.5. A one-negative four-row component has a connected three-row maximal cofactor, so Lemmas 3.3 and 3.5 give its rigidity as well. The same arguments apply to its endpoint extensions, which preserve negative positions. The leading-coefficient hypothesis is Lemma 3.6; apply Lemma 3.7. □

4. Classification through four negatives

4.1. **Four rows and the two-negative lift.** The four-row classification needed below is stated first. Its proof is in Appendix A; it is an algebraic case calculation, not a census.

Proposition 4.1. *For a connected diagram with exactly four nonempty rows, every connected mate is its rotation, except for the complete classes $\{F, F^*, H, H^*\}$ with profiles*

$$\begin{aligned} F &= (w, b, b, b + w - c; c, c, w), \\ H &= (w, b, b + w - c, b; c, w, c), \end{aligned} \qquad 0 \leq c < w, \quad b > c + w. \tag{4.1}$$

All four members are distinct and have exactly three negative indices.

At $(c, w, b) = (0, 1, 2)$, the displayed family gives the eight-cell pair

$$(1, 2, 2, 3; 0, 0, 1), \qquad (1, 2, 3, 2; 0, 1, 0).$$

Theorem 4.2. *A connected diagram with at most two negative indices in its full row board is rotation-rigid, in every row count and with common betas allowed.*

Proof. Work zero-free by Lemma 3.1. The cases through four rows are Lemma 3.5 and proposition 4.1. Above four rows the maximal cofactor has $m = n - 1 \geq 4$ positive rows. A split after row k would require $k(m - k) \geq m - 1 \geq 3$ negatives. Thus it is connected and has at most two negatives; its endpoint extensions retain those same negative positions. Induction and Lemma 3.3 apply, including to the arbitrary mate after factorization of its maximal cofactor. □

4.2. **Columns and three corners.**

Theorem 4.3. *Let D be connected with $n \geq 2$ nonempty rows. If $x_n \geq y_2$, or if $x_{n-1} \geq y_1$, every connected mate is a rotation. Common betas are allowed, and there is no bound on the number of negatives.*

Proof. Rotate to use the first condition, and homogenize. Through four rows there are at most two negatives. A connected maximal cofactor remains in the same single-column class, as do its endpoint extensions, so induction and Lemma 3.3 apply. If disconnected, only its first subdiagonal can separate: the cofactor is a singleton followed by a dense block. Corollary 3.8 applies. Descend by Lemma 3.1. □

Lemma 4.4. *Let R_1, R_2 be the two displayed four-row profiles in* (4.1)*, with $1 \leq c < w$, $b > c + w$. No positive border of R_1 is Schur-equal to a positive border of R_2, including either orientation of each cofactor.*

Proof. Put $t = b - c$, $d = w - c$, and $S = 3t + w$. Both cofactor spans are S. The border readings give common unordered margins. In fixed orientations the parent rows are

$$(w + u, b + d, 2b - c, 2b - c, b + d + v),$$
$$(w + u', b + d, 2b - c, 2b - c + d, b + v').$$

Canceling common entries in their multisets gives: at the same margins, $v = t$; at interchanged margins, either $u = v = t$ or $(u, v) = (2t, 2t - d)$. These exhaust the two possible matches for the fixed entry $2b - c + d$.

For $v = t, u < t$, the marker compares top extensions of the two children. Every row of these extensions exceeds w, whereas an exceptional four-row diagram with overlap multiset $\{c, c, w\}$ has a row w. They cannot be rotations either, since their overlap words differ even up to reversal. For $v = t, u > t$, the two bottom extensions have different row multisets: after canceling w, b, one contains b and the other only larger values. At the unequal swapped margins the marker compares a bottom extension still containing w with a top extension whose rows all exceed it.

At $u = v = t$, put $M = 4t + w$, $T = 3b - 2c$. The eight-term four-row expansion of the endpoint cofactors gives

$$\begin{aligned}[h_M](s_{\mathfrak{B}_{t,t}R_1} - s_{\mathfrak{B}_{t,t}R_2}) = -h_w\big((h_c h_{b+d} - h_w h_b)h_T \\ + (h_b^2 - h_c h_{2b-c})h_{2b-c+d} \\ + h_w h_{2b-c}^2 - h_b h_{b+d} h_{2b-c}\big).\end{aligned}$$

The square coefficients cancel. Since $c < w < b < b + d < 2b - c < 2b - c + d < T < M$, the double coefficient is

$$[h_M h_T](s_{\mathfrak{B}_{t,t}R_1} - s_{\mathfrak{B}_{t,t}R_2}) = -h_w(h_c h_{b+d} - h_w h_b) \neq 0. \tag{4.2}$$

Rotating whole parents handles either orientation of the children. □

Proposition 4.5. *Let D be connected with $n \geq 5$ rows and*

$$x_{n-2} \geq y_1, \qquad x_{n-1} \geq y_2, \qquad x_n \geq y_3.$$

Then D is rotation-rigid. Common betas are allowed.

Proof. Homogenize first. The maximal cofactor is connected, zero-free and has $n - 1$ positive rows: its subdiagonals are $x_{i+2} - y_i > 0$. For five rows, Proposition 4.1 identifies its complete class, and Lemma 4.4 excludes crossing between the exceptional orbits. After aligning the literal cofactor, a margin swap would equate its opposite endpoint extensions. These have the same ordered overlap word, whereas the two exceptional orbits in (4.1) have inequivalent overlap words. Thus the extensions are rotations, forcing a symmetric child. For $n \geq 6$, the child retains the three displayed inequalities one order down; so do its endpoint extensions. Induction, Lemma 3.3, and Lemma 3.1 finish the proof. □

4.3. **Square and hook.** A two-row block $Q = \langle A, B, d\rangle$ has beta lists $(A + B - d, B; B - d, 0)$ and polynomial

$$f_Q = h_A h_B - h_{A+B-d} h_d, \qquad 0 \leq d < \min(A, B).$$

Its rotation exchanges A, B. Superscripts $\uparrow, \downarrow$ continue to denote endpoint extension.

Theorem 4.6 (Square rigidity)**.** *Let F be connected with $n \geq 5$ rows and negative positions exactly*

$$\{(n-1, 1), (n-1, 2), (n, 1), (n, 2)\}.$$

Then F is rotation-rigid. Common betas are allowed.

Proof. First take five rows. Its maximal cofactor consists of two two-row blocks $Q = \langle A, B, d\rangle$, $R = \langle C, D, e\rangle$. Its maximum ordinary degree is four, even when a block has a unit. Hence an equal mate's cofactor has four positive rows. Unique factorization forces the same two blocks up to order and rotation. This recovers the square pattern on the mate without assuming it.

Now work zero-free, so $d, e > 0$. Set $q_1 = A + B - d$, $P = C + D - e$, $s_1 = D - e$, $\gamma = P + g$, $W = v + \gamma$, with $u, v, g > 0$. All sources have the unique form

$$\begin{aligned}X &= (W + q_1 + u, W + q_1, W + B, v + P, v + D), \\ Y &= (W + B - d, W, v + s_1, v, 0).\end{aligned} \tag{4.3}$$

Its row word is $(A + u, q_1, B + C + g, P, D + v)$, its overlap word is (A, B, C, D), and its span is $N = q_1 + P + u + v + g$. Rotation sends (Q, R, u, v, g) to (R^*, Q^*, v, u, g).

Put

$$\Phi(t) = \det \begin{pmatrix} h_{A+u} & h_{q_1+u} & h_{t+q_1+u} \\ h_A & h_{q_1} & h_{t+q_1} \\ h_d & h_B & h_{t+B} \end{pmatrix}.$$

For $t > 0$ this is a connected three-row Schur polynomial. If $\sigma = s_{Q^{\uparrow u}}$, $\tau = s_{\langle A+u,q_1,A\rangle}$, expansion gives

$$\Phi(t) = h_{t+q_1+u} f_Q - h_{t+q_1}\sigma + h_{t+B}\tau. \tag{4.4}$$

The three nonzero bottom-row-pair minors give

$$s_F = f_R\Phi(W) - s_{R^{\downarrow v}}\Phi(\gamma) + s_{\langle P,D+v,D\rangle}\Phi(\gamma - s_1). \tag{4.5}$$

All three arguments are positive. The corner coefficient is $f_Q f_R$. The next label $M = N - \min(u,v)$ has coefficient

$$[h_M]s_F = -\begin{cases} s_{Q^{\uparrow u}} f_R, & u < v, \\ f_Q s_{R^{\downarrow v}}, & v < u, \\ s_{Q^{\uparrow u}} f_R + f_Q s_{R^{\downarrow v}}, & u = v. \end{cases} \tag{4.6}$$

Every coefficient label is below M.

Orient both parents so $u \le v$, $u' \le v'$. The marker fixes $u = u'$. With strict margins, unique factorization identifies the unextended factor and then the orientation of Q; consequently Q and $W = N - q_1 - u$ agree. This remains true when $f_Q = f_R$, by canceling that factor. Strict and tied margins cannot mix. If the two factors differ, divisibility by the old factors distinguishes them; if they agree, cancellation leaves one endpoint extension against a sum of two, whose coefficients at $h_{q_1+u}h_d$ are -1 and -2.

At tied margins the order of unequal factors can be aligned by global rotation. An orientation change has marker difference $E_Q f_R + f_Q E_R$, where

$$E_Q = h_{A+u}h_B - h_{B+u}h_A, \qquad E_R = h_C h_{D+u} - h_D h_{C+u},$$

and either summand may be absent. A nonzero E_Q is not divisible by f_Q: both have ordinary degree two and different weighted degrees. Reduction modulo the distinct irreducible factors forces each orientation to agree. For equal factors put $p = h_{A+u}h_B$, $r = h_A h_{B+u}$. The four marker values, after their common term is removed, are $p+r, p+r, 2p, 2r$. Their only nontrivial coincidence is the global rotation pair, unless $A = B$, when the block is already symmetric. The span then fixes g.

It remains to treat strict margins, with Q, u, W fixed. Subtract the common first term of (4.5). If $v' < v$, the largest possible label in the remaining difference is $K = \max(N - v', P + v)$. When the first is larger, its private coefficient is $-s_{R'^{\downarrow v'}} f_Q \neq 0$. When the second is larger, vanishing would require $h_e\Phi(\gamma) = h_D\Phi(\gamma - s_1)$, impossible by unique factorization of the connected cubics and $e < D$.

At equality of those two labels, vanishing forces

$$f_Q \mid h_D\Phi(\gamma - s_1) - h_e\Phi(\gamma).$$

The two column syzygies of the displayed 3-by-2 matrix give $h_A\sigma \equiv h_d\tau \pmod{f_Q}$, with $\tau \not\equiv 0$. More generally, $f_Q \mid h_{c_1}\Phi(t_1) - h_{c_2}\Phi(t_2)$ implies

$$\begin{aligned} f_Q \mid\ & h_d h_{c_2} h_{t_2+q_1} - h_d h_{c_1} h_{t_1+q_1} \\ & - h_A h_{c_2} h_{t_2+B} + h_A h_{c_1} h_{t_1+B}. \end{aligned} \tag{4.7}$$

Indeed multiply by h_A and cancel τ in the integral domain $\mathbb{Q}[h]/(f_Q)$. For $(c_1,t_1) = (D, C+g)$, $(c_2,t_2) = (e, P+g)$, the unique top label on the right is $P + g + q_1$, with coefficient $h_d h_e$. This label exceeds all labels of f_Q. A cubic divisible by the quadratic f_Q would have at that label a scalar multiple of its two-term polynomial, not a single nonzero monomial. Thus the tie is impossible, and $v' = v$.

Only $R' = R^*$ remains. If $C \neq D$, comparing $P+v$ with $N-v$ gives respectively $h_C\Phi(D+g) = h_D\Phi(C+g)$, or $(h_D h_{C+v} - h_C h_{D+v})f_Q = 0$; both are impossible. At the tie use (4.7) with $(c_1,t_1) = (D, C+g)$, $(c_2,t_2) = (C, D+g)$. Its unique top coefficient is $h_d h_C$ or $-h_d h_D$, giving the same contradiction. Thus $R' = R$ and $g' = g$.

For $n \ge 6$, the maximal cofactor is connected: a negative at its subdiagonal would require simultaneously $i \le 2$ and $i + 2 \ge n - 1$, which is impossible. It is the square one order down;

endpoint extensions preserve that exact negative set. Induction and Lemma 3.3 prove rigidity. Homogenization supplies common betas. □

Remark 4.7. The full-board degree assertion in the proof is a maximum, not homogeneity. For $X = (13, 12, 9, 6, 5)$, $Y = (9, 7, 4, 2, 0)$, the maximal cofactor is

$$(h_3h_2 - h_5)(h_2h_3 - h_4h_1),$$

with ordinary degrees three and four. Its maximum still recovers four nonempty cofactor rows.

Proposition 4.8. *A connected diagram with $n \geq 5$ rows and negative set*

$$\{(n-2, 1), (n-1, 1), (n, 1), (n, 2)\}$$

or its rotation is rotation-rigid, with common betas allowed.

Proof. For a zero-free five-row source the maximal cofactor is a singleton and a three-row connected component with one negative. Apply Corollary 3.8. For larger sources its maximal cofactor is connected and is the hook one order down. Induction and Lemma 3.3 apply; the endpoint extensions remain in the same class. Homogenization handles units. In particular, at five rows the full chain is only

$$0 = y_5 < y_4 < y_3 \leq x_5 < y_2 \leq x_4 < x_3 < y_1 \leq x_2 < x_1;$$

none of its three weak inequalities is silently made strict before homogenization. □

Proof of Theorem 1.1(i)–(iii). Since D is nonempty and connected, s_D is nonconstant and irreducible by [1]. An empty diagram contributes 1, whereas every nonempty connected component, including a single row or column, has a nonconstant irreducible Schur function. Component factorization and $s_H = s_D$ therefore force H to have exactly one nonempty component. This reduction also applies to parts (iv) and (v).

By Lemma 2.3, the negative sites form a partition, rotation conjugates it, and $\nu \leq (n-1)(n-2)/2$. At three negatives the shapes are a column, a row, or the three-cell staircase. For $n \geq 5$, use Theorem 4.3 and proposition 4.5; at four rows use Proposition 4.1. At four negatives $n \geq 5$; up to rotation their shapes are a column, a square, or a hook. Apply Theorems 4.3 and 4.6 and proposition 4.8. The first assertion is Theorem 4.2. □

5. Five negatives

Theorem 5.1 (Five-row five-negative classification)**.** *Let D be connected with five nonempty rows and exactly five full-board negative indices. Common betas are allowed. Its complete connected Schur class is its rotation orbit unless it is one of the following classes. For integers $1 \leq q \leq b < g$, put*

$$X = (b + 4g, b + 3g, b + 2g, b + g, b),$$

(5.1)

i	$Y(F_i)$	$Y(H_i)$
1	$(3g, g+q, g, q, 0)$	$(2g+q, 2g, g, q, 0)$
2	$(3g+q, g+q, g, q, 0)$	$(2g+q, 2g, g+q, q, 0)$.

The complete class is $\{F_i, F_i^*, H_i, H_i^*\}$. All four members are distinct.

5.1. **One opened cell and the sixteen-plus-eight expansion.** A five-row connected board has no negative entry on or above the first subdiagonal. There are six positions below that subdiagonal. If exactly five are negative, the remaining position must be one of $(3, 1), (4, 2), (5, 3)$: a nonnegative entry farther southwest would force at least two of these six positions to be nonnegative. Call the three openings E, M, F, respectively. Rotation fixes M and interchanges E, F. Equivalently, the negative row-count words are $(0, 0, 0, 2, 3), (0, 0, 1, 1, 3), (0, 0, 1, 2, 2)$.

In this section retain the unit variable. For a board with n nonempty rows write

$$P_D = \det(H_{X_i - Y_j}) = H_0^n s_D(H_1/H_0, H_2/H_0, \ldots), \qquad H_j = 0 \quad (j < 0),$$

in $\mathbb{Q}[H_0, H_1, \ldots]$, with all nonnegative-index variables independent. Equality of the original polynomials is equivalent to equality of these homogeneous polynomials, since the row count is read by Lemma 2.1.

Connected-skew-Schur irreducibility [1] also holds after this homogenization. Indeed, the diagonal monomial is the unique minimum-square-weight monomial and has coefficient one, so the ordinary degree is exactly n and $H_0 \nmid P_D$. A factorization of the homogeneous polynomial has homogeneous factors. After setting $H_0 = 1$, irreducibility over $\mathbb{Q}$, obtained from the corrected integral statement by Gauss's lemma, makes one factor constant. A homogeneous factor specializing to a nonzero constant is a scalar power of H_0. That power is zero because $H_0 \nmid P_D$. Thus P_D is irreducible. The coefficient-one minimum-square monomial fixes scalar associates whenever connected factors are compared below.

For a five-negative board let $A^{(0)}$ be its matrix with the single opened second-subdiagonal position $(k+2, k)$ set to zero, and put $\tau = X_{k+2} - Y_k \geq 0$. This is a change of one matrix position, not a specialization of every occurrence of H_τ. Multilinearity gives

$$P_D = \det A^{(0)} + H_\tau C_{k+2,k}(A^{(0)}), \qquad k \in \{1, 2, 3\}, \tag{5.2}$$

where C_{ij} denotes the signed cofactor. The first determinant is

$$\det A^{(0)} = \sum_{S \subseteq \{1,2,3,4\}} (-1)^{|S|} \prod_{i \in S} H_{z_i} \prod_{[a,b] \in \mathcal{I}(S)} H_{w_{a\cdots b}}, \qquad w_{a\cdots b} = \sum_{i=a}^{b} r_i - \sum_{i=a}^{b-1} z_i.$$

Here $\mathcal{I}(S)$ joins adjacent rows exactly across the edges in S. To prove the expansion, if the first row uses column j, rows $2, \ldots, j$ must use columns $1, \ldots, j-1$; remove that interval and repeat. These are the sixteen interval terms. After the opened row and column are deleted, the four remaining rows have nested supports of sizes $4, 4, 3, 2$. The last row has two choices, the next has two remaining choices, and the first two rows have two matchings. Thus the second term of (5.2) has eight supported permutation terms. Interchanging the first two remaining rows pairs their signs. The full expansion has twelve terms of each sign, before collection. Repeated labels and common betas may merge or cancel terms; no argument below treats the two layers as separately invariant.

5.2. **Corner readings and a prime obstruction.** After orienting an end opening as E, use the normal forms

$$M = (x, y, L, c, b;\ a, u, v, p), \qquad E = (x, L, y, c, b;\ u, v, a, p), \qquad L = u + v - \tau. \tag{5.3}$$

Their complete validity conditions are

$$u, v > \tau \geq 0, \qquad a, p \geq 0, \qquad b > p,$$

$$M: \quad x > a, \quad y > a + u, \quad c > v + p; \qquad E: \quad x > u, \quad y > a + v, \quad c > a + p.$$

The three second-subdiagonal indices are $z_i + z_{i+1} - r_{i+1}$; these inequalities make exactly the indicated one nonnegative. Strict ordering then forces every more southwestern entry to be negative.

Put $q = b - p$, $w = c + q$, and define

$$\begin{aligned} B &= H_u H_v - H_\tau H_L, & F_s(z) &= H_s H_z - H_\tau H_{z+s-\tau}, \\ T(z) &= H_b H_z - H_p H_{z+q}, & G_s(z) &= H_s T(z) - H_\tau T(z + s - \tau), \\ Q_s(z) &= H_x F_s(z) - H_a F_s(z + x - a). \end{aligned}$$

The symbol B in these subsections denotes the displayed binomial, not a beta list. It is primitive and linear in its unique largest variable H_L, so it is irreducible and has no coordinate-variable factor. The same argument applies to $F_s(z)$ for $s, z > \tau$ and to $T(z)$ for $z > p$. On an M-board, $G_u(c), Q_v(y)$ are the connected three-row determinants $(u, c, b; \tau, p)$, $(x, y, v; a, \tau)$. Indeed, $c > v+p > \tau+p$ and $y > a+u > a+\tau$, and the remaining endpoint inequalities are inherited from that board. They are therefore irreducible after homogenization. These cubic identifications are not asserted for arbitrary E-parameters. Every candidate is required to satisfy its own pattern inequalities. The single opening supplies the prime factor B; closing it gives the product $H_u H_v$, so this factorization argument does not extend to the six-negative mask.

Let $m = X_1$, and put $\delta = X_1 - X_2$, equal to $x-a$ for M and $x-u$ for E. Direct complementary expansion gives

$$C_{15} = H_a H_p B, \qquad C_{14} = -H_a H_b B,$$
$$C_{25}^{M} = -H_x H_p B, \qquad C_{25}^{E} = -H_a H_p F_v(x),$$

$$N = m - \min(q, \delta), \qquad [H_N](P_D - H_m C_{15}) = \begin{cases} C_{14}, & q < \delta, \\ C_{25}, & \delta < q, \\ C_{14} + C_{25}, & q = \delta. \end{cases} \tag{5.4}$$

The largest generator occurs only at $(1,5)$. Its nonzero cofactor reads $m, \tau, L, \{u, v\}, \{a, p\}$. Only $(1,4), (2,5)$ can carry the next index N, and these positions cannot occur together: row five would have no supported column left. Their coefficients in (5.4) are nonzero, including at ties.

For M, divide the second coefficient algebraically by B, then set all remaining variables to one. The values are -1 in a strict regime and -2 at a tie. Rotation interchanges q, δ. For E, writing $C = C_{15}$ and P for the second coefficient, the normalized greatest common divisors are

$$\gcd(C, P) \sim \begin{cases} H_a B, & q < \delta, \\ H_a H_p, & \delta < q, \\ H_a, & q = \delta. \end{cases}$$

Their ordinary degrees are $3, 2, 1$. At a tie the parenthesis in $-H_a(H_b B + H_p F_v(x))$ is divisible by neither H_p nor B: $b > p$, and $F_v(x)$ is an irreducible quadratic distinct from B, since $x > u$.

Lemma 5.2 (Prime obstruction)**.** *Suppose $u > v > \tau \geq 0$, $b > p \geq 0$, and $c > v + p$. Put $d = c + u - v$. Then*

$$B \nmid H_u T(c) - H_v T(d).$$

Proof. Write $P = H_u T(c) - H_v T(d)$, and suppose $B \mid P$. We have $d > u + p$. Set $H_u = H_\tau = 0$. The divisor becomes zero, while

$$P \longmapsto H_v\big(H_p H_{d+q} - H_b H_d\big),$$

with H_p, H_b also specialized if their indices are u or τ. The factors H_v, H_d, H_{d+q} survive. If both monomials in parentheses survive, they are distinct: $p < b$ and $p < d$, so $\{p, d+q\} \neq \{b, d\}$. Exactly one surviving monomial is also impossible. Thus both must vanish, forcing $p = \tau, b = u$. Now set $H_v = H_\tau = 0$ instead. Again B becomes zero, but P becomes $H_u^2 H_c \neq 0$, since $c > v + \tau$. This contradiction proves the assertion.

In particular, no inequality $d > b$ is assumed. If $d \leq b$, then $b \geq d > u > \tau$, so the first specialization already leaves the nonzero term $-H_v H_b H_d$, or $-H_v H_b^2$ when $d = b$. Its only possible companion cannot cancel it by the same multiset comparison. This includes $\tau = 0$, where H_τ is the independent variable H_0. □

5.3. **Same-opening rigidity.**

Proposition 5.3. *Equal five-negative M-boards are identical or rotations. Equal five-negative E-boards, both oriented with opening $(3,1)$, are identical after translation.*

Proof. All comparisons are between collected homogeneous polynomials. A supported complementary minor is nonzero: its order-preserving matching is supported and uniquely minimizes square weight. This justifies using a unique largest remaining matrix index throughout the proof.

Middle opening with both endpoints fixed. Suppose first that x, a, b, p agree. The row multiset and the first cofactor leave only

$$\{y, c\} = \{\eta, d\}, \qquad \{u, v\} = \{U, V\}. \tag{5.5}$$

Put $A(z) = H_x H_z - H_a H_{z+x-a}$. Subtract the common contribution

$$B\big(H_a H_p H_m - H_a H_b H_{m-q} - H_x H_p H_{m-\delta} + H_x H_b H_{m-\delta-q}\big).$$

Its four permutation sets are disjoint, since row five already uses one of columns four and five. The remainder is

$$\begin{aligned}\mathcal{R} = {} & H_L A(y)T(c) - H_v A(y)T(c+u-\tau) \\ & - H_u A(y+v-\tau)T(c) + H_\tau A(y+v-\tau)T(c+u-\tau).\end{aligned} \tag{5.6}$$

Its maximal candidate positions are $(1,3),(3,5)$, with indices $y+v-\tau+\delta, c+u-\tau+q$ and cofactors $H_a G_u(c), H_p Q_v(y)$.

If only the core is interchanged, orient $u > v$. A unique maximum cannot vanish, so equality forces $y+\delta = c+q$. The coefficient at the resulting contact is

$$H_p H_x F_v(y) - H_a H_b F_v(c).$$

Unique factorization gives $p = a, x = b, y = c$, exactly rotation.

If only y, c are interchanged, orient $y > c$. The maximum forces $v+\delta = u+q$, with coefficient $H_a G_u(c) - H_p Q_v(c)$. Irreducibility gives $a = p$ and equality of the cubics. Their row multisets give $\{u,b\} = \{x,v\}$. The alternative $u = v, b = x$ is rotation. Otherwise $x = u, b = v$. The remaining cubic comparison is between $(u,c,v;\tau,p)$ and $(u,c,v;p,\tau)$. If $u \neq v$ and $p \neq \tau$, the four noncommon high indices

$$c+u-p, \quad c+u-\tau, \quad c+v-p, \quad c+v-\tau$$

have a unique maximum with nonzero coefficient. Hence $p = \tau$. For the resulting candidate pair

$$(u,y,L,c,v;\tau,u,v,\tau), \qquad (u,c,L,y,v;\tau,u,v,\tau),$$

the full difference is $-B\mathcal{S}$, where

$$\begin{aligned}\mathcal{S} = {} & H_u(H_c H_{y+v-\tau} - H_y H_{c+v-\tau}) \\ & - H_v(H_c H_{y+u-\tau} - H_y H_{c+u-\tau}) \\ & + H_\tau(H_{c+v-\tau} H_{y+u-\tau} - H_{c+u-\tau} H_{y+v-\tau}).\end{aligned} \tag{5.7}$$

For $u > v$, its largest-generator coefficient is $-F_v(c)$ at $H_{y+u-\tau}$; for $v > u$, it is $F_u(c)$ at $H_{y+v-\tau}$. Neither vanishes. Thus this distinct candidate is not an equality.

If both pairs are interchanged, put $h = y+v$, $k = c+u$. The four candidate indices are

$$h+\delta-\tau, \quad k+q-\tau, \quad k+\delta-\tau, \quad h+q-\tau.$$

For $h \neq k$ and $\delta \neq q$ the maximum is unique. If $h \neq k$, equality therefore forces $\delta = q$; the contact coefficient is, up to sign, $(H_a H_b - H_p H_x)F_u(c)$, or its counterpart with $F_v(y)$. It forces $a = p, b = x$, giving rotation. If $h = k$ but $\delta \neq q$, equality of the active cubic cofactors reads $\{u,c\} = \{v,y\}$. The crossed assignment requires $c = v, y = u$, contrary to validity; the other assignment is the unchanged profile. At $h = k, \delta = q$, the two maximal positions can occur together. Both determinants have the same quadratic term $H_a H_p H_\tau H_\nu^2$, where $\nu = h+\delta-\tau$. After canceling that term, the linear coefficient of the difference is

$$(H_a H_b - H_p H_x)(H_u H_c - H_v H_y).$$

The second factor is nonzero for a distinct candidate, again by $c > v, y > u$. The first forces rotation. This proves the assertion with both endpoints fixed.

Middle opening with only the bottom endpoint fixed. Now assume only a, b, p agree, and write the mate as

$$M' = (\xi,\eta,L,d,b;\ a,U,V,p), \qquad \{\xi,\eta,d\} = \{x,y,c\}, \quad \{U,V\} = \{u,v\}.$$

After subtracting the two common top-right terms, the maximal candidates are

$$R = x+y+v-a-\tau, \quad R' = \xi+\eta+V-a-\tau, \quad Z = c+q+y-\tau, \quad Z' = d+q+\eta-\tau.$$

Let $\epsilon, \epsilon', \zeta, \zeta'$ be the indicators of their attaining the maximum. Its collected coefficient is

$$H_a\big(\epsilon G_u(c) - \epsilon' G_U(d)\big) - H_p B\big(\zeta H_x - \zeta' H_\xi\big).$$

There is no quadratic term: $(1,3),(2,5)$ cannot occur together. If a top candidate is active, equality forces $B \mid \epsilon G_u(c) - \epsilon' G_U(d)$. A single active top candidate is impossible by irreducibility

of the cubic. Thus both are active, and the row sum gives $d = c+V-v$. A genuine core reversal, oriented with $u > v$, would imply

$$B \mid G_u(c) - G_v(c+u-v) = H_u T(c) - H_v T(c+u-v),$$

contrary to Lemma 5.2. Hence $d = c, U = u, V = v$. Only x, y can now interchange. First-column expansion reduces that equality to $H_x \mathcal{A}_y = H_y \mathcal{A}_x$, where $\mathcal{A}_z$ is the connected four-row determinant $(z, L, c, b; u, v, p)$. Coordinate-factor-freeness excludes $x \neq y$. If no top candidate is active, both right candidates are active and force $x = \xi$. Both endpoints are fixed, so the preceding argument applies.

For unrestricted M/M, rotate both boards so $q \leq \delta$. The second-corner evaluation separates strict and tied regimes. In the strict regime, C_{15}, C_{14} give the products $H_a H_p, H_a H_b$. Their reduced Laurent quotient H_b/H_p reads b, p, since $b > p$, and then a. In the tied regime, $m - N$ reads the common value $q = \delta = d_0$. Rotate one board if necessary to align $\{a, p\}$. Then $x = a + d_0, b = p + d_0$, so both endpoints agree. This proves the middle-opening assertion.

End opening: a joint-anchor identity. Put

$$A_0 = H_x H_L - H_u H_{x+v-\tau}, \qquad J_z = A_0 H_z - F_v(x) H_{z+u-\tau} + B H_{x+z-\tau}.$$

For $z > v$, J_z is the irreducible connected three-row determinant $(x, L, z; u, v)$. The first three row betas are $X_1 = m, X_2 = m - x + u, X_3 = m - x + \tau$. Direct expansion gives

$$P_E = J_y T(c) + H_a \sum_{i=1}^{3} \lambda_i (H_p H_{X_i} - H_b H_{X_i - q}), \qquad (\lambda_1, \lambda_2, \lambda_3) = (B, -F_v(x), A_0). \tag{5.8}$$

The six indicated permutation sets are disjoint. If x, u, v, τ, m and a, b, p agree, subtracting them gives $J_y T(c) = J_{y'} T(c')$. The irreducible factors have degrees three and two, so factorization reads c, then y. Thus joint agreement of these anchors forces identity.

End opening with the upper prefix fixed. Assume x, u, v, τ, m agree. The remaining invariant data are $\{a, p\}$ and $\{y, c, b\}$. Subtract the common contribution of the first three entries in column five. The remainder is

$$\mathcal{R}_E = J_y T(c) - H_a H_b J_{y+c-a}.$$

Its largest index is $N_1 = \max(m - q, w)$, with coefficient

$$\begin{cases} -H_a H_b B, & m - q > w, \\ -H_p J_y, & w > m - q, \\ -H_a H_b B - H_p J_y, & w = m - q. \end{cases}$$

These are nonzero, and divisibility by B holds exactly in the first regime; otherwise the irreducible quadratic would divide the irreducible cubic J_y. If both boards are top-only, the products $H_a H_b, H_a H_p$ recover the bottom endpoint and the joint-anchor identity applies. Otherwise both have $w = w' = N_1$.

If $a' = a, p' = p$, common X_3, w gives $y' = y$. Interchanging c, b leaves $T(c)$ unchanged and would require

$$H_b J_{y+c-a} = H_c J_{y+b-a}.$$

The cubic factors have no coordinate divisor, so $b = c$, and the board is unchanged. If instead $a' = p, p' = a$, with $a \neq p$, then $y' = y + p - a$. Let e, e' indicate whether the top candidate also attains N_1. Equality gives

$$H_p J_y + e H_a H_b B = H_a J_{y'} + e' H_p H_{b'} B.$$

Setting $H_a = 0$ forces $e' = 1$, and setting $H_p = 0$ forces $e = 1$, by coordinate-factor-freeness. Both boards are therefore tied, so $q = q'$, $c' = c$, $b' = b + a - p = x + y - \tau$. Canceling common rows gives $\{y, b\} = \{y', b'\}$. Since both entries changed, $y = b'$, forcing $x = \tau$, contrary to $x > u$. Thus the upper-prefix route gives identity.

End opening with the bottom edge fixed. Assume a, b, p, c agree. Only x, y and u, v can be interchanged. The top-row contributions in columns three, four, and five agree and may be subtracted.

For a core interchange, orient $u > v$ and put $z = w+y-a$. The maximum is $\max(x, z)+u-\tau$, so equality forces $x = z$. The contact coefficient is

$$K = F_v(y)T(c) - H_aH_bF_v(c + y - a).$$

With $J_2 = c + y + v - a - \tau$, its largest index is $\max(w, J_2)$. The coefficients for $J_2 > w, w > J_2, w = J_2$ are respectively

$$H_aH_bH_\tau, \qquad -H_pF_v(y), \qquad -H_pF_v(y) + H_aH_bH_\tau.$$

The last evaluates to one at all remaining variables equal to one; all three are nonzero.

For a free-row interchange, orient $y > x$. The maximum compares $y+v-\tau$ and $w+y-a+u-\tau$. Equality requires $w = a + v - u$, hence $v > u + b$. Its coefficient is

$$\begin{aligned} K = {} & F_u(x)T(c) - H_aH_bF_u(c + x - a) \\ & + H_aH_pH_uH_{w+x-a} - H_aH_pH_xH_v. \end{aligned}$$

Its largest remaining index is the maximum of $x + v - u$ and $x + v - \tau - q$, with coefficients $H_aH_pH_u$, $H_aH_bH_\tau$, or their sum at a tie. None vanishes. For both interchanges, the maximum compares $w + y - a + u - \tau$ and $w + x - a + v - \tau$. Each exceeds the opposite top-column-two index by $w - a > 0$. Equality forces $y + u = x + v$, with coefficient

$$-H_aH_p(H_xH_v - H_yH_u).$$

Its vanishing gives only the identical candidate: the crossed assignment is excluded by $x > u, y > v$. This proves bottom-edge rigidity.

End opening with only the bottom endpoint fixed. Assume a, b, p agree. Write

$$E' = (\xi, L, \eta, d, b;\ U, V, a, p), \qquad \{\xi, \eta, d\} = \{x, y, c\}, \quad \{U, V\} = \{u, v\}.$$

After the two common top-right terms are subtracted, the maximal candidates are

$$R = x + y - \tau, \quad R' = \xi + \eta - \tau, \quad Z = c + q + y + u - a - \tau, \quad Z' = d + q + \eta + U - a - \tau.$$

With indicators for the maximum, its coefficient is

$$B\big(\epsilon T(c) - \epsilon' T(d)\big) - H_aH_p\big(\zeta F_v(x) - \zeta' F_V(\xi)\big).$$

If a top candidate is active, setting $H_a = 0$ excludes a single active top term. Both top indices agree, so the row sum gives $c = d$ and bottom-edge rigidity applies. Otherwise both right candidates are active and $F_v(x) = F_V(\xi)$. The multiset intersection of $\{u, v\}$ and $\{x, v\}$ is exactly $\{v\}$, since $x > u$. It reads $x = \xi, u = U, v = V$, and the joint-anchor identity applies.

Finally, the three gcd degrees separate unrestricted E/E equality into equal corner regimes. For $q < \delta$, the first two coefficients read a, b, p, so the bottom-endpoint route applies. For $\delta < q$, division by the known $-H_aH_p$ reads $F_v(x)$, which together with B reads x, u, v, τ in order. The common m supplies the upper prefix. At $q = \delta$, the degree-one gcd reads a; the first coefficient then reads p, B. Reducing the second-coefficient equality modulo H_p reads b, and subtraction reads $F_v(x)$. The joint-anchor identity finishes. We have proved

$$P_M = P_{M'} \implies M' \in \{M, M^*\}, \qquad P_E = P_{E'} \implies E' = E. \tag{5.9}$$

□

5.4. **Mixed openings and arithmetic progression.**

Proposition 5.4. *If a five-negative M-board and a five-negative end-opening board are Schur-equal, translations and rotations put them into exactly one of the two pairs in* (5.1).

Proof. Orient the end opening as E, and use the notation of (5.3). For M, the second coefficient in (5.4) is always divisible by B. For E, it is divisible by B exactly when $q < \delta$, since $F_v(x)$ is an irreducible quadratic distinct from B. Thus the E-corner is strict with $q < \delta$. The M-corner cannot tie: after division by B, evaluation at all variables equal to one gives -2, whereas the E-coefficient gives -1. Rotate M if necessary so that it too has $q < \delta$. The first two coefficients

now give common a, b, p through the reduced quotient H_b/H_p. Consequently the general mixed pair is

$$\begin{aligned} M = (x, y, L, c, b;\ a, u, v, p), \qquad E' = (\xi, L, \eta, d, b;\ U, V, a, p), \\ \{\xi, \eta, d\} = \{x, y, c\}, \qquad \{U, V\} = \{u, v\}. \end{aligned} \tag{5.10}$$

No equality $d = c$ is assumed.

Subtract the common two-corner polynomial $H_a B(H_p H_m - H_b H_{m-q})$. The maximal remaining candidates are

$$R = x + y + v - a - \tau, \quad R' = \xi + \eta - \tau, \quad Z = w + y - \tau, \quad Z' = d + q + \eta + U - a - \tau.$$

Let J be their maximum, and write $\epsilon, \epsilon', \zeta, \zeta'$ for the corresponding indicators. The coefficient in the difference is

$$K = \epsilon H_a G_u(c) - \epsilon' BT(d) - \zeta H_x H_p B + \zeta' H_a H_p F_V(\xi). \tag{5.11}$$

The positions $(1, 3), (2, 5)$ cannot occur together, so this is the whole coefficient, not a derivative containing a further H_J. At least one top and one right candidate must attain J. A right-only tie would require $H_x B = H_a F_V(\xi)$, impossible by coordinate-factor-freeness and $x > a$. A top-only tie would equate $H_a G_u(c)$ with the coordinate-factor-free product $BT(d)$. Single active terms have nonzero cofactors.

Set $H_p = 0$. If only R' is the active top term, the specialization is $-H_b H_d B \neq 0$. If only R is active, it is $H_a H_b F_u(c)$, which vanishes only for $a = p$. If both top terms are active, their indices and the row sum give $d = c + a - v$. The specialization condition is

$$H_p \mid H_a H_u H_c - H_a H_\tau H_{c+u-\tau} - H_d H_u H_v + H_d H_\tau H_L.$$

Here $c, d, c + u - \tau > p$. Consider the possible identifications in this order. If $p = a$, the residual is $-H_d B \neq 0$. If $p = \tau$, equality requires $H_a H_c = H_d H_v$, hence $a = v, d = c$. If $p = u$, it requires $H_a H_{c+u-\tau} = H_d H_L$, hence $a = L, d = c + u - \tau$. If $p = v$ but none of the preceding cases holds, the three surviving terms evaluate to one at all remaining variables equal to one. If $p = L$, they evaluate to minus one. If none of these identifications holds, the original residual would have to vanish. But $H_a F_u(c) = H_d B$ is impossible by coordinate-factor-freeness and $d > a$. Thus exactly the following possibilities remain:

$$\begin{array}{c|llll} \text{I} & R > R', & a = p, & & \\ \text{II} & R = R', & a = v, & \tau = p, & d = c, \\ \text{III} & R = R', & p = u, & a = L, & d = c + u - \tau. \end{array} \tag{5.12}$$

The order of the specialization cases includes all coincident-index possibilities, including $p = 0$.

Regime I. Divide (5.11) by H_p. If Z is absent, the result is $G_u(c) + H_p F_V(\xi)$, nonzero modulo H_p. Therefore Z is active, and its contact gives

$$w = x + v - p, \qquad x = c + b - v > b.$$

If Z' is absent, the coefficient is $G_u(c) - H_x B$. Its largest possible index is $\max(x, x + L - p)$. The coefficients for $L > p, L < p, L = p$ are respectively

$$H_p H_\tau, \qquad -B, \qquad 2H_p H_\tau - H_u H_v,$$

all nonzero. Hence Z' is active too. The right tie and row sum give $\xi = x + U - p$. The highest shifted terms cancel, leaving

$$K/H_p = H_b F_u(c) - H_x B + H_p(H_V H_\xi - H_u H_w).$$

If $U = v$, then $\xi = w$ and the last term vanishes; unique factorization would force $b = x$, impossible. Thus $U = u, V = v$ and $u \neq v$. The remaining equation is

$$\begin{aligned} 0 = {} & H_b H_u H_c - H_b H_\tau H_{c+u-\tau} - H_x H_u H_v + H_x H_\tau H_L \\ & + H_p H_v H_{x+u-p} - H_p H_u H_{x+v-p}. \end{aligned}$$

Put $A = c + u - \tau = x + L - b$, $C = x + u - p$, $D = x + v - p$. Every largest-index candidate is among A, C, D, x, with uncollected coefficients $-H_b H_\tau, H_p H_v, -H_p H_u, -B$.

First suppose $\max(u,v) > p$. If $u < v$, the largest candidates A, D have negative nonzero coefficients, so the equation is impossible. Otherwise $u > v$ and $u > p$, since $u \neq v$ was already established. Thus $C > D, x$; equality forces $A = C$ and $H_b H_\tau = H_p H_v$. Hence $b = v, \tau = p$, and the equation reduces to

$$H_p(H_c H_L - H_u H_{c+v-p}) = 0,$$

which forces $c = u$, since $v > p$.

For completeness suppose $\max(u,v) \leq p$. If $L > b$, then A is a unique maximum. If $L < b$, the coefficient at x is $-B$, $H_\tau H_L$, or $H_\tau H_L - 2H_u H_v$, according as neither core index equals p, $u = p$, or $v = p$. None vanishes. If $L = b$, that coefficient is nonzero except possibly when $u = p > v$. In that last case the whole polynomial becomes

$$H_p(H_L H_c - H_p H_{c+v-\tau}),$$

which is nonzero because $c > p + v > p$ and $v > \tau$. Therefore Regime I forces

$$b = v, \quad \tau = p, \quad c = u, \quad x = u.$$

Since $\xi = 2u - p > u$, the row multiset then gives $y = \xi = 2u - p$, $\eta = d = u$.

Regime II. Here $a = v, \tau = p, d = c$. Dividing the contact coefficient by H_p gives

$$H_L T(c) - H_v T(c + u - p) - \zeta H_x B + \zeta' H_v F_V(\xi).$$

Without Z, reduction modulo H_v leaves $H_L T(c) \neq 0$. Thus Z is active and $w = x$. Without Z', the unique largest variable H_{w+u-p} has coefficient $H_p H_v \neq 0$, so Z' is active and $\xi = w + U - v$. The choice $U = v$ leaves

$$H_b(H_L H_c - H_v H_{c+u-p}) \neq 0,$$

since $c > v$. Thus $U = u, V = v$, and the residual is

$$H_b(H_L H_c - H_v H_{c+u-p}) + H_v(H_v H_\xi - H_u H_w).$$

Modulo H_v, this is $H_b H_L H_c$, with $c, L > v$, so $b = v$. The residual then becomes $H_v(H_L H_c - H_u H_{c+v-p})$, forcing $c = u$. Hence $x = u + v - p$, $\xi = y = 2u - p$, and $\eta = x$.

Regime III. Here $p = u, a = L, d = c + u - \tau$, and the contact coefficient divided by H_u is

$$H_L T(c) - H_v T(d) - \zeta H_x B + \zeta' H_L F_V(\xi).$$

Reduction modulo H_L forces Z to be active. Its contact gives $x = c + b - \tau$. Since $b > u$, we have $x > d > c$. As $d \in \{x, y, c\}$, necessarily $d = y$ and $\{\xi, \eta\} = \{x, c\}$. Without Z', direct cancellation leaves the unique largest variable H_x with coefficient $H_L H_\tau \neq 0$. Therefore Z' is active, giving $\xi = w + U - v$, where $w = c + b - u$. If $U = v$, then $c < \xi = w < x$, impossible. If $U = u$, then $\xi = c + b - v < x$, so $\xi = c, b = v > u, \eta = x$. The residual is

$$2H_L H_v H_c - H_L H_u H_{c+v-u} - H_v^2 H_{c+u-\tau}.$$

Since $c > u + v$ and $v > u > \tau$, its largest index is one or both of $c + v - u, c + u - \tau$, with negative nonzero coefficients. Regime III is impossible.

The necessary mixed parameters are therefore exactly

$$\begin{gathered} \tau = p, \quad b = v, \quad c = u, \quad y = 2u - p, \quad u > v + p, \\ (\xi, \eta, d; U, V) = (y, x, c; u, v), \\ (a, x) = (v, u + v - p) \quad \text{or} \quad (a, x) = (p, u). \end{gathered} \tag{5.13}$$

Set $g = u - p$ and replace the family parameters b, q by $v, v - p$. The domain becomes $1 \leq q \leq b < g$, and the profile recurrence gives exactly (5.1).

□

Proof of the identities in (5.1). Reverse the five rows, which has positive sign, and index them by $k = 0, \ldots, 4$. For integers $j \geq 0$, put

$$U_j = (H_{b+(k-j)g})_{k=0}^4, \qquad V_j = (H_{b+(k-j)g-q})_{k=0}^4.$$

Let S_0 be the down-shift on five-vectors. Since $b < g$ and $q \le b$, the negative-index convention gives $S_0U_j = U_{j+1}$ and $S_0V_j = V_{j+1}$. For arbitrary columns,

$$\sum_{i=1}^{5} \det(z_1, \dots, S_0 z_i, \dots, z_5) = 0,$$

by taking the linear coefficient in $\det((I + tS_0)Z) = \det Z$. Apply this identity to the two seeds

$$(U_2, V_1, U_1, V_0, U_0), \qquad (V_2, V_1, U_1, V_0, U_0).$$

In each case three summands have a repeated column. The two remaining summands give

$$\det(U_3, V_1, U_1, V_0, U_0) = \det(V_2, U_2, U_1, V_0, U_0),$$
$$\det(V_3, V_1, U_1, V_0, U_0) = \det(V_2, U_2, V_1, V_0, U_0).$$

These are precisely the two identities in (5.1). The proof retains H_0, so it includes $q = b$. □

Completion of the proof of Theorem 5.1. By Lemma 2.2, every connected mate has five rows and five negatives. Proposition 5.3 excludes every nonrotation same-opening equality, including two end openings after orientation. Proposition 5.4 gives all remaining pairs and proves their identities.

Their parameters are unique. From the primitive quadratic factor in the first cofactor, recover

$$b = \min(u, v), \qquad g = \max(u, v) - \tau, \qquad q = b - \tau.$$

The singleton pair is $\{b, b-q\}$ in family 1 and $\{b-q, b-q\}$ in family 2; these are distinct since $q > 0$. Within each family F_i has middle opening, and H_i, H_i^* have the two different end openings. Also $F_i \neq F_i^*$: its row betas form an arithmetic progression, whereas its column betas do not, as their first gap is larger than g and their last gap is $q < g$. Thus the four members are distinct. Same-opening rigidity and the unique mixed parameters exclude additional members. All arguments were in the faithful homogeneous polynomial ring, including the common-beta boundary. □

5.5. **Endpoint and higher-row exclusion.** For reference, zero-free five-negative boards have the following border coordinates, up to rotation. The notation in this subsection is local. Put $Q = \langle A, B, d\rangle$, $0 < d < \min(A, B)$, $P = A + B - d$, and $f_Q = h_A h_B - h_P h_d$. Then

$$\begin{aligned} T_1 &= (s+u, p+\ell, A+a, P, B+v; s, p, A, B), \\ &\quad s, p, u, v > 0, \quad \ell > s, \quad a > p; \\ T_2 &= (s+u, A+L, P, B+R, r+v; s, A, B, r), \\ &\quad s, r, u, v > 0, \quad L > s, \quad R > r. \end{aligned} \tag{5.14}$$

Here T_1^* has opening E and T_2 has opening M. The local u, v are exterior margins, equal to (q, δ) on T_1^* and (δ, q) on T_2. Expansion of $\Phi_Q(a, v) = s_{\mathfrak{B}_{a,v}Q}$ gives

$$\begin{aligned} \Phi_Q(a, v) &= h_{P+a+v}(h_A h_B - h_P h_d) \\ &\quad - h_{P+v}(h_{A+a}h_B - h_{P+a}h_d) + h_{B+v}(h_{A+a}h_P - h_{P+a}h_A). \end{aligned} \tag{5.15}$$

With $C_Q(p; \ell, a, v) = h_{p+\ell}\Phi_Q(a, v) - h_p\Phi_Q(a+\ell, v)$, first-column and last-row expansion yield

$$\begin{aligned} s_{T_1} &= h_{s+u}C_Q(p; \ell, a, v) - h_s C_Q(p; \ell + u, a, v), \\ s_{T_2} &= h_{s+u}h_{r+v}\Phi_Q(L, R) - h_s h_{r+v}\Phi_Q(L+u, R) \\ &\quad - h_{s+u}h_r\Phi_Q(L, R+v) + h_s h_r \Phi_Q(L+u, R+v). \end{aligned} \tag{5.16}$$

Lemma 5.5. *For a connected five-row diagram C with exactly five full-board negatives, common betas allowed, and $t > 0$, $s_{C^{\uparrow t}} = s_{C^{\downarrow t}}$ if and only if $C = C^*$.*

Proof. Both endpoint extensions are connected and have the same negative positions as C: only the first matrix row or the last matrix column changes, and those indices are positive before extension. They therefore have the same opening. By Proposition 5.3, equality makes them

identical or rotations. They are not identical for $t > 0$, since their ordered first row lengths differ. Hence

$$C^{\uparrow t} = (C^{\downarrow t})^* = (C^*)^{\uparrow t}.$$

Injectivity of the first-row extension gives $C = C^*$. Conversely, that symmetry and rotation invariance give equality of the extensions. The same-opening proposition already includes common betas. □

Lemma 5.6. *No positive border over either rotation orbit of F_i in* (5.1) *is Schur-equal to a positive border over the H_i orbit, for $i = 1, 2$ and $1 \le q \le b < g$.*

Proof. Every positive endpoint extension of an exceptional five-row member is rotation-rigid by Theorem 5.1. To check it leaves the exceptional list, note that one beta list of every exceptional member is arithmetic. Its displayed column list is not arithmetic, and its first three gaps are not all equal. A top extension makes the row gaps $(g + t, g, g, g)$, so neither list remains arithmetic. After a bottom extension is normalized, its row list stays arithmetic, but a displayed F-type member would require $y_3 = g$, whereas the extension has $y_3 = g + t$; an H-type would require $y_2 = 2g$, whereas the extension has $y_2 = 2g + t$. The negative patterns distinguish these two types.

No extension of F_i can be an extension of H_i, or its rotation: for equal directions literal equality can be undone; for different directions one row list is arithmetic and the other is not. Under rotation the bottom row-beta gap of an H_i extension is its original top column-beta gap $q < g$, whereas an F_i extension has bottom row-beta gap g. Thus Lemma 3.3 excludes all unequal-margin cross-orbit borders.

At tied margins (t, t), canceling common parent rows leaves respectively

$$\{g + t, 3g - q\} = \{2g - q + t, 2g\}, \qquad \{g - q + t, 3g - q\} = \{2g - q + t, 2g - q\}.$$

Each forces $t = g$. In the unnormalized row order $b + kg$, $k = 5, 4, \ldots, 0$, put $a_k = h_{b+kg}$, $c_k = h_{b+kg-q}$; negative k give zero. The four row vectors of the two pairs are

$$\begin{array}{l|l} \mathfrak{B}_{g,g}F_1 & (a_{k-3}, c_{k-1}, a_{k-1}, c_k, a_k, a_{k+1}) \\ \mathfrak{B}_{g,g}H_1 & (c_{k-2}, a_{k-2}, a_{k-1}, c_k, a_k, a_{k+1}) \\ \mathfrak{B}_{g,g}F_2 & (c_{k-3}, c_{k-1}, a_{k-1}, c_k, a_k, a_{k+1}) \\ \mathfrak{B}_{g,g}H_2 & (c_{k-2}, a_{k-2}, c_{k-1}, c_k, a_k, a_{k+1}). \end{array}$$

For $b > q$, the two alphabets consist of distinct independent generators, since $0 < q < g$. For $a_0^3 a_5 c_2 c_4$, the three a_0-positions are forced to be $(3, 1), (5, 3), (6, 5)$ in the first source and $(4, 2), (5, 3), (6, 5)$ in the first mate. On each side a_5 then forces $(2, 6)$. The two remaining factors in the source are c_4, c_2; in the mate the possibilities are c_3^2 and $c_1 c_5$. Thus only the source contributes, with column word $(2, 6, 1, 4, 3, 5)$.

For $a_0^2 a_5 c_1^2 c_4$, the a_0-positions are $(5, 3), (6, 5)$ in the second source and $(4, 2), (6, 5)$ in its mate. Again a_5 forces $(2, 6)$. The factor c_4 then forces $(1, 2)$ in the source and $(1, 3)$ in the mate. The remaining source choices give $c_0 c_2$ or a zero entry, whereas the mate gives c_1^2, with column word $(3, 6, 1, 2, 4, 5)$. Both contributing permutations have six inversions. Consequently

$$\begin{aligned} [a_0^3 a_5 c_2 c_4](s_{\mathfrak{B}_{g,g}F_1} - s_{\mathfrak{B}_{g,g}H_1}) &= 1, \\ [a_0^2 a_5 c_1^2 c_4](s_{\mathfrak{B}_{g,g}F_2} - s_{\mathfrak{B}_{g,g}H_2}) &= -1. \end{aligned} \tag{5.17}$$

At $b = q$, only c_0 becomes a unit. The displayed monomials have ordinary degree six, so no term containing that unit can affect them. Alternatively Γ replaces the parameters by $(2b+1, 2g, 2q)$ and the margins by $(2u, 2v)$. □

Proof of Theorem 1.1(iv). The five-row base is the algebraic Theorem 5.1. Work zero-free for larger rows. If a six-row source has connected maximal cofactor, its complete class is given by that base. Cross-orbit borders are excluded by Lemma 5.6, and same-orbit margin swaps by Lemma 5.5.

A disconnected $(n-1)$-row cofactor with three components would use at least $2n-5$ negatives. With only five negatives and $n \ge 6$, it has two components, using at least $n - 2$ negatives. The

only possibilities are a six-row parent with split $1+4$ and one internal negative, or a seven-row parent with split $1+5$ and no internal negative, including their reversals. The first is Corollary 3.8; the second is Theorem 4.3. For $n \geq 8$ no split is possible.

All other larger-row steps have a connected child. Induction identifies its literal diagram up to rotation, and also gives rigidity of its endpoint extensions, which retain exactly five negatives. Lemma 3.3 then closes the step. Homogenization supplies all common-beta sources. This proves the higher-row assertion from Theorem 5.1 and the endpoint and border lemmas, with no bound on area or row count. □

6. Staircase components and residual matching

6.1. **Closure and the base statement.** The negative partition of $\mathcal{T}_n$ is $(3,2,1)$. By Lemma 2.3 this pattern is stable under rotation. Stability under arbitrary Schur equality requires the stronger overlap conditions in the following lemma.

Lemma 6.1. *Let D be connected with $n \geq 5$ rows and full negative set $\mathcal{T}_n$ of (1.1). Every connected Schur-equal mate has n rows and the same negative set. Common betas are allowed.*

Proof. Use the overlap criterion in the proof of Lemma 2.2: an entry at row distance $k-1$ is negative exactly when the corresponding k-row intersection has size at most $k-2$. In a staircase, every such intersection has size at least $k-1$ for $2 \leq k \leq n-3$, and at most $k-2$ for $k \geq n-2$. The invariant overlap partitions, padded using the common row count, preserve both universal conditions. Thus they recover the whole staircase mask, not merely its negative count. □

At five rows the staircase is exactly an upper Hessenberg board. In profile coordinates its condition is

$$z_i \geq 0, \qquad r_i > z_{i-1} + z_i, \qquad z_0 = z_5 = 0. \tag{6.1}$$

Zero-freeness is exactly $z_i \geq 1$, a condition we call *thickness*. For general order k, an upper Hessenberg board has the interval expansion

$$s_{(r;z)} = \sum_{S \subseteq \{1,\dots,k-1\}} (-1)^{|S|} \prod_{i \in S} h_{z_i} \prod_{[a,b] \in \mathcal{I}(S)} h_{w_{a\cdots b}}, \qquad w_{a\cdots b} = \sum_{i=a}^{b} r_i - \sum_{i=a}^{b-1} z_i. \tag{6.2}$$

Here $\mathcal{I}(S)$ joins adjacent rows exactly across the edges in S. The interval-block argument used for (5.2) applies in every order and proves this formula. At five rows there are sixteen indexed summands; coinciding monomials are collected with their signs. A thick profile is *separated* when

$$\{z_1, z_2, z_3, z_4\} \cap \{w_{a\cdots b} : 1 \leq a \leq b \leq 5\} = \varnothing;$$

otherwise it has a *contact*. No separation hypothesis is imposed on the classification below.

Theorem 6.2 (Staircase base)**.** *For a connected five-row staircase, with common betas allowed, every connected mate is its rotation unless the pair is in one of the following complete four-member classes. Add d to every entry of both words in each displayed profile:*

	F : *rows; gaps*	H : *rows; gaps*
M1	$(a,b,a,a+b,b;t,0,t,t)$	$(a,a+b,b,a,b;t,t,0,t)$
M2	$(a,b,a,a+b-t,b;0,t,0,0)$	$(a,a+b-t,b,a,b;0,0,t,0)$
M3	$(a,b,b,b,a+b;0,0,0,a)$	$(a,b,a+b,b,b;0,a,0,0)$
M4	$(a,b,b,a+b,a+b;0,0,a,a)$	$(a,b,a+b,a+b,b;0,a,a,0)$.

(6.3)

For M1 and M2 the integer parameters satisfy $d,t \geq 0$, $a,b > t+d$, $a \neq b$. For M3 and M4 they satisfy $d \geq 0$, $a \geq 1$, $b > a+d$. Each class is $\{F, F^*, H, H^*\}$. M1 and M2 coincide at $t=0$.

6.2. **Safe cuts and specialized content.** For a thick profile $D = (L; g)$, write $\mathcal{G}(D)$ for its gap multiset and $\mathcal{W}(D)$ for its interval-index set. The letters L, g denote the same row and gap data as r, z in (6.2). A contact is an element of $\mathcal{G}(D) \cap \mathcal{W}(D)$. All generator specializations keep these numerical indices fixed. Throughout the staircase arguments, write

$$R_t(a, b) = h_a h_b - h_t h_{a+b-t},$$

$$T_t(a, b, c) = h_a h_b h_c - h_t h_{a+b-t} h_c - h_t h_a h_{b+c-t} + h_t^2 h_{a+b+c-2t}.$$

These are the two-row and constant-gap three-row polynomials on their thick domains. When $z = \min \mathcal{G}(D)$, put $J_k(D) = [h_z^k] s_D$.

Lemma 6.3 (The minimum-gap interface)**.** *Let $z = \min \mathcal{G}(D)$, with multiplicity m. Then $z \notin \mathcal{W}(D)$. Cutting every z-seam gives thick components D_j, and contracting those seams gives a thick profile D/z, with*

$$s_D\big|_{h_z=0} = \prod_j s_{D_j}, \qquad [h_z^m] s_D = (-1)^m s_{D/z}.$$

The first factorization recovers the component polynomials and their row counts, but not their order. At five rows the numbers of uncollected terms in the respective minimum-gap layers are

$$\#\{\text{terms of } h_z\text{-degree } j\} = \binom{m}{j} 2^{4-m}, \qquad \sum_{j=0}^{m} \binom{m}{j} 2^{4-m} = 16. \tag{6.4}$$

For thick profiles A, B, let u, v be their adjacent endpoint rows and let α, β be the gaps incident to those rows inside A, B, respectively; an absent gap is zero. A positive joining gap q is admissible here when

$$u > \alpha + q, \qquad v > q + \beta.$$

Thus the joined profile $A \cdot_q B$ is thick. Replacing u, v by $u + v - q$ gives the thick contraction $A \odot_q B$, and

$$s_{A \cdot_q B} = s_A s_B - h_q s_{A \odot_q B}.$$

Connectedness of the joined profile alone is not the hypothesis of this identity.

This qualification is substantive. For a connected but non-Hessenberg join,

$$s_{(2,2,2;1,1)} - \big(h_2 s_{(2,2;1)} - h_1 s_{(2,3;1)}\big) = h_3^2 - h_2 h_4 \neq 0.$$

The applications in the staircase proof retain the stronger joining inequalities.

Proof. Every row exceeds every incident gap and therefore exceeds z. Extending an interval adds $L_i - g$, which is positive, so every interval index exceeds z. In (6.2), setting $h_z = 0$ keeps every minimum seam, and taking its highest power contracts every minimum seam. No interval factor contributes another h_z. Contraction preserves the strict row inequalities: the merged row exceeds the sum of its two remaining exterior gaps. Connected irreducibility [1] and ordinary degree recover the component polynomials and their sizes. Choosing which of the m minimum seams to contract proves (6.4). Partitioning the interval expansion according to whether the joining seam is contracted proves the final identity. □

Fix a positive integer q. A *q-component* is a component obtained after every q-seam has been cut. Orient it to put an endpoint adjoining a cut at the bottom.

Write $A = (a_1, \ldots, a_n; r_1, \ldots, r_{n-1})$, and put $r_0 = r_n = 0$. The conditions used below are

$$a_i > r_{i-1} + r_i \quad (1 \le i \le n), \qquad r_i \ge 1,\ r_i \neq q \quad (1 \le i < n), \qquad a_n > q + r_{n-1}.$$

For a singleton the last condition is $a_1 > q$. An additional inherited boundary condition at the other endpoint is retained whenever present. Put $P_A = s_A|_{h_q=0}$.

Lemma 6.4 (Monomial content and one primitive factor)**.** *Every q-component has a unique factorization*

$$P_A = G_A K_A,$$

where G_A is a monic monomial in its internal gap variables and K_A is irreducible. Every variable in G_A has index less than q. Through four rows the complete content table is as follows. For two rows,

$$G_{(a,b;r)} = \begin{cases} h_r, & a = q, \\ 1, & a \neq q. \end{cases}$$

For three rows, put $u = a + b - r$. Then

$$G_{(a,b,c;r,s)} = \begin{cases} h_r, & a = q, \\ h_s, & a \neq q \text{ and } [(b = q,\ r = s) \text{ or } (u = q,\ a = s)], \\ 1, & \text{otherwise.} \end{cases}$$

For four rows,

$$G_{(a,b,c,d;r,s,t)} = \begin{cases} h_r h_s, & a = q,\ c = q,\ s = t, \\ h_r, & a = q \text{ and not } (c = q,\ s = t), \\ h_s, & a \neq q \text{ and at least one of } \begin{cases} b = q,\ r = s, \\ c = q,\ s = t, \\ a + b - r = q,\ a = s, \end{cases} \\ 1, & \text{otherwise.} \end{cases}$$

The three conditions in the third line are alternatives; simultaneous occurrence does not multiply their contributions. A singleton has $G_A = 1$.

Proof. Let T be the full interval index. It exceeds every other index and, by the inherited boundary, exceeds q. Thus

$$P_A = (-1)^{n-1} \left(\prod_i h_{r_i} \right) h_T + B, \qquad h_T \text{ does not occur in } B.$$

Set $G_A = \gcd(\prod_i h_{r_i}, B)$. After division, the polynomial is primitive and linear in h_T, so Gauss's lemma proves irreducibility. This also covers $B = 0$.

For the content calculation at four rows use

$$u = a + b - r, \quad v = b + c - s, \quad w = c + d - t,$$
$$x = a + b + c - r - s, \quad y = b + c + d - s - t, \quad T = a + b + c + d - r - s - t.$$

The exact expansion is

$$\begin{aligned} s_A = {} & h_a h_b h_c h_d - h_r h_u h_c h_d - h_s h_a h_v h_d - h_t h_a h_b h_w \\ & + h_r h_s h_x h_d + h_r h_t h_u h_w + h_s h_t h_a h_y - h_r h_s h_t h_T. \end{aligned} \tag{6.5}$$

The analogous three-row expansion has four terms. Since $c > q + s$ in three rows, its only possible q-indices are $a, b, a + b - r$. Setting an internal gap variable to zero in those four terms gives exactly the three-row table.

If $a = q$ in four rows, near-concatenation gives

$$P_A = -h_r P_{(q+b-r,c,d;s,t)}.$$

The new first row exceeds q, so the three-row table gives the first two lines of the four-row table.

Suppose $a \neq q$. Every possible variable divisor is a gap variable. A gap value $r \neq s$ cannot divide P_A: after setting $h_r = 0$, the first-row cut leaves a three-row factor whose largest term survives. If $r = t \neq s$, both end seams are cut and the middle two-row factor is still nonzero. Similarly a value $t \neq s$ cannot divide. For the latter assertion the only exceptional largest-term case is $x = q$; the remaining prefix is

$$h_a h_b h_c - h_r h_u h_c - h_s h_a h_v.$$

Its positive monomial cannot cancel a negative one because $u > \max(a, b)$, $v > \max(b, c)$. If setting $h_t = 0$ kills the positive term, then $a = t$ or $b = t$, since $c > t$; the first negative term survives. Hence only h_s can divide.

Cutting the middle seam gives

$$P_A\big|_{h_s=0} = \Big(R_r(a,b)\big|_{h_q=h_s=0}\Big)\Big(R_t(c,d)\big|_{h_q=h_s=0}\Big).$$

The first factor vanishes exactly when $b=q, r=s$ or $a=s, u=q$. The second vanishes exactly when $c=q, s=t$, because $d>q+t$ and $w>d$. In each two-term factor a surviving positive term cannot cancel its negative term, whose full interval exceeds both rows. This proves the divisibility conditions, including coincident gaps.

Only repeated gap values require a multiplicity check. If $r=s\neq t, b=q$, the coefficient of the first power of h_s is

$$h_u(-\overline{h}_c h_d + h_t h_w) + h_a(-h_v h_d + h_t h_y).$$

Compare u and y. In their tie, $a=w$, and the coefficient at that common largest index is $-\overline{h}_c h_d + 2h_t h_w \neq 0$. If $s=t\neq r, c=q$, that coefficient is

$$h_d(-\overline{h}_a h_v + h_r h_x) + h_w(-\overline{h}_a \overline{h}_b + h_r \overline{h}_u).$$

When $a=s$ the surviving terms are positive. Otherwise compare x, w; their tie gives $u=d$ and the nonzero coefficient $2h_r h_u - h_a\overline{h}_b$, since $u>\max(a,b)$. For $u=q, a=s=t\neq r$, the first coefficient is $h_d(h_b\overline{h}_c + h_r h_x)\neq 0$. Bars here mean specialization at both $h_q=h_s=0$. If all three gaps coincide, a surviving one-seam term has the common negative sign of all such terms; hence the first coefficient is nonzero. This proves the multiplicities in the table.

Finally, suppose a variable h_v with $v>q$ divided P_A. Cut all internal v-seams after setting $h_v=0$. Every resulting component has an endpoint adjacent to such a seam, so its full interval exceeds $v>q$. Its largest-index coefficient is an internal gap monomial containing neither h_v nor h_q. Each specialized factor is therefore nonzero, contrary to the asserted divisibility. Internal gaps cannot equal q, proving the low-gap assertion. □

Normalize primitive factors by

$$\widehat{K}_A = (-1)^{n-1} P_A / G_A.$$

Their largest-index coefficients are positive monomials with numerical coefficient one. Thus normalized associates are equal. In the remaining staircase arguments, a bar denotes specialization at $h_q=0$, and

$$\varepsilon(P) = P\big|_{h_j=1\ (j\neq q)}, \qquad \|P\|_{h,1} = \sum_{\lambda} \big|[h_\lambda]P\big| \quad \text{for } P\in\mathbb{Q}[h_j : j\neq q].$$

The sum is over distinct commutative monomials after collection. If all q-seams of a five-row profile are cut, its aggregate extracted content is recovered directly from the restriction:

$$\prod_A G_A = \prod_{1\le j<q} h_j^{\nu_{h_j}(s_D|_{h_q=0})}.$$

Here ν_{h_j} denotes the exponent of the variable divisor. A linear primitive has full interval index greater than q, and a nonlinear primitive has no variable divisor. This identifies the aggregate content, not its ownership by individual components.

6.3. **Matching the recovered primitives.**

Lemma 6.5 (Endpoint coefficients of a four-row restriction)**.** *Let $A=(a,b,c,d;r,s,t)$ be a q-component with $a\neq q$, and put*

$$x=a+b+c-r-s, \qquad y=b+c+d-s-t, \qquad T=a+b+c+d-r-s-t, \qquad M=\max(x,y).$$

The two largest indices occurring in P_A are T, M, and

$$-\frac{[h_M]P_A}{[h_T]P_A} = \begin{cases} h_a/h_r, & y>x,\\ h_d/h_t, & x>y,\\ h_a/h_r + h_d/h_t, & x=y. \end{cases}$$

The strict case recovers one ordered endpoint row/gap pair; the tie recovers the unordered two such pairs, with multiplicity. The same ratio is obtained from $\widehat{K}_A$.

Proof. Every proper interval index is at most M, and all shorter intervals and gap indices are strictly below M. Moreover $T > M$, $y > d > q$, and $x = q$ therefore implies $y > x$. The coefficients in (6.5) are

$$[h_T]P_A = -h_r h_s h_t, \qquad [h_M]P_A = \mathbf{1}_{M=x} h_r h_s h_d + \mathbf{1}_{M=y} h_s h_t h_a.$$

Both displayed contributions survive and have the same sign. Each reduced Laurent monomial identifies its ordered pair because $a > r$ and $d > t$; equal contributions retain coefficient two. The content has all its indices below $q < M$, so division by it does not change the ratio. Finally, if $x > y$, then $a - r > d - t > q$. Rotation still satisfies the inherited boundary condition and puts the recovered endpoint first. This also holds at a tie, when both endpoint excesses exceed q. □

Lemma 6.6 (Residuals through three rows and cross-size matching)**.** *Among q-components with at most four rows, the quadratic normalized primitives are exactly*

$$\begin{aligned} B_r(a,c) &= h_r h_{a+c-r} - h_a h_c, \\ M_{q,r}(a,c) &= h_r h_{a+c+q-2r} - h_{a+q-r} h_c - h_a h_{c+q-r}, \\ I_{q;r,s}(c) &= h_{q-s+r} h_c - h_s h_{q+c-2s+r} + h_r h_{q+c-s}. \end{aligned}$$

Their matching keys are respectively $(r, \{a, c\})$, $(r, \{a, c\})$, and (r, s, c). Different types cannot match. Their possible origins, through four rows, are

component condition	G_A	$\widehat{K}_A$
$(a, b; r),\ a \neq q$	1	$B_r(a, b)$
$(q, b, c; r, s)$	h_r	$B_s(q + b - r, c)$
$(a, q, c; r, r),\ a \neq q$	h_r	$M_{q,r}(a, c)$
$(s, q - s + r, c; r, s)$	h_s	$I_{q;r,s}(c)$
$(q, b, q, d; r, s, s)$	$h_r h_s$	$M_{q,s}(q + b - r, d)$.

All origins retain their thick and inherited boundary conditions. A linear primitive h_T comes from the singleton (T), with $T > q$, or from $(q, T - q + r; r)$, with $1 \leq r < q$ and $T > 2q$. The stronger bound belongs only to the two-row origin.

A primitive three-row cubic determines its profile up to rotation. A three-row primitive cubic and a four-row primitive cubic can match only by endpoint contraction: the four-row component is

$$(q, b, c, d; r, s, t),$$

and the three-row component is $(q + b - r, c, d; s, t)$ or its rotation. The four-row content is h_r.

Proof. Sending every surviving generator to one gives the quadratic values $0, -1, 1$, respectively. Their largest coefficients recover r. The negative monomial of B recovers its unordered row pair. For M, after removing the largest-index monomial, the smallest index is $\min(a, c)$ and the largest is $\max(a, c) + q - r$. For I, the smaller index in its negative monomial is s, and its largest index is $q + c - s$. The content table proves the origin list. These readings allow coincident same-sign monomials.

For a primitive three-row cubic the possible contact types are untouched, middle-row contact $b = q, r \neq s$, and interval contact $a + b - r = q, a \neq s$. Their pairs (augmentation, norm) are $(0, 4), (-1, 3), (1, 3)$. An untouched cubic is a full Schur function, so Lemma 3.5 applies. For a middle contact, the unequal gap labels distinguish the two negative monomials. After ordering the gaps, a nontrivial match would require both

$$a' = c - (q - r), \qquad a' = c + (q - s),$$

which is impossible. For an interval contact, its surviving diagonal recovers $c > q$ and $\{a, b\}$, where $a, b < q$. Then $r = a + b - q$, the largest coefficient recovers s, and the negative monomial recovers a. This proves cubic rigidity.

For a four-row cubic beginning with q, the endpoint identity in Lemma 6.4 proves the asserted contraction rule. Suppose instead $a \neq q$. The content table leaves $b = q, r = s$, $c = q, s = t$,

or $u = q, a = s$, where $u = a + b - r$. Expanding (6.5) after division by the content gives the following norms:

$$\begin{array}{c|c} b = q,\ r = s & 6,\ \text{or } 5 \text{ if } c = q \\ c = q,\ s = t & 6,\ \text{or } 5 \text{ if } b = q \text{ or } u = q \\ u = q,\ a = s & 6,\ \text{or } 5 \text{ if } c = q \text{ or } v = q. \end{array}$$

There is one exception to this display:

$$c = q, \quad s = t, \quad d = r, \quad a = q + r - s.$$

To check the exception and completeness of the norm reading, in the right-middle case the six normalized terms are

$$\begin{aligned} &h_a h_{b+q-s} h_d + h_a \overline{h}_b h_{q+d-s} - h_r h_{a+b+q-r-s} h_d \\ &\quad - h_r \overline{h}_{a+b-r} h_{q+d-s} - h_s h_a h_{b+q+d-2s} + h_r h_s h_{a+b+q+d-r-2s}. \end{aligned}$$

The only possible opposite-sign coincidence pairs the first term with the fourth. Since $a, b+q-s > r$, it forces $d = r$; then $a + b - r > a$ forces $a = q + r - s$.

For the left-middle case $b = q, r = s$, the normalized primitive is

$$\overline{h}_c h_u h_d + h_a h_v h_d - h_r h_x h_d - h_t h_u h_w - h_t h_a h_y + h_r h_t h_T.$$

The full-interval term is private. Canceling shared factors in the other positive–negative comparisons excludes all but $h_a h_v h_d = h_t h_u h_w$: one uses $x > \max(u, c, a, v)$, $w > \max(c, d)$, $y > \max(v, d)$, and $c, d, v > t$. In the remaining comparison t must match a, because $v, d > t$. Then $w > d$ forces $w = v$, giving $d = q + t - r$, contrary to $d > q + t$. Only the first term can be deleted by an additional q-contact, namely $c = q$.

For interval extraction $u = q, a = s$, the normalized primitive is

$$-h_b \overline{h}_c h_d + h_s \overline{h}_v h_d + h_t h_b h_w - h_r h_x h_d - h_s h_t h_y + h_r h_t h_T.$$

Here $r < s < b < q$, $c > s+t$, $d > q+t$, and $w > \max(x, d)$. The full-interval term is private. For the term $h_s h_v h_d$, canceling its shared factor with each negative term would respectively require a row b or c to equal s, would equate the distinct least indices s, r, or would equate $\{v, d\}$ with $\{t, y\}$. For $h_t h_b h_w$, the comparisons with the first and fifth terms would require $\{c, d\} = \{t, w\}$ or $\{b, w\} = \{s, y\}$; the comparison with the fourth is impossible because $w > \max(r, x, d)$. The only possible deletions are $c = q$ or $v = q$, not both. Thus these two cases have the stated norms even when same-sign monomials coincide.

On the exceptional locus put $\delta = q - s$. The normalized cubic is

$$h_a^2 h_b - h_r^2 h_{b+2\delta} - h_s h_a h_{b+r+q-2s} + h_r h_s h_{b+r+2q-3s}.$$

Its norm is four, but it cannot be a three-row cubic. Its augmentation would force the untouched type, whose row multiset would be $\{a, a, b\}$. The negative monomial $h_r^2 h_{b+2\delta}$ contains none of those row values: $r < a, b < b + 2\delta$ and $a < b + 2\delta$. Every negative three-row term has an endpoint-row factor. This contradiction disposes of the exception and proves the cross-size result. □

Theorem 6.7 (Size-preserving four-row residual matching)**.** *Let A, B be four-row q-components whose normalized primitive factors have the same degree, either three or four. Their factors agree exactly in the corresponding cases below. The quadratic case is already covered by Lemma 6.6.*

For cubic primitives, either both components are endpoint lifts

$$(q, U - q + r, c, d; r, s, t), \qquad 1 \le r < q,$$

of the same primitive three-row profile $(U, c, d; s, t)$, up to rotation of that contracted profile, or the components are identical, rotations, or members of the rotation quadruple

$$(a, q, a, a + 2q - 2z; z, z, z) \sim (a, a + q - z, q, a + q - z; z, z, z), \quad a, q > 2z, \quad a \neq q.$$

The endpoint lifts may use different values of r. All inherited boundary conditions remain in force.

For quartic primitives, the possibilities are identity, rotation, a full four-row equality from (4.1) *untouched by* $h_q = 0$*, or the rotation quadruple*

$$\begin{aligned} A &= (a, q, a, a+2q-r-s; r, s, r), \\ B &= (a, a+q-s, q, a+q-r; r, r, s), \end{aligned} \qquad a, q > r+s, \quad r \neq s, \quad a \neq q. \tag{6.6}$$

For the full-four-row alternative, "untouched" means that q *is not an interval index of either component.*

Proof. Use the indices u, v, w, x, y, T of (6.5). For $a \neq q$, Lemma 6.5 recovers a, r after orienting a strict comparison with $y > x$, and both endpoint pairs at a tie.

For the remaining readings, let $e_j = \xi^j$ for $j \neq q$, $e_q = 0$, and define

$$\mathcal{D}(P) = \sum_{j \neq q} \left.\frac{\partial P}{\partial h_j}\right|_{h_i = 1 \ (i \neq q)} e_j, \qquad \mathcal{D}_0 = e_x + e_y - e_v - e_T.$$

The contact set records exactly which of a, b, c, u, v, x equal q. Expansion gives

contact set	$\mathcal{D}(P_A)$
$\varnothing$	$\mathcal{D}_0$
b	$\mathcal{D}_0 + e_t + e_w - e_c - e_d$
c	$\mathcal{D}_0 + e_r + e_u - e_a - e_b$
u	$\mathcal{D}_0 + e_c + e_d - e_t - e_w$
uc	$e_x + e_y - e_T - e_a - e_b - e_t - e_v - e_w$
bc	$e_r + e_t + e_x + e_y + e_u + e_w - e_v - e_T$
uv	$e_x + e_y - e_T + e_a + e_s + e_c + 2e_d - e_t - e_w$.

If $G_A = h_g$, then

$$\mathcal{D}(\widehat{K}_A) = -\mathcal{D}(P_A) - \varepsilon(\widehat{K}_A) e_g.$$

The possible nonendpoint contact sets and augmentations of P_A are

$\varnothing, b, c, u$	0
v, uv, bc	1
x, uc	-1.

These are identities of collected polynomials, not distinct-index approximations.

Cubic primitives. The endpoint class and its separation from nonendpoint components follow from Lemma 6.6. For nonendpoint cubics the content table and the preceding derivative give

contact	extraction condition	$\varepsilon(\widehat{K}_A)$	least nonzero derivative coefficient
b	$r = s$	0	-1 at t
c	$s = t$	0	-1 at r
u	$a = s$	0	$+1$ at t
bc	$r = s$ or $s = t$	-1	-1 at the nonduplicated outer gap
uv	$a = s$	-1	$+1$ at t.

In the all-equal bc case the indicated index is that common gap. For uc, the augmentation is one and the part below q is

$$\mathcal{D}(\widehat{K}_A)_{<q} = e_a + e_b + e_t - e_s.$$

Thus interval-extraction and middle-row-extraction mechanisms cannot cross-match.

For the uc case, the endpoint reading is strict and recovers a, r, hence $b = q + r - a, c = q$. Subtracting $e_a + e_b$ from the low derivative leaves $e_t - e_s$. Its nonzero case recovers both gaps; if it is zero, the largest-index coefficient $h_r h_t$ recovers their common value. Then $d = T - 2q + s + t$. This covers both extraction conditions $a = s$ and $s = t$, including their intersection.

For interval extraction $u = q, a = s$ outside that case, the strict endpoint reading recovers $a = s, r$. If also $v = q$, then $b = q - s + r, c = 2s - r$, and T fixes d. For an isolated u-contact, the least derivative recovers t. The restored polynomial $P_A = h_s \widehat{K}_A(-1)$ has its surviving diagonal

as unique minimum-square monomial, recovering the row multiset. Only c, d can exchange. Their difference is

$$h_s\left[h_d(h_r h_{q+c-s} - h_a h_{b+c-s}) - h_c(h_r h_{q+d-s} - h_a h_{b+d-s})\right].$$

If $c > d$, its largest index is $q + c - s$, with nonzero coefficient $h_s h_r h_d$; the reverse inequality is symmetric.

For middle extraction with a nonsquare largest coefficient $h_\alpha h_\beta$, the endpoint gap r and the least derivative index identify whether the duplicated gap is on the left or right. If that least index differs from r, the contact is on the left, $r = s$, and the index is t. Otherwise the contact is on the right, $s = t$, and the other factor is the duplicated gap. Thus G_A and the ordered gap triple are recovered. In the left-middle case with $c \neq q$, direct expansion gives

$$\widehat{K}_A = -h_d M_{q,r}(a, c) - h_t h_u h_w - h_t h_a h_y + h_r h_t h_T.$$

Here $u = a + q - r$, $w = T - a - q + 2r$, and $y = T - a + r$ are known. Subtracting the last three terms therefore recovers $-h_d M_{q,r}(a, c)$. Its quadratic is primitive and linear in its largest generator h_x, with coefficient h_r: all indices in its other two terms exceed r. Gauss's lemma proves irreducibility using only $a, c > r$ and $q > 2r$; the additional boundary $c > q + r$ is not required here. Hence d, then c, is recovered. In the right-middle case put $\eta = a - r > 0$. The exact derivative identity

$$\mathcal{D}(P_A) - e_r + e_a - e_y + e_T = (\xi^\eta - 1)(\xi^b + \xi^v)$$

recovers b, v, since $v > b$, and then s, t, d. The bc case is immediate once its gaps are recovered. At an endpoint tie the two endpoint pairs are already known, and the same readings fix the middle data. This includes the norm-four locus $c = q, s = t, d = r, a = q + r - s$: its derivative and the last identity are unchanged by the cancellation.

The remaining cubic case has all gaps equal to z and a single middle-row contact. For this calculation, rotate to $(A, q, C, D; z, z, z)$, with $A, C, D \neq q$. This rotation preserves the specialized determinant, but need not preserve $D > q + z$. We use only $A, D > z$, $C, q > 2z$, and the fact that the second row is the only index equal to q. No q-component lemma is applied in this temporary orientation. Put $\delta = q - z$. Direct expansion gives, at $h_z = 0$,

$$h_D(h_{A+\delta} h_C + h_A h_{C+\delta}).$$

For $A \neq C$, the quadratic is primitive and linear in its largest generator, so it recovers $\{A, C\}$, and the linear factor recovers D. Swapping A, C changes the remaining primitive by

$$\begin{aligned} -h_z\big[&h_{A+\delta} h_{C+D-z} + h_A h_{C+D+\delta-z} \\ &- h_{C+\delta} h_{A+D-z} - h_C h_{A+D+\delta-z}\big]. \end{aligned}$$

For $A \neq C$, its private largest index is $\max(A, C) + D + \delta - z$; the strict bound needed here is exactly $D > z$, not the discarded inherited boundary. For $A = C = \lambda$, the specialization is $2h_\lambda h_{\lambda+\delta} h_D$. Another decomposition into such a shifted pair is possible only when $D = \lambda + 2\delta$, replacing the repeated row by $\lambda+\delta$ and the endpoint by λ, or conversely. These are precisely the constant-gap quadruple in the theorem, retaining only the orientations satisfying the original component hypotheses.

Quartic primitives. Now $G_A = 1$. If no row equals q, the diagonal monomial survives as the unique minimum-square monomial, with coefficient one. Merging two neighboring interval blocks of indices U, V across a gap g increases square weight by

$$g^2 + (U + V - g)^2 - U^2 - V^2 = 2(U - g)(V - g) > 0.$$

If exactly one row is deleted at $h_q = 0$, every minimum surviving term has negative sign. This distinguishes x from uc, and untouched or u-contact profiles from single-middle contacts. For bc, the two minimum candidates are

$$-h_s h_a h_{2q-s} h_d, \qquad h_r h_t h_u h_w.$$

They cannot cancel because quartic primitivity requires $s \neq r, t$. A negative minimum, or a tie, cannot match a surviving diagonal. If the positive candidate is uniquely minimal, matching a

v- or uv-contact would force its middle-row multiset to be $\{r, t\}$, the two factors below q. Its thickness inequality $L_2 + L_3 > g_1 + 2g_2 + g_3$ contradicts that equality.

For $x = q$, the endpoint reading recovers a, r, and the diagonal recovers the unique row $d > q$. Then $t = q + d - T$ and the gap multiset recover the gaps. After subtracting the fixed terms, equality reduces to equality of

$$R_r(a, b)R_t(c, d), \qquad P_A = R_r(a, b)R_t(c, d) - h_s h_a R_t(v, d) - h_r h_s h_t h_T.$$

Quadratic factorization fixes the ordering; a crossed match requires $r = t, a = d$, impossible because $a < q < d$. For uc, the endpoint reading gives a, r, hence $b = q + r - a, c = q$. The low derivative $-e_a - e_b - e_t$ recovers t, and then s, d.

For $v = q$, including uv, the surviving diagonal and strict endpoint reading recover a, r, d, since $b, c < q < d$. Then $t = a + q + d - r - T$, the gap multiset, and the row multiset recover $s, \{b, c\}$. Removing fixed terms leaves

$$\overline{R_r(a, b)}R_t(c, d) = \overline{R_r(a, c)}R_t(b, d).$$

Only the first quadratic can specialize. If only one side does so, the factor degrees are $1 + 1 + 2$ against $2 + 2$, impossible. Otherwise quadratic matching gives identity or $r = t, a = d$, which is rotation. The tied endpoint case already has both endpoint pairs. For bc, the low derivative is $e_r + e_t$; together with the endpoint reading it recovers both endpoints and outer gaps, and the middle rows are q, q.

An untouched polynomial has least derivative index v, exceeding every gap. An isolated u-contact has least derivative index t, with coefficient -1, so these cases cannot mix. For the latter, the endpoint reading and derivative recover a, r, t, then s, b. The diagonal recovers $\{c, d\}$, and the displayed interval-contact swap defect excludes a nontrivial exchange. For two untouched profiles, Proposition 4.1 supplies exactly the full-function alternatives.

It remains to compare single-middle contacts. In the strict orientation, a, r are recovered. For $b = q$, the least derivative index is t, with coefficient one; for $c = q$, it is r, also with coefficient one. Two left-middle profiles therefore have the same gap triple. Removing their fixed terms leaves

$$h_d(h_r h_s h_x - h_r h_u h_c - h_s h_a h_v).$$

The cubic factor is primitive and linear in h_x: here $r \neq s$, $u > q > s$, $v > q > r$, and $a > r, c > s$. It is irreducible, so d, then c, are fixed. Two right-middle profiles are fixed by the preceding derivative identity $(\xi^{a-r} - 1)(\xi^b + \xi^v)$.

For a mixed comparison, write the source as $(a, q, c, d; r, s, r)$. Equality of the least derivative indices forces its last gap to be r. The mate is (a, b, q, e), with gaps (r, s, r) or (r, r, s), and $c + d = b + e$. Remove the common full and suffix terms. For the first gap order the only possible largest residual indices and coefficient contributions are

	first index	second index
source	$a + q + c - r - s:\ h_r h_s h_d$	$c + d - r:\ h_r^2 h_{a+q-r}$
mate	$a + b + q - r - s:\ h_r h_s h_e$	$q + e - r:\ -h_r R_r(a, b).$

A strict second mate maximum has augmentation zero, unlike the positive source contribution. At a mate tie, canceling its negative monomial requires $a = s, b = e$, and matching the result forces $b = q$, contrary to a single-middle contact. The first mate maximum therefore dominates; matching forces $d = e, c = b$. The remaining equality is

$$h_{a+q-r}R_r(c, d) = h_{d+q-r}R_r(a, c),$$

forcing $a = d$, contrary to the strict orientation.

For the swapped gap order the mate contributions are

$$a + b + q - 2r:\ h_r^2 h_e, \qquad q + e - s:\ -h_s R_r(a, b).$$

If the second is maximal, including a tie, its coefficient is nonzero modulo h_r, unlike every source coefficient. Thus the first dominates. The source's first contribution cannot match it modulo

h_r^2, so the source's second dominates and $e = a + q - r$. In the remainder, setting $h_s = 0$ and $h_r = 0$ respectively gives

$$\{c, d\} = \{a, b + q - r\}, \qquad \{q + c - s, d\} = \{b, q + e - s\}.$$

The first matching $c = a, d = b + q - r$ forces $b = a + q - s$, giving (6.6). The other forces $a = b, d = a$, after which the remaining terms require

$$\{2a + 2q - 2r - s, a\} = \{2a - r, a + 2q - r - s\}.$$

Both entries on the right exceed a, a contradiction. At a tie of the endpoint readings, the same derivative forces equal outer gaps, and the known endpoint pairs make the mixed comparison a rotation.

Finally, for $U = a + q - r$, $V = a + q - s$, and $D = a + 2q - r - s$, direct expansion gives the full identity

$$s_{(a,q,a,D;r,s,r)} - s_{(a,V,q,U;r,r,s)} = h_q\big(h_a R_r(a, D) - h_U R_r(a, V)\big).$$

It proves sufficiency of the quartic family and, at $r = s$, of the nonendpoint cubic family. The endpoint identities prove the remaining sufficiency statements. □

These matching results recover possible components and their contractions. They do not fix the order or orientations of components in a parent.

7. Staircase assembly and higher-row exclusion

7.1. **Compatibility across cuts.** For the following illustration put $L = w_{1234}$, $R = w_{2345} > L$, $T = w_{345}$, and $\kappa = \max(L, T)$. The third supported label has collected coefficient

$$\begin{aligned}[h_\kappa]s_D = &\ -\mathbf{1}_{\kappa=L} h_{z_1} h_{z_2} h_{z_3} h_{r_5} \\ &+ \mathbf{1}_{\kappa=T} h_{z_3} h_{z_4} (h_{r_1} h_{r_2} - h_{w_{12}} h_{z_1}).\end{aligned} \tag{7.1}$$

The binomial is nonzero because $w_{12} > r_1, r_2, z_1$. At $L = T$, both contributions are retained and their sum has augmentation -1. No ordering regime is assumed invariant.

Example 7.1. The equal profiles

$$(2, 4, 4, 4, 5; 1, 1, 1, 2), \qquad (2, 4, 5, 4, 4; 1, 2, 1, 1)$$

have $(L, R, T) = (11, 13, 10)$ and $(11, 13, 11)$, respectively. On the second side a positive and a negative third-label term cancel, leaving $-h_5 h_1^3$, the coefficient on the first side. The proofs below use the collected coefficients and permit this change of regime.

Lemma 7.2 (End-to-interior seam matching)**.** *Suppose a gap q occurs once in the common gap multiset of two equal thick five-row profiles. If its seam is at an end on one side and in the interior on the other, then, after independent rotations, the pair is*

$$(q, b, b, b, u; z, z, z, q) \sim (q, b, u, b, b; z, q, z, z), \qquad u = b + q - z, \quad q > z, \quad b > q + z.$$

No restriction on the other three gaps is assumed. In particular this classifies the $4 + 1$ against $2 + 3$ comparison, including its endpoint-contact subcase.

Proof. Orient the q-cuts as $4 + 1$ and $2 + 3$, and write the first profile as $(a, b, c, d, e; r, s, t, q)$. The four-row primitive cannot have degree four. The two-row component on the other side must extract its gap and become linear: otherwise that side has two nonlinear primitives, whereas the first side has one. Thus its first row is q.

If the common nonlinear primitive is cubic, Lemma 6.6 forces $a = q$ and identifies the other three-row component with $(u, c, d; s, t)$ or its rotation, where $u = q + b - r$. The low-gap content fixes the mate's first gap as r, and its linear primitive gives $e = q + B - r$, where B is its second row. The row multisets reduce to $\{b, e\} = \{B, u\}$. Since $u > b$, this forces $B = b, e = u$. If the common primitive is quadratic, the four-row table forces $a = c = q, s = t$. Its type is $M_{q,s}(u, d)$, whose three-row origin has middle row q and both gaps s. The same content and row-multiset argument again gives $B = b, e = u$.

Every candidate is therefore one of

$$\begin{aligned} F &= (q,b,c,d,u;r,s,t,q),\\ H_+ &= (q,b,u,c,d;r,q,s,t),\\ H_- &= (q,b,d,c,u;r,q,t,s), \qquad u=b+q-r, \end{aligned}$$

with

$$q>r,\quad b>r+\max(q,s),\quad c>s+t,\quad d>q+t,\qquad q\notin\{s,t\}.$$

Put $J=R_r(q,b)$, and

$$A_0=b+c+d+q-r-s-t,\quad B_0=b+c+d-r-t,\quad D_0=2b+c+q-2r-s.$$

After division by h_q, the only potentially largest indices in $s_F-s_{H_-}$ are A_0,B_0, with coefficient contributions $-h_bh_sh_t$, $h_bh_qh_t$. All others are strictly below their maximum. Since $A_0-B_0=q-s\neq 0$, this difference is nonzero.

For $(s_F-s_{H_+})/h_q$, the complete list of possible largest contributions is

$$\begin{array}{c|c} A_0 & -h_bh_sh_t\\ B_0 & h_tJ\\ D_0 & h_dh_rh_s. \end{array}$$

Coincident indices require addition of their displayed contributions. If $s>q$, then $B_0>A_0$. Its only possible cancellation is a tie with D_0, whose coefficient has all-ones value one and is nonzero. Hence $s<q$. Now $A_0>B_0$, so equality forces $A_0=D_0$ and $h_rh_d=h_th_b$. Since $b>r,d>t$, $t=r,d=b$.

After this substitution the possible largest contributions are

$$\begin{array}{c|c} 2b+c-2r & h_rJ\\ 2b+c-r-s & -h_sJ\\ 2b+q-2r & -h_rR_r(b,c). \end{array}$$

If $s<r$, a necessary tie gives $c=q+s-r$; the tied coefficient has private largest factor h_{b+q-r} with coefficient h_sh_r. If $s>r$, the necessary tie gives $c=q$; after that cancellation, the next largest index is $2b+q-r-s$, with coefficient $-h_sJ$. Both are nonzero. Thus $s=r$.

Finally put $\alpha=2b+q-2r$, $\beta=b+c+q-2r$. The entire remaining difference is

$$-h_rh_q\left[h_\alpha R_r(b,c)-h_\beta R_r(b,b)+h_u(h_bh_{b+c-r}-h_ch_{2b-r})\right].$$

If $c>b$, the index β is uniquely largest; if $c<b$, α is uniquely largest. Both coefficients are nonzero. For $c=b$, the bracket vanishes identically. This proves necessity and sufficiency, with $z=r$. All the displayed contributions follow by grouping (6.2); the strict inequalities displayed above bound every omitted index below the respective maximum. □

Lemma 7.3 (The two-and-two cross-pattern)**.** *Let $q>z\geq 1$. A thick equality between profiles with gap words (z,z,q,q) and (z,q,q,z) is, after rotating the second profile, exactly*

$$(q,b,b,u,u;z,z,q,q)\sim(q,b,u,u,b;z,q,q,z),\qquad u=b+q-z,\quad b>q+z.$$

Proof. Write the source as $(a,b,c,d,e;z,z,q,q)$. The safe z-cut and three-row rigidity leave only

$$H_1=(a,c,d,e,b;z,q,q,z),\qquad H_2=(b,c,d,e,a;z,q,q,z).$$

Their common validity gives $c,e>q+z,d>2q$. The q-cut on the source has a three-row component and two singletons; the mate has two two-row components and a singleton. A primitive cubic on the source cannot match. The three-row content table therefore forces $a=q$ or $b=q$. If both equal q, the source retains a quadratic while the mate has only linear factors. If $b=q,a\neq q$, its primitive quadratic is of type M, with augmentation -1; the mate's surviving quadratic is of type B, with augmentation zero. Thus $a=q,b\neq q$.

Put $u=q+b-z$, $v=q+c-z$. For H_2, the specialization would give

$$h_eR_z(u,c)=h_{q+e-z}R_z(b,c),$$

forcing the unequal linear indices $e = q + e - z$, impossible. For H_1, it gives

$$h_e R_z(u, c) = h_v R_z(e, b).$$

Hence $e = v$ and $\{u, c\} = \{e, b\}$. Since $u > b$, $c = b, e = u$. Only the row d remains:

$$F_d = (q, b, b, d, u; z, z, q, q), \qquad H_d = (q, b, d, u, b; z, q, q, z).$$

Set $\alpha = 2b + q - 2z$, $\lambda = b + d - z$, $\mu = b + d - q$, and $v_0 = 2b - z$. The exact defect is

$$s_{F_d} - s_{H_d} = -h_z h_q \left[-h_\alpha R_q(b, d) + h_\lambda R_q(b, u) + h_u(h_d h_{v_0} - h_u h_\mu)\right].$$

For $d > u$, λ is uniquely largest, with coefficient $R_q(b, u) \neq 0$. For $d < u$, α is uniquely largest, with coefficient $-R_q(b, d) \neq 0$. At $d = u$, $\lambda = \alpha, \mu = v_0$, and the bracket vanishes identically. □

Lemma 7.4 (Two terminal cuts)**.** *An equality for which a unique q-seam is at an end on both sides is rotation-trivial. Equivalently, after putting both q-seams last, equal five-row profiles are identical.*

Proof. Write the profiles as $A \cdot_q (e)$, $B \cdot_q (f)$, with $e, f > q$. Every four-row primitive has degree at least two. Low-gap content and unique factorization of the restriction therefore give

$$e = f, \qquad G_A = G_B, \qquad \widehat{K}_A = \widehat{K}_B.$$

The full row multiset also recovers the component row multisets. The residual family (6.6), including its constant-gap cubic case, cannot satisfy those row multisets: after removing a, q, it would require

$$\{a, a + 2q - r - s\} = \{a + q - r, a + q - s\},$$

whose right entries both exceed a.

If the four-row components are full equals, near-concatenation requires their q-contractions with (e) to be full equals too. A nonrotation pair from (4.1) cannot persist: its gap values and unchanged middle rows fix its parameters, while contraction increases the last row by $e - q > 0$. For a component and its rotation, the contraction gap words are reverses, so they cannot give that end/middle exception. Direct matching makes the component symmetric; matching by rotation would require $e = q$.

For endpoint cubic lifts, Lemma 6.6 and theorem 6.7 and the row multiset leave only

$$(q, b, c, u, e; r, s, t, q), \qquad (q, b, c, u, e; r, t, s, q), \qquad u = b + q - r.$$

Put $J = R_r(q, b)$ and

$$\Psi(v) = h_v\big[h_{b+c+q-r-v} R_r(b, e) - h_{b+c-v} R_q(u, e) - h_{b+c+e-r-v} J\big].$$

The full difference is $h_q(\Psi(s) - \Psi(t))$. If $s < t$, its unique largest index is $b + c + e - r - s$, with coefficient $-h_q h_s J \neq 0$; the other inequality is symmetric. Thus $s = t$. For quadratic primitives the four-row origin is $(q, b, q, d; r, s, s)$. Its matching key recovers $s, \{b + q - r, d\}$, and content recovers r. The additional row pair $\{b, d\}$ fixes b, d in order by subtracting sums of squares. All possibilities are now identical. □

7.2. **Interior matching.**

Lemma 7.5 (Adjacent endpoint exchange)**.** *For thick profiles of at least three rows, exchanging just the first two rows while keeping the gap word fixed preserves the Schur function only when those rows are equal. The same holds for the last two rows.*

Proof. Two applications of near-concatenation give

$$s_{(a,b,c,\ldots;r,s,\ldots)} - s_{(b,a,c,\ldots;r,s,\ldots)} = -h_s \left[h_a s_{(b+c-s,\ldots)} - h_b s_{(a+c-s,\ldots)}\right].$$

When $a > b$, the second determinant has a strictly larger full-interval index than every index in the first summand and than a, b, s. Its largest coefficient is nonzero. The other inequality is symmetric; rotation proves the last-two assertion. □

Lemma 7.6 (A contact-blind quadratic operator)**.** *Let $e > q \geq 1$ and $K > q$ be integers, and put $\eta = e - q$. Let $\mu : \mathbb{Z} \longrightarrow \mathbb{Q}$ have finite support satisfying*

$$\operatorname{supp}\mu \subseteq \{s \in \mathbb{Z} : 0 < s < K - q\}, \qquad H_N(\mu) = \sum_s \mu_s h_s h_{N-s}.$$

Then

$$h_e H_K(\mu) - h_q H_{K+\eta}(\mu) = 0 \quad \Longrightarrow \quad \mu = 0.$$

Proof. Reduce modulo h_q. Since the quotient ring is an integral domain and $e \neq q$, $H_K(\mu)$ is divisible by h_q. For every support point, $K - s > q$. Thus its only possible summand containing h_q has $s = q$; the reflected point $K - q$ is excluded. Substitution and cancellation give

$$H_K(\mu) = \mu_q h_q h_{K-q}, \qquad H_{K+\eta}(\mu) = \mu_q h_e h_{K-q}.$$

If q is outside the support range, $\mu_q = 0$ already. Otherwise $K > 2q$. In the second equality, the coefficient of $h_q h_{K+\eta-q}$ is μ_q on the left and zero on the right: its reflected support point exceeds $K - q$, and neither e nor $K - q$ equals q. Hence $\mu_q = 0$. Both quadratic polynomials vanish. Comparing their monomials, with μ extended by zero, gives

$$\mu_s = -\mu_{K-s}, \qquad \mu_s = -\mu_{K+\eta-s}.$$

At a reflection midpoint the coefficient equation gives $\mu_s = 0$. Composition gives $\mu_{s+\eta} = \mu_s$, and finite support implies $\mu = 0$. □

The strict upper support bound cannot be dropped. At $K = q + e$, the excluded measure δ_e gives $h_e(h_e h_q) - h_q h_e^2 = 0$. Below the bound follows from the actual thickness inequalities; no support point equal to q is excluded.

Lemma 7.7 (Interior-to-interior reconstruction)**.** *Let $q > z \geq 1$, and let*

$$F = (a, b, c, d, e; z, q, z, z), \qquad H = (A, B, C, D, E; z, q, z, z)$$

be thick five-row Hessenberg profiles. If $s_F = s_H$, then $F = H$, or, after interchanging the profiles,

$$\begin{aligned} F &= (p, m, p, t, m; z, q, z, z), \\ H &= (m, p, m, t, p; z, q, z, z), \end{aligned} \qquad t = p + m - q,$$

where $p \neq m$ and $p, m > q + z$. The nonrotation alternative is separated. No band condition or specialized four-row matching theorem is used in the proof.

Proof. Thickness means exactly

$$a > z, \qquad b, c > q + z, \qquad d > 2z, \qquad e > z,$$

and the capitalized counterparts. Set

$$x = a + b - z, \quad y = c + d + e - 2z, \qquad x' = A + B - z, \quad y' = C + D + E - 2z.$$

The minimum-cut coefficients are

$$J_0(F) = h_a h_d h_e R_q(b, c), \tag{7.2}$$

$$J_3(F) = -R_q(x, y), \tag{7.3}$$

$$J_2(F) = h_e R_q(x, c + d - z) + h_{d+e-z} R_q(x, c) + h_a R_q(b, y). \tag{7.4}$$

The three terms in the last line contract the pairs $\{1, 3\}$, $\{1, 4\}$, and $\{3, 4\}$ of minimum seams. The two monomials of $R_q(b, c)$ are relatively prime, since $q < \min(b, c)$ and $b + c - q > \max(b, c)$. Its monomial gcd and quotient, and then the highest coefficient, therefore give

$$\{b, c\} = \{B, C\}, \quad \{a, d, e\} = \{A, D, E\}, \quad \{x, y\} = \{x', y'\}. \tag{7.5}$$

All braces denote multisets. The local bounds used for extraction are

$$x > \max(a, b, q), \qquad y > \max(c, d, e, c + d - z, d + e - z, q).$$

In the notation $\bar{P} = P|_{h_q=0}$, full equality also gives

$$\overline{s_F} = \overline{R_z(a, b)}\,\overline{T_z(c, d, e)} = \overline{R_z(A, B)}\,\overline{T_z(C, D, E)}. \tag{7.6}$$

This is the cut at the q-seam. The proof uses this equality of full specializations, not just equality of J_0, J_2, J_3.

Ordered top pair. Suppose $x' = x$, $y' = y$. If $B = b$, then $A = a, C = c$. After subtracting the fixed last term in (7.4), the remainder is

$$h_x H_{y+z}(\mu) - h_q H_{y+z+x-q}(\mu), \qquad \mu = \delta_e + \delta_{d+e-z},$$

where δ_v denotes unit mass at v. With $K = y + z$, the two complements are

$$K - e = c + d - z > q, \qquad K - (d + e - z) = c > q.$$

Thus Lemma 7.6 applies to the difference of the source and mate measures, with multiplier h_x. It recovers $\{e, d+e-z\}$. Since $d > 2z$, the smaller value is e; hence $E = e, D = d$. This includes $x = y$.

Otherwise $B = c \neq b, C = b, A = a + b - c \neq a$. The singleton multiset makes $A = d$ or $A = e$, so $y = x + e - z$ or $y = x + d - z$; in particular, $y > x$. Reduce (7.4) modulo h_q before extracting h_y. The other factors have indices below y, and

$$\bar{h}_a h_b = \bar{h}_A h_c.$$

Both sides cannot vanish without $a = A = q$. Monomial matching therefore gives $a = c = p$, $b = A = m$, with $p \neq m$ and $p, m > q + z$. Up to interchanging the profiles, the remaining singleton allocations are

	F	H
T_1	(p, m, p, m, t)	(m, p, m, p, t)
T_2	(p, m, p, t, m)	(m, p, m, t, p)
T_3	(p, m, p, m, t)	(m, p, m, t, p).

All have gap word (z, q, z, z); coincidences of t with p, m, q are retained. For $X = p + m - z$, their defects are

$$\begin{aligned}
T_1: &\quad \overline{J_2(F) - J_2(H)} = h_X(h_p h_{m+t-z} - h_m h_{p+t-z}),\\
T_2: &\quad J_2(F) - J_2(H) = h_q\big(h_{p+2m-q-z} h_{p+t-z} - h_{2p+m-q-z} h_{m+t-z}\big),\\
T_3: &\quad \overline{J_2(F) - J_2(H)} = h_X(h_X \bar{h}_t - h_m h_{p+t-z}).
\end{aligned}$$

The first is nonzero because $t - z > 0, p \neq m$. The second vanishes only for $p = m$ or $t = p+m-q$. In the third, $t = q$ leaves a nonzero term; otherwise matching forces $t = m$, since $X > m$. This is T_2 with $t = m$, which would force $p = q$. Only the stated family remains.

Swapped top pair. The locus $x = y$ was already treated. Otherwise interchange the profiles so $x > y$, and suppose $x' = y, y' = x$. Modulo h_q, extraction of h_x gives

$$\bar{h}_e h_{c+d-z} + \overline{h_{d+e-z}} h_c = \bar{h}_A h_B. \tag{7.7}$$

Here $c, c + d - z, B > q$. Evaluate the surviving variables at one. Both left terms cannot vanish, since that would give $e = q = d + e - z$, contrary to $d > 2z$. If their monomials coincide they still contribute twice. Hence

$$A \neq q, \qquad e = q \text{ or } d + e - z = q,$$

with exactly one alternative true.

Put $\delta = q - z > 0$, and define

$$\begin{aligned}
N_\delta(c, e) &= h_{c+\delta} h_e + h_c h_{e+\delta} - h_z h_{c+e+\delta-z},\\
I(c, d, e) &= h_e R_z(c, d) + h_z^2 h_{c+q-z}.
\end{aligned}$$

The entire specialization needed here is

condition	$\overline{T_z(c, d, e)}$
$e = q$	$-h_z R_z(c, d + \delta)$
$e \neq q,\ d = q$	$-h_z N_\delta(c, e)$
$d + e - z = q$	$I(c, d, e)$
none of these	$T_z(c, d, e)$.

(7.8)

The first row includes $d = e = q$. The table is exhaustive because $c, c+d-z, c+d+e-2z > q$: only $d, e, d+e-z$ can equal q. Its primitive factors are irreducible in $\mathbb{Q}[h_j : j \geq 1, j \neq q]$. Each is linear in its private largest variable, with coefficient $-h_z$ for R_z, N_δ, and h_z^2 for I, T_z. The remaining polynomial is relatively prime to h_z: modulo h_z it is a nonzero product, or for N_δ a sum of two positive monomials. Gauss's lemma proves irreducibility after specialization. No preservation-of-irreducibility assumption is used. The respective primitive degrees are $2, 2, 3, 3$, and

$$\varepsilon(R_z) = 0, \qquad \varepsilon(N_\delta) = 1.$$

These readings also distinguish the quadratic types when smaller indices coincide.

If $d + e - z = q$, then $d, e < q$, and (7.7) forces

$$A = e, \qquad B = c + d - z = b, \qquad C = c.$$

Indeed $B > c$, so (7.5) identifies it with b. The strict inequality $x > y$ gives $a > e$, and $\{D, E\} = \{a, d\}$. If $a \neq q$, the mate's tail is untouched by specialization: its only possible interval contact is $a + d - z > e + d - z = q$. In (7.6), factor degrees force

$$R_z(a, b) = R_z(e, b),$$

since each side has exactly one irreducible quadratic and one irreducible cubic. The factors have the same leading normalization. It follows that $a = e$, a contradiction. If $a = q$, the source's left factor is $-h_z h_{q+b-z}$, and the mate's tail is $-h_z$ times an irreducible quadratic. After canceling $-h_z$, irreducible factor degrees would be $1+3$ against $2+2$, which is impossible. This separately excludes the row contact that accompanies the interval contact.

It remains that $e = q$. First $a \neq q$: otherwise the source specialization has monomial gcd $h_z^2 h_x$, while the mate's left factor, with $A \neq q$, is an irreducible quadratic and its tail has monomial gcd at most h_z. The source specialization is consequently

$$-h_z R_z(a, b) R_z(c, d + \delta).$$

The mate must have monomial content h_z and two quadratic primitives. Its middle-contact primitive is excluded by $\varepsilon(N_\delta) = 1$, whereas both source quadratics have augmentation zero. Thus $E = q$. Canceling $-h_z$, the distinct largest indices x, y, swapped on the mate, match the quadratic factors in (7.6) and give

$$\{A, B\} = \{c, d + \delta\}, \qquad \{C, D + \delta\} = \{a, b\}.$$

Together with (7.5), these imply

$$\{A, D\} = \{a, d\}, \qquad \{A, D + \delta\} = \{a, d + \delta\}.$$

Subtracting square sums gives $D = d$, then $A = a$. If $B = b$, the top pair was not genuinely swapped. Otherwise $B = c, C = b, b \neq c$, and the factor matching gives $a = d + \delta$. The sole remaining pair is

$$(d + q - z, b, c, d, q; z, q, z, z), \qquad (d + q - z, c, b, d, q; z, q, z, z).$$

Its full second-coefficient defect is

$$J_2(F) - J_2(H) = h_q\big(h_{b+d+q-2z} h_{c+d-z} - h_{b+d-z} h_{c+d+q-2z}\big). \tag{7.9}$$

For $v = b+d-z, w = c+d-z$, its bracket is $h_{v+\delta} h_w - h_v h_{w+\delta}$. Since $\delta > 0$, monomial matching forces $v = w$, hence $b = c$, a contradiction.

For sufficiency in the ordered family, join $(p, m; z)$ and $(p, t, m; z, z)$ in either order across q. Both functions equal

$$s_{(p,m;z)} s_{(p,t,m;z,z)} - h_q s_{(p,t,t,m;z,z,z)}.$$

Rotating the second joining gives the stated profile. Every row exceeds q, since $t = p + m - q > p, m > q$; every nonsingleton interval exceeds its constituent rows. Thus this nonrotation family is separated. □

Remark 7.8. The extractions of h_x and h_y above follow the specialization $h_q = 0$; before it, terms carrying h_q can have larger indices. The cases $x = y$, $a = q$, $e = q$, $d = e = q$, $d+e-z = q$, and $t = q$ are retained. The injectivity bound comes from $c > q$ and $c + d - z > q$, not from a comparison of d with b, c or from any global band hypothesis. For example, the simultaneous row and interval contact occurs in

$$(7, 13, 9, 5, 3; 1, 7, 1, 1), \qquad (3, 13, 9, 5, 7; 1, 7, 1, 1).$$

These have equal J_0, J_3, but after setting $h_7 = 0$ and canceling $-h_1$ their full-function comparison would be

$$h_{19}\big(h_3h_5h_9 - h_1h_3h_{13} + h_1^2h_{15}\big) = R_1(3, 13)R_1(9, 11).$$

The irreducible factor degrees are $1 + 3$ and $2 + 2$, respectively. This is the $a = q, d + e - z = q$ branch, not an excluded degeneration.

7.3. **Assembly of the five-row classes.**

Lemma 7.9 (All remaining five-row placements)**.** *Every equality of thick five-row profiles is identity, rotation, or a member of the following rotation quadruples:*

(7.10)

	F	H
A1	$(p, m, m, m, m + p - z; z, z, z, p)$	$(p, m, m + p - z, m, m; z, p, z, z)$
A2	$(p, m, m, m + p - z, m + p - z; z, z, p, p)$	$(p, m, m + p - z, m + p - z, m; z, p, p, z)$
B	$(p, m, p, p + m - w, m; z, w, z, z)$	$(p, p + m - w, m, p, m; z, z, w, z)$.

For A1 and A2, $z \geq 1, p > z, m > p + z$. For B, $z, w \geq 1, p, m > z + w, p \neq m$.

Proof. Classify by the multiplicity of the minimum gap z. Every safe cut below identifies irreducible full component polynomials, not merely their specialized residuals.

A unique minimum. If the unique minimum is at an end, its safe cut has factor degrees $1, 4$. An interior cut has degrees $2, 3$, so irreducibility forces the minimum seam of an equal mate to be at an end as well. Lemma 7.4, with $q = z$, now gives rotation-triviality.

If the unique minimum is interior, normalize

$$F = (a, b, c, d, e; r, z, s, t), \qquad r, s, t > z.$$

The cut gives a two-row and a three-row component, each up to reversal. A left-only reversal is excluded by Lemma 7.5. The other candidates are

$$(a, b, e, d, c; r, z, t, s), \qquad (b, a, e, d, c; r, z, t, s).$$

Their contractions have gap words (r, s, t), (r, t, s). A nonrotation four-row comparison would be (4.1). Its unique larger gap would then move between positions three and four in the original profiles. Lemma 7.2 would require the other three gaps, including z, to coincide, contrary to their strict inequalities. Thus contractions match directly or by rotation. Direct matches are trivial. A tail-only reversal followed by rotation forces a symmetric tail. Reversing both pieces and rotating the contractions gives exactly

$$r = s = t, \quad c = a, \quad e = b, \quad d = a + b - z.$$

These are B, with repeated gap r, exceptional gap z, and rows $a, b > r + z$. They are separated.

Two minima and distinct other gaps. Let the gaps be $\{z, z, r, q\}$, with $r, q > z$ distinct. Neither unique upper gap can change end/interior status by Lemma 7.2. Together with the degrees in the safe z-cut, this fixes the gap word up to rotation. The four orders, allowing the names r, q to exchange, are

gaps	cut sizes	contracted rows
(z, z, r, q)	$(1, 1, 3)$	$(a + b + c - 2z, d, e)$
(z, r, z, q)	$(1, 2, 2)$	$(a + b - z, c + d - z, e)$
(z, r, q, z)	$(1, 3, 1)$	$(a + b - z, c, d + e - z)$
(r, z, z, q)	$(2, 1, 2)$	$(a, b + c + d - 2z, e)$.

The first two orders have different factor degrees; in the third both upper gaps are interior, and in the fourth both are at ends. Distinct gaps identify the two quadratic factors and orient a recovered three-row factor once the gap word is fixed. In the first order only the first two singleton rows can exchange, and Lemma 7.5 excludes it. In the other three orders, the highest coefficient is a three-row function with the fixed asymmetric gap word (r, q). Its ordered rows in the table therefore agree. Combined with the recovered components, they fix every original row.

Two minima and equal other gaps. Write $q > z$. Up to rotation the four orders are $P = zzqq$, $Q = zqzq$, $R = zqqz$, $S = qzzq$. The safe-cut degrees are $(1, 1, 3)$ for P, R, and $(1, 2, 2)$ for Q, S. The highest coefficients have the contracted triples

$$\begin{array}{c|c} P & (a+b+c-2z, d, e; q, q) \\ Q & (a+b-z, c+d-z, e; q, q) \\ R & (a+b-z, c, d+e-z; q, q) \\ S & (a, b+c+d-2z, e; q, q). \end{array}$$

For P/P, the top coefficient fixes the orientation of the recovered three-row component: its nontrivial reversal would require $a + b = 2z$. Only the first-two exchange remains, and is rigid. For R/R, rotate the mate to align its middle triple. An endpoint exchange is allowed by the top coefficient only when $a = e$ or $b = d$, giving identity or full rotation. For S/S, align the two recovered quadratic components by rotation. A single component reversal is an endpoint exchange. Reversing both and comparing the top triples gives identity or full rotation.

For Q/Q, the safe z-cut fixes the singleton a. If $a = q$, the q-restriction is

$$-h_z h_{q+b-z} R_z(c, d) h_e.$$

It recovers $\{c, d\}$ and $\{q + b - z, e\}$. The row multiset also recovers $\{b, e\}$, so subtraction of sums of squares fixes b, e individually. If $a \neq q$, the q-cut has two irreducible quadratics and the unique singleton h_e, recovering e. The highest z-coefficient fixes its contracted triple in order: reversal with the same last row e forces the first row to equal e and changes nothing. Hence b is fixed. Removing the known $R_z(a, b)$ from the q-cut then recovers $\{c, d\}$. In both cases only c, d can exchange. The safe constant term $h_a R_q(b, c) R_q(d, e)$ shows that a nontrivial exchange requires $b = e$. For this remaining comparison, the full difference is

$$\begin{aligned} h_z h_q \big[& h_{a+b+c-q-z} R_q(b, d) - h_{a+b+d-q-z} R_q(b, c) \\ & + h_{a+b-z}(h_c h_{b+d-q} - h_d h_{b+c-q}) \big]. \end{aligned}$$

If $c > d$, its private largest index is $a + b + c - q - z$; if $d > c$, it is $a + b + d - q - z$. The coefficients are nonzero, since $a > z$ and $b, c, d > q + z$. Thus $c = d$.

For Q/S, the safe cut forces the middle row of S to equal a, the singleton of Q. If $a \neq q$, the q-cut has nonlinear degrees $2, 2$ on Q and degree three on S, impossible. If $a = q$, it compares an ordinary quadratic primitive B with a middle-contact primitive M, of augmentations zero and minus one. This is also impossible. Only P/R remains, and Lemma 7.3 gives A2 without any extra contact hypothesis.

Three minima. The unique $q > z$ is at an end or in the interior. End/end is Lemma 7.4, and end/interior is Lemma 7.2. After independently rotating the profiles, interior/interior has gap word (z, q, z, z) on both sides. Lemma 7.7 gives identity or the separated commutation family B, including the balanced top pair and every contact specialization.

Four minima. All gaps equal z. Set $\alpha_i = L_i - z > 0$. If r_α is the ribbon Schur function, then

$$s_D = h_z^5 \Phi_z(r_\alpha), \qquad \Phi_z(h_j) = h_{j+z}/h_z \quad (j > 0).$$

This substitution is injective in the localization at h_z, and commutes with reversal. By [2], a nonrotation ribbon equality requires reversal choices in at least two nonsymmetric composition factors. Each has at least two rows and three cells. The identity $\ell(\alpha \circ \beta) = \ell(\alpha) + |\alpha|(\ell(\beta) - 1)$ forces at five rows an outer factor $(1, 2)$ or $(2, 1)$ and a two-row inner factor; row-preserving one-part factors are absorbed into the inner parts. Their profiles are B with $w = z$. Thickness gives $p, m > 2z$.

The coverage just proved can be recorded without equation-space tasks:

(7.11)

minimum-gap multiplicity	possible nonrotation classes
1	B with a uniquely smaller gap
2, other gaps distinct	$\varnothing$
2, other gaps equal	A2
3	A1 or B
4	B with equal gaps.

A1 and A2 are sufficient by the two branch lemmas. For B put $E = (p, m; z)$. Its two profiles are $E \cdot_w (E \odot_w E)$ and $(E \odot_w E) \cdot_w E$, the latter rotated if necessary. Both functions equal $s_E s_{E\odot_w E} - h_w s_{E\odot_w E\odot_w E}$. This proves the classification and all its sufficiency assertions. □

Completion of the proof of Theorem 6.2. By Lemma 6.1, an arbitrary connected mate of a thick five-row staircase has the same full board pattern. The minimum-square cofactor reading in Lemma 2.1 recovers its positive gap multiset, so it is thick as well. Lemma 7.9 therefore leaves exactly (7.10). The dictionary to (6.3) is

M1	$(p, m, z, w) = (a + d, b + d, d + t, d)$
M2	$(p, m, z, w) = (a + d, b + d, d, d + t)$
M3, M4	$(p, m, z) = (a + d, b + d, d)$.

For B use M1 when $z \geq w$, with $(a, b, t, d) = (p-w, m-w, z-w, w)$, and M2 when $w > z$, with $(a, b, t, d) = (p-z, m-z, w-z, z)$. Both rows and gaps agree, including the long row $p+m-w$, and the domain becomes exactly $a, b > t+d$. For A1 and A2 take $(a, b, d) = (p-z, m-z, z)$. The four members are distinct under the stated nonrotation inequalities. The sorted row multisets have multiplicities $(1, 3, 1)$, $(1, 2, 2)$, and $(2, 2, 1)$ for A1, A2, and B, respectively, so these families cannot mix. In A1 and A2 the three row values recover $p, m, k = m + p - z$, then z. In B the repeated rows recover $\{p, m\}$, the largest row k gives $w = p + m - k$, and the gap multiset recovers z. Exchanging p, m rotates the same quadruple. Thus no additional mate can arise from another parameter choice; at $z = w$ the M1 and M2 descriptions give the same quadruple.

For unit boards, Lemmas 3.1 and 6.1 put both images in the thick domain. All bare template parameters on a Γ-image are odd, so a, b, t in the dictionary are even and d is odd. Replace them by $a/2, b/2, t/2, (d-1)/2$. Their inequalities give exactly the integer domains in the theorem, including $d = 0$. Injectivity returns the full original class. □

Corollary 7.10 (Reduced minors for the contact families)**.** *Every nonrotation equality between thick five-row profiles with a contact is represented by one of the following reduced Toeplitz-minor coincidences, of orders three and two respectively.*

Proof. The B profiles in (7.10) are separated: every row exceeds both gap values, and every interval index is at least each of its constituent row lengths. Thus Lemma 7.9 leaves only A1 and A2. In those two families write $q = p$, $b = m$, and set $\sigma = b - z$, $\delta = q - z$. Then $0 < \delta < q < \sigma$. Rotate both displayed profiles and normalize the last column beta to zero. Their common row-beta list is

$$X = (q + 4\sigma, q + 3\sigma, q + 2\sigma, q + \sigma, q).$$

Their decreasing column-beta lists are the decreasing arrangements of

$$\{j\sigma : j \in I\} \ \cup \ \{\delta + j\sigma : j \in J\},$$

with masks

	I	J	I'	J'
A1	$\{0, 3\}$	$\{0, 1, 2\}$	$\{0, 1\}$	$\{0, 2, 3\}$
A2	$\{0, 2, 3\}$	$\{0, 1\}$	$\{0, 1, 2\}$	$\{0, 3\}$.

The primed masks describe the second member. The sets I, J may intersect: an element in their intersection is a common shift level in two different residue sequences, not a repeated column. Complements below are taken in $\{0, 1, 2, 3, 4\}$.

In the localization at h_q, put

$$U(t)=\sum_{j=0}^{4}h_{q+j\sigma}t^j,\qquad V(t)=\sum_{j=0}^{4}h_{z+j\sigma}t^j,\qquad C(t)=\frac{V(t)}{U(t)}=\sum_{j=0}^{4}c_jt^j\pmod{t^5}.$$

Let $T(C)_{ij}=c_{i-j}$ for $0\le i,j\le 4$, with $c_j=0$ for $j<0$, and define $T(U)$ similarly. In increasing row-beta order, columns from the two levels are shifted columns of $T(U)$ and $T(V)$. Multiplication by $T(U)^{-1}$, followed by expansion along the coordinate columns, gives, including the column-ordering signs,

$$s_D=\begin{cases}-h_q^5\det T(C)[I^c,J], & D \text{ in A1},\\ h_q^5\det T(C)[I^c,J], & D \text{ in A2}.\end{cases}$$

The primed masks give the same formula for the mate. For $\iota(j)=4-j$, both rows of the mask table satisfy

$$I'=\iota(J^c),\qquad J'=\iota(I^c).$$

Transposition and simultaneous reversal of the row and column orders therefore identify the reduced minors. Explicitly, they are

$$\det\begin{pmatrix}c_1&c_0&0\\c_2&c_1&c_0\\c_4&c_3&c_2\end{pmatrix}=\det\begin{pmatrix}c_2&c_0&0\\c_3&c_1&c_0\\c_4&c_2&c_1\end{pmatrix}$$

for A1, and

$$\det\begin{pmatrix}c_1&c_0\\c_4&c_3\end{pmatrix}=\det\begin{pmatrix}c_3&c_0\\c_4&c_1\end{pmatrix}$$

for A2. Row and column index sets in these minors are in increasing order. Since $U(0)=h_q$, the required inverse exists in the localization, and the localization map is injective. No specialization at $h_q=0$ is used in this reduction.

The Toeplitz-minor correspondence and the row/column reversal are classical; see [3]. Here the preceding classification supplies the additional conclusion that equality forces one of these charts. No claim that an arbitrary contact profile admits a chart is needed. □

7.4. **Endpoint extensions and borders.** The minimum-gap classification in Theorem 6.2 now supplies the complete list of possible nonliteral five-row children. The coefficients here test their endpoint extensions and borders directly, including every tie of the largest candidate indices.

Lemma 7.11. *For every connected diagram C with $n\ge 5$ rows and full negative set $\mathcal{T}_n$, common betas allowed, and every integer $t>0$,*

$$s_{C^{\uparrow t}}=s_{C^{\downarrow t}}\quad\Longleftrightarrow\quad C=C^*.$$

Proof. For five rows, unequal endpoint row lengths are separated by the row multiset. If they both equal a but the exterior gaps differ, the largest proper interval span is $\kappa=N-a+\max(z_1,z_4)$. After cancellation of the full-interval term, its shifted holder has label $\kappa+t$ and nonzero coefficient $-h_ah_{z_1}h_{z_2}h_{z_3}$ if $z_4>z_1$, or $h_ah_{z_2}h_{z_3}h_{z_4}$ if $z_1>z_4$.

For tied exterior data write $C=(a,x,y,z,a;b,c,d,b)$, $q=a-b>0$. The exact domain is $b,c,d\ge 0, x>b+c, y>c+d, z>b+d$. Put $X=x+y-c$, $Y=y+z-d$, and $K_s=h_ah_s-h_bh_{s+q}$. Define

$$\begin{aligned}F_{u,v}=\;&h_{a+t}(h_{u+q}h_v-h_uh_{v+q})-h_a(h_{u+q+t}h_v-h_uh_{v+q+t})\\&+h_b(h_{u+q+t}h_{v+q}-h_{u+q}h_{v+q+t}).\end{aligned}$$

Expansion of the sixteen terms gives the full identity

$$D_C(t)=h_b(h_yF_{x,z}-h_cF_{X,z}-h_dF_{x,Y}).$$

Its largest possible label is $L = \max(X, Y) + q + t$, with coefficient

$$[h_L]D_C(t) = \begin{cases} h_b h_c K_z, & X > Y, \\ -h_b h_d K_x, & Y > X, \\ h_b(h_c K_z - h_d K_x), & X = Y. \end{cases}$$

Each strict coefficient is nonzero, since $s > b$ makes $K_s \neq 0$. At the tie, nonsymmetry forces $x \neq z$; the generator $h_{\max(x,z)+q}$ then occurs in just one summand and is above the other coefficient labels. Thus the coefficient is again nonzero. If $x = z, c = d$, the profile is symmetric.

For a zero-free source with $n \geq 6$, the maximal cofactor is connected and has pattern $\mathcal{T}_{n-1}$. Put $u = x_1 - x_2, v = y_{n-1}$, $\kappa = N - \min(u, v)$, and call the normalized child K. Formula (3.8), with its cofactor bounds, gives

$$[h_{\kappa+t}]D_C(t) = \begin{cases} (-1)^n s_{K^{\downarrow v}}, & u > v, \\ (-1)^{n+1} s_{K^{\uparrow u}}, & v > u, \\ (-1)^{n+1} D_K(u), & u = v. \end{cases}$$

The strict coefficients are nonzero. At the tie induction leaves only $K = K^*$, which together with $u = v$ means $C = C^*$. Finally homogenize, replacing the extension length by $2t$, to cover all common betas. This induction concerns endpoint differences only; it does not infer literal cofactor equality from equal polynomials. □

Lemma 7.12. *For every pair in* (6.3)*, no positive connected border over the F rotation orbit is Schur-equal to a positive connected border over the H rotation orbit.*

Proof. We give the endpoint and tied-border coefficients explicitly. They follow by substitution in (6.2), combining equal labels before extraction. Put $A = h_{a+d}$, $B = h_{b+d}$, and use

	c	ℓ	κ	σ
M1	$d + t$	$a + b + d - t$	$2a + 2b + d - 2t$	t
M2	d	$a + b + d$	$2a + 2b + d - t$	0
M3, M4	d	$a + b + d$	$a + 3b + d$	0.

Set $C = h_c$, $K = Ch_\ell - AB \neq 0$; its negative is a connected two-row Schur polynomial. For $\Delta_\epsilon(z) = s_{F^{\epsilon z}} - s_{H^{\epsilon z}}$, the largest-label coefficients are

	$[h_{\kappa+z}]\Delta_\uparrow(z)$	$[h_{\kappa+z}]\Delta_\downarrow(z)$
M1, M2	C^2K	$-C^2K$
M3	$-h_d^2 K$	not needed
M4	$-Ah_dK$	not needed.

All other labels are smaller under the stated family inequalities. For M1 and M2 the opposite extension directions are separated by row multisets, because $a \neq b$. For M3 and M4 the distinct endpoint lengths $a + d, b + d, a + b + d$ separate every comparison except top against top. Thus all positive endpoint cross-comparisons are unequal. By Lemma 3.3, this excludes borders with unequal margins.

At equal margins (q, q), orient the children as displayed. Their corner and square-marker coefficients cancel, and the linear coefficient at h_{S+q} is $E_q = \Delta_\uparrow(q) + \Delta_\downarrow(q)$. For M1 and M2 set $q_0 = \min(a, b) - \sigma > 0$, $A_q = h_{a+d+q}$, $B_q = h_{b+d+q}$. Its largest label is $L = \max(\kappa, \kappa + q - q_0)$, and

(7.12)

regime	$[h_L]E_q$
$q < q_0$	$C^2(A_qB - AB_q)$
$q > q_0,\ a > b$	$-BCK$
$q > q_0,\ a < b$	ACK
$q = q_0,\ a > b$	$AC(B^2 - Ch_{2b+d-\sigma})$
$q = q_0,\ a < b$	$BC(Ch_{2a+d-\sigma} - A^2)$.

For verification of the maximum, every remaining label in the interval expansion is strictly dominated by one of $\kappa, 2a + b + d - t + q, a + 2b + d - t + q$. Ordering these gives the three

regimes. At $q = q_0$ both contributions are retained, giving the last two rows rather than either strict formula. The first binomial is nonzero for $a \neq b, q > 0$; the strict branches contain $K \neq 0$; the tied branches have a generator strictly above $b + d > c$, or $a + d > c$. Every coefficient is nonzero.

For M3 and M4 the parent row multiset forces $q = b$: after common rows are canceled it gives $\{2b, a + b + q\} = \{a + 2b, b + q\}$. At that value the largest label of E_b is $4b + d$, with coefficient $-Ah_dK$ for M3 and $-A^2K$ for M4. The inequality $b > a + d$ makes these isolated and nonzero. All labels of E_q are below the outer marker $S + q$. The formulas allow $d = 0$ with $h_0 = 1$; alternatively Γ sends (a, b, t, d) to $(2a, 2b, 2t, 2d + 1)$ and doubles the border margins. Thus every unit case descends as well. □

Proposition 7.13. *Every connected source with at least six rows and full negative set $\mathcal{T}_n$ is rotation-rigid against arbitrary connected mates. Common beta entries are allowed.*

Proof. For a zero-free six-row source the maximal cofactor is a connected five-row staircase. An arbitrary mate has a positive-row connected cofactor by Lemma 2.1 and irreducibility. Lemma 6.1 and theorem 6.2 identify its complete class. Cross-orbit borders are excluded by Lemma 7.12; after same-orbit alignment, Lemma 7.11 excludes a nontrivial margin swap. At larger row counts the cofactor is the connected staircase one order down. Induction and the same endpoint assertion apply without an upper row bound. Homogenization supplies common betas. The five-row input is the algebraic classification in Lemma 7.9; all subsequent steps use the displayed endpoint and border coefficients. □

7.5. **Verification.** Both finite-order classifications and their higher-row deductions use algebraic arguments in the text. No execution record, feasibility answer, or exclusion certificate is a premise.

The supplement at `https://github.com/johnnyshields/papers` contains auxiliary checks by signed permutations and column-subset dynamic programming. In formal-label checks, numerical substitution is followed by collection: equal numerical indices do not remain independently labeled generators. The cancellation in Example 7.1 is permitted by the proof, not assigned to an additional search branch. Finite sample boxes test identities, units, ties, and implementations; they do not establish coverage of the parameter domains.

8. Disconnected cofactors and rigidity

8.1. **Endpoint coefficients.** The notation $D_Q(t) = s_{Q^{\uparrow t}} - s_{Q^{\downarrow t}}$ continues from Section 3. By Theorem 4.2, every one-negative component and its endpoint extensions are rotation-rigid. Thus

$$D_Q(t) = 0 \quad \Longleftrightarrow \quad Q = Q^*, \qquad t > 0. \tag{8.1}$$

The same assertion holds for dense components by Lemma 3.5.

Lemma 8.1. *Let $Q \neq Q^*$ be connected and zero-free, either dense of order $m \geq 2$, or one-negative of order $m \geq 3$. Then the nonzero, shift-independent leading-coefficient assertion of* (3.9) *holds for every $t > 0$.*

Proof. The dense case and orders three and four are Lemma 3.6. For a one-negative component of order at least five, use (3.8). Every cofactor index is at most κ, so an untied top has one nonzero minor as coefficient. At $C_2 = S - D_{m-1}$, put $d = D_{m-1}$. The coefficient is $(-1)^{m+1}D_R(d)$, where

$$R = ((C_2, \ldots, C_m); (D_1, \ldots, D_{m-1})).$$

This child is connected and one-negative. By (8.1), vanishing would make R symmetric. Together with the equal exterior gaps d, that would make Q symmetric. All minors are fixed as t varies. □

Lemma 8.2. *Let $Q = (C, D)$ be connected and zero-free with four rows and exactly two negative indices, normalized by $C_1 = S, D_4 = 0$. For every $t > 0$,* (3.9) *holds with $\kappa = \max(C_2, S - D_3)$ and a nonzero coefficient independent of t.*

Proof. In (3.8), all cofactor indices are at most κ. The coefficient of $h_{\kappa+t}$ is M_{13}, $-M_{24}$, or $M_{13} - M_{24}$, according as $S - D_3$ is greater than, less than, or equal to C_2. Each strict case is a minor with positive sorted diagonal and is nonzero.

At a tie put $d = D_3 = S - C_2 > 0$ and $R = ((C_2, C_3, C_4); (D_1, D_2, D_3))$. It has three positive rows and exactly two negatives. A connected three-row board has at most one negative; three singleton components would cost three. Thus R consists of a singleton of length $s > 0$ and a connected dense two-row component T. Directly, $M_{13} - M_{24} = -D_R(d)$. For the two possible component orders,

$$D_R(d) = h_{s+d}s_T - h_s s_{T^{\downarrow d}}, \qquad D_R(d) = h_s s_{T^{\uparrow d}} - h_{s+d}s_T,$$

respectively. Endpoint extension preserves the split. Modulo h_s, either expression is, up to sign, $h_{s+d}s_T \neq 0$: the two coordinate variables are distinct and the connected degree-two factor has no coordinate-variable divisor. This proves the tied case. □

Lemma 8.3. *If Q is connected and zero-free, has at least five rows and at most four negatives, then Q and its positive endpoint extensions are rotation-rigid against arbitrary connected mates. For nonsymmetric Q,* (3.9) *also holds.*

Proof. The rigidity assertions follow from Theorems 4.2, 4.3 and 4.6 and propositions 4.5 and 4.8; their only four-row exceptions are (4.1). Endpoint extensions preserve negative positions.

Use (3.8). Only a tied top level needs consideration. Its coefficient is an endpoint difference of the maximal cofactor R, with the common exterior gap as extension parameter. If R is connected, it has at most four negatives. The only possible nonrotation ambiguity is the four-row three-negative family

$$(w, b, b, b + w - c; c, c, w), \qquad (w, b, b + w - c, b; c, w, c), \qquad c < w.$$

Opposite endpoint extensions retain the same overlap word. The displayed words, including their reversals, are distinct, so these extensions cannot pass between the two exceptional rotation orbits. Equality therefore implies symmetry of R, hence of the tied parent Q.

If R is disconnected, Lemma 3.2 leaves orders $1 + 3$ or $2 + 2$ when R has four rows, and only $1 + 4$ when it has five rows. In the unequal-order cases, factorization separates the extended first component from the unchanged last component by ordinary degree; opposite endpoint extensions cannot agree. In the $2+2$ case, equality forces the two factors to exchange. Two-row rigidity then identifies the components as reverse partners, making R, and thus Q, symmetric. At six or more cofactor rows a split costs at least five negatives. This exhausts the tied cases without using the threshold theorem. □

8.2. **A saturated negative column.**

Corollary 8.4. *Let Q be connected and zero-free of order $m \geq 2$, with at most two negative indices in its full row board. For all positive integers s, u, v, g, every parent in* (3.6) *is rotation-rigid against arbitrary connected mates.*

Proof. The component and its positive endpoint extensions are rotation-rigid by Theorem 4.2; endpoint extension preserves every negative position. For nonsymmetric Q, (3.9) follows from Lemma 8.1 in the dense and one-negative cases. With two negatives, its order is at least four: use Lemma 8.2 at order four and Lemma 8.3 at every larger order. Thus every hypothesis of Lemma 3.7 holds. □

Theorem 8.5. *Let F be connected with $n \geq 6$ nonempty rows. Suppose that, after a possible rotation, the first column of its full row board has all $n-2$ negative entries permitted by connectedness, and at most two negative entries lie outside that column. Then every connected skew diagram H with $s_H = s_F$ is F or F^*, up to translation. Common beta entries are allowed.*

Proof. The hypothesis is preserved by Lemma 3.1, so first work zero-free. Use the indicated orientation, and write the full beta lists as X, Y, normalized by $y_n = 0$. The column condition is $x_3 < y_1$. The maximal cofactor has a positive first singleton row, and its remaining block

$$Q = (x_3, \ldots, x_n)/(y_2, \ldots, y_{n-1})$$

has $m = n-2 \geq 4$ positive rows and at most two negatives. Its diagonal entries are subdiagonal entries of the original board, hence positive. A disconnection after row k would require a negative lower-left rectangle with $k(m-k) \geq m-1 \geq 3$ entries. Thus Q is connected and zero-free.

Put

$$s = x_2 - y_1, \qquad u = x_1 - x_2, \qquad v = y_{n-1}, \qquad g = y_1 - x_3.$$

These parameters are positive. Normalize Q by subtracting v from both lists. Its span is $S = x_3 - v$, so the offset in (3.6) is $z = v + S + g = y_1$. Substitution recovers exactly the lists of F. Apply Corollary 8.4. Finally, homogenization transports an arbitrary full-board equality at its common row count, preserves the hypothesis, and descends the rotation conclusion by injectivity and rotation covariance. □

The negative count is $n-2$, $n-1$, or n, and is unbounded. For every $n \geq 6$, the partition $(3, 1^{n-3})$ is realizable by Theorem 2.5; it satisfies the column hypothesis and has n negatives. The lower row bound cannot be dropped: the first source at $(b, g, q) = (2, 3, 1)$ in (5.1) has three first-column negatives and two outside, but has a nonrotation mate.

8.3. **Two nonlinear components.**

Lemma 8.6. *Suppose the zero-free maximal cofactor has two connected components Q, R, each with at least two rows. Assume that each component and its endpoint extensions are rotation-rigid against arbitrary connected mates. Assume also that every nonsymmetric component satisfies* (3.9)*, with a nonzero coefficient independent of the extension parameter. Then the parent is rotation-rigid against arbitrary connected mates.*

Proof. Normalize $Q = (C, D)$, $C_1 = A, D_p = 0$, and $R = (E, F)$, $E_1 = S, F_r = 0$. Factorization recovers their functions and hence their literal diagrams up to order and rotation. Every parent of the ordered components has lists

$$\begin{aligned} &z = v + S + g, \\ &X = (z + A + u, z + C_1, \ldots, z + C_p, v + E_1, \ldots, v + E_r), \qquad u, v, g > 0. \\ &Y = (z + D_1, \ldots, z + D_p, v + F_1, \ldots, v + F_r, 0), \end{aligned} \tag{8.2}$$

Its span is $N = A + S + u + v + g$. At $N - \min(u, v)$, the coefficient after corner removal, up to one common sign, is

$$\begin{cases} s_{Q^{\uparrow u}} s_R, & u < v, \\ s_Q s_{R^{\downarrow v}}, & v < u, \\ s_{Q^{\uparrow u}} s_R + s_Q s_{R^{\downarrow u}}, & u = v. \end{cases} \tag{8.3}$$

Divisibility by the old irreducible factors identifies the regimes. If $s_Q = s_R$, cancel that factor: a strict marker has one endpoint-extension contribution, whereas a tied marker has two. At their common largest index, the tied coefficient is exactly twice the strict one, since both extensions have the same signed maximal cofactor. At a tie the span recovers g.

If the two old factors differ, rotate to align their order and subtract two proposed tied markers. Reduction modulo s_Q shows that s_Q divides the difference of the two proposed top extensions of Q. Such a difference, if nonzero, is homogeneous of ordinary degree p and weighted degree $|Q| + u$. Divisibility by s_Q, of ordinary degree p and weighted degree $|Q|$, would force a scalar quotient and hence $u = 0$, a contradiction. Thus the top extensions agree. The same argument modulo s_R equates the bottom extensions. Endpoint rigidity fixes each component orientation. For equal old factors, cancel their common factor. Of the four orientation choices, a whole-parent rotation pair gives $s_{Q^{\uparrow u}} + s_{Q^{\downarrow u}}$; the other two give $2s_{Q^{\uparrow u}}$ and $2s_{Q^{\downarrow u}}$. Endpoint rigidity separates these three values unless the component is symmetric.

For strict margins rotate to have $u < v$. The marker identifies the ordered Q, its orientation, and u. Suppose two remaining candidates have $v_- < v_+$. Subtract the common initial last-column terms. At $K = \max(N - v_-, S + v_+)$, the coefficient of their difference is

$$\mathbf{1}_{K=N-v_-}(-1)^{p+r} s_Q s_{R_-^{\downarrow v_-}} + \mathbf{1}_{K=S+v_+}(-1)^{r+1} s_T. \tag{8.4}$$

The second cofactor has subdiagonal indices

$$C_1 - D_1, \ldots, C_p - D_p, E_2^+ - F_1^+, \ldots, E_r^+ - F_{r-1}^+ > 0.$$

Thus T is connected, and its span $N - v_+$ is below K. All indices of the product coefficient are also below K. Strict maxima are nonzero, and at a tie irreducibility separates s_T from the product. The values v, g are fixed.

Only reversal of R remains. It is literal if $R = R^*$. Otherwise, write κ_R, c_R for its offset and nonzero coefficient in (3.9). For $0 \le i \le p$, let A_i be the determinant with column list D and row list $(A+u, C_1, \ldots, C_p)$ with entry i deleted, counting from zero. Then $A_0 = s_Q$, $A_1 = s_{Q\uparrow u}$. Put

$$w_0 = g + A + u, \qquad w_i = g + C_i \quad (i \ge 1).$$

Laplace expansion expresses the orientation difference as a signed sum of $A_i \mathcal{E}_R(w_i, v)$. For $v < w_0$ the $i = 0$ term has a private largest index. For $v > w_0$, the coefficient at $S + v$ has the private moving index $\kappa_R + w_0$. At $v = w_0$ the $i = 0$ term vanishes, leaving, up to sign,

$$T = \sum_{i=1}^{p} (-1)^{p-i} A_i D_R(w_i). \tag{8.5}$$

Write

$$L = A + u, \qquad t_0 = \kappa_R + g + A, \qquad B_Q = \det(h_{C_i - D_j})_{2 \le i \le p,\ 1 \le j < p}.$$

The sorted diagonal of B_Q is positive, so $B_Q \neq 0$, and $[h_L]A_1 = (-1)^{p+1} B_Q$. For $i \ge 2$, the top index of $D_R(w_i)$ is below t_0. Also $t_0 > A$, whereas every index in $[h_L]A_i$ is below A. Extracting the largest indicated coefficients in (8.5) gives

$$\begin{array}{c|c} t_0 > L & [h_{t_0}]T = (-1)^{p-1} A_1 c_R \neq 0 \\ t_0 = L & [h_L^2]T = B_Q c_R \neq 0 \\ t_0 < L & [h_L h_{t_0}]T = B_Q c_R \neq 0. \end{array} \tag{8.6}$$

No row-order restriction was placed on Q beyond $p \ge 2$. This excludes the last orientation change. □

8.4. **Minimum realizing row count six.**

Theorem 8.7. *Let F be connected with $\nu(F) = 6$ and $n_{\min}(\rho(F)) = 6$. Every connected skew diagram H satisfying $s_H = s_F$ belongs to $\{F, F^*\}$. The two members are distinct, and common beta entries are allowed.*

Proof. First work zero-free with six rows. The maximal cofactor has five positive rows and is disconnected by Lemma 3.4. Three components cost at least seven negatives. The component budget (3.3) therefore leaves only

component orders	cross cost	internal negatives
$1 + 4$	4	2
$2 + 3$	6	0.

For $1 + 4$, the component and its endpoint extensions are rigid by Theorem 4.2; Lemma 8.2 supplies Lemma 3.7's remaining hypothesis. For $2 + 3$, both components and their endpoint extensions are dense and rigid by Lemma 3.5. Their endpoint coefficients satisfy Lemma 8.1, so Lemma 8.6 applies. Both lifting lemmas compare with arbitrary connected mates.

For $n > 6$, the maximal cofactor R is connected, has one fewer row, and retains the negative partition. Induction applies to it and to its positive endpoint extensions. An equal mate is zero-free by Lemma 2.1; factorization of the maximal coefficient identifies its connected cofactor with R or R^*. After rotation aligns the literal cofactor, Lemma 3.3 recovers the unordered margins. A strict swap equates $R^{\uparrow t}$ and $R^{\downarrow t}$. They are not literal equals, so induction makes them rotations; undoing the common extension gives $R = R^*$, and the swap rotates the parent.

Lemma 3.1 descends the zero-free conclusion to full boards at their common row count. Finally, the negative partitions in this stratum are

$$(4,2),\ (2,2,1,1),\ (4,1,1),\ (3,1,1,1),\ (3,3),\ (2,2,2).$$

Each differs from its conjugate, so Lemma 2.3 gives $F \neq F^*$. No negative partition was imposed on the mate. □

9. Dense endpoint differences

The final placement comparison needs more than a nonzero leading coefficient. It needs the endpoint difference of a dense four-row component not to consist solely of that term.

Lemma 9.1. *If $T \neq T^*$ is dense with three rows, then*

$$s_{T^{\uparrow t}}(x,y,z) - s_{T^{\downarrow t}}(x,y,z) \neq 0 \qquad (t>0).$$

Proof. Write its normalized lists as $A_1 > A_2 > A_3 > B_1 > B_2 > 0$, with last column beta zero, and put

$$\lambda = (A_1 - A_3 - 2, A_2 - A_3 - 1, 0), \qquad \mu = (B_1 - 2, B_1 - B_2 - 1, 0).$$

The three-variable alternant formula gives

$$s_T(x,y,z) = (xyz)^{A_3 - B_1 + 2} s_\lambda(x,y,z) s_\mu(x,y,z). \tag{9.1}$$

Top extension adds t to the first part of λ, and bottom extension adds t to the first part of μ.

For fixed $t > 0$, the ratio $s_{(p+t,q,0)}/s_{(p,q,0)}$ determines $(p,q,0)$. To see this, write $(p,q,0) = (a+b,b,0)$ and $H_j = h_j(x,y)$. At $z=0$, the ratio is H_{a+t}/H_a. Setting $x=1$ gives

$$\frac{1-y^{a+t+1}}{1-y^{a+1}},$$

whose first nonconstant term has degree $a+1$, determining a. The branching rule gives

$$(xy)^{-b} s_{(a+b,b)}(x,y,z) = \sum_{i=0}^{a} \sum_{j=0}^{b} (xy)^{-j} H_{a-i+j} z^{i+j}. \tag{9.2}$$

For $b < b'$, the first difference in the cross-multiplied ratios occurs at z^{b+1}: the extra summand at that degree has $i=0, j=b+1$. After multiplying by $(xy)^{b+1}$, its coefficient is

$$H_{a+t}H_{a+b+1} - H_a H_{a+t+b+1} = (xy)^{a+1} H_{t-1} H_b \neq 0.$$

Thus b is determined as well. Equality of the two extended functions in (9.1) forces $\lambda = \mu$, or $A_1 = A_3 + B_1 = A_2 + B_2$, exactly the beta reflection equations for $T = T^*$. □

Lemma 9.2. *Let $Q \neq Q^*$ be dense with normalized four-row lists*

$$C_1 = S > C_2 > C_3 > C_4 > D_1 > D_2 > D_3 > D_4 = 0. \tag{9.3}$$

Put $\kappa = \max(C_2, S - D_3)$. There is a nonzero polynomial c_Q, independent of $t > 0$, such that

$$\max \operatorname{index} D_Q(t) = \kappa + t, \qquad [h_{\kappa+t}] D_Q(t) = c_Q, \tag{9.4}$$

and

$$D_Q(t) \neq h_{\kappa+t} c_Q. \tag{9.5}$$

Proof. In (3.8), all cofactor indices are at most κ. For unsigned minors M_{ij} of Q,

$$c_Q = \begin{cases} -M_{24}, & C_2 > S - D_3, \\ M_{13}, & C_2 < S - D_3, \\ M_{13} - M_{24}, & C_2 = S - D_3. \end{cases}$$

In the strict cases this is a nonzero dense three-row function, and remains nonzero in three variables. At the tie put $\alpha = S - C_2 = D_3$, and let T be the normalized maximal cofactor of Q. Then $c_Q = -D_T(\alpha)$. The child T is dense with three rows. Since Q is its border with equal margins, nonsymmetry of Q implies nonsymmetry of T. Lemma 9.1 shows that $c_Q(x,y,z) \neq 0$.

Every dense four-row determinant vanishes in three variables. Indeed, over $\mathbb{Q}(x,y,z)$,

$$h_j(x,y,z) = \sum_{a \in \{x,y,z\}} \frac{a^{j+2}}{\prod_{b \in \{x,y,z\} \setminus \{a\}} (a-b)} \qquad (j \geq 0).$$

Thus a matrix $(h_{C_i-D_j}(x,y,z))$ with positive indices has rank at most three, by separating the factors a^{C_i+2} and a^{-D_j} in each summand. This applies to both endpoint extensions of Q, whereas

$$h_{\kappa+t}(x,y,z)c_Q(x,y,z) \neq 0.$$

Their difference therefore cannot be just its leading term. This proves (9.5) as well as the nonvanishing in (9.4). □

Lemma 9.3. *For a dense four-row Q, write $f = s_Q$, $f_x = s_{Q^{\uparrow x}}$, and $f^{\,y} = s_{Q^{\downarrow y}}$. If $x > \ell > 0$, $u > 0$, and $v = x + u$, then*

$$h_{\ell+u}f_x - h_{\ell+v}f - h_\ell D_Q(v) \neq 0, \tag{9.6}$$

$$h_{\ell+u}D_Q(x) - h_\ell D_Q(v) \neq \pm h_{\ell+v}f. \tag{9.7}$$

Proof. For symmetric Q, the first assertion is factorization of two different connected endpoint functions against coordinate factors, and the second is immediate. Suppose $Q \neq Q^*$. Write

$$B_Q = \det(h_{C_i-D_j})_{2\le i\le 4,\ 1\le j\le 3}, \qquad [h_S]f = -B_Q.$$

For (9.6), the possible largest indices are $S+x, \ell+v, \kappa+v$. If $\ell < \kappa$, a tie of the first and third has coefficient $-h_{\ell+u}B_Q - h_\ell c_Q$, nonzero modulo h_ℓ. If $\ell > \kappa$, a tie of the first and second has coefficient $-h_S B_Q - f$, nonzero modulo h_S. If $\ell = \kappa$, the remaining coefficients are $-f - h_\ell c_Q$ or

$$-h_S B_Q - f - h_\ell c_Q.$$

The first is nonzero modulo h_ℓ; the second contains the diagonal monomial of f, all of whose indices are strictly below κ, so that monomial cannot cancel. A strict maximum has a nonzero coefficient.

The left side of (9.7) has largest index $\kappa + v$, with coefficient $-h_\ell c_Q$. If $\ell < \kappa$, a matching largest index on the right is possible only at $S = \kappa + v$, where its coefficient is not divisible by h_ℓ. If $\ell > \kappa$, the right side has a larger index. If $\ell = \kappa$, the right side has either a larger index, a coefficient f not divisible by h_ℓ, or a nonzero square of its top generator. None can agree with the left side. Only (9.4), not the lower remainder, has been used. □

10. A dense four-row component and two singletons

Fix Q as in (9.3), and retain $f, f_t, f^{\,t}, D_Q$ from Section 9. Write $\Phi(x,y) = \Phi_Q(x,y)$.

Lemma 10.1. *Let a connected zero-free diagram have maximal-cofactor function $h_\ell h_m f$, where $\ell, m > 0$ and $f = s_Q$ is dense of order four. Every connected Schur-equal diagram, after a possible rotation, has one of the following two forms, allowing Q or Q^*. Put*

$$L = \ell + g_1, \quad R = m + g_2, \quad \ell, m, g_1, g_2, u, v > 0, \quad N = S + L + R + u + v.$$

The end placement E, with component order $(\ell), (m), Q$, has lists

$$\begin{aligned} X_E &= (N, N-u, S+R+v, v+C_1, v+C_2, v+C_3, v+C_4), \\ Y_E &= (N-u-\ell, S+R+v-m, v+D_1, v+D_2, v+D_3, v, 0). \end{aligned} \tag{10.1}$$

The middle placement M, with order $(\ell), Q, (m)$, has lists

$$\begin{aligned} X_M &= (N, N-u, v+R+C_1, v+R+C_2, v+R+C_3, v+R+C_4, v+m), \\ Y_M &= (N-u-\ell, v+R+D_1, v+R+D_2, v+R+D_3, v+R, v, 0). \end{aligned} \tag{10.2}$$

Both are connected and zero-free. All possible component orders are represented, including both singleton orders.

Proof. The row-count reading gives seven rows. Factorization of $[h_N]s_D = h_\ell h_m f$ recovers exactly two one-row factors and one connected four-row factor. Nonnegative-index rigidity in [7] identifies the latter as Q or Q^*, against arbitrary mates. Up to rotation, it lies at an end or between the singletons. Reading the exterior row and column gaps gives $u, v > 0$; the two component separations give $g_1, g_2 > 0$. These are exactly (10.1) and (10.2). Conversely their subdiagonal entries are the positive rows of the indicated components, and their separating

entries are negative. The strict lists and positive diagonal give the asserted diagrams. No hypothesis about a mate's negative partition has been made. □

For these forms, set

$$\begin{aligned} A_L &= h_{\ell+u}h_{m+L} - h_\ell h_{m+L+u}, \\ U_Q(x) &= h_{\ell+u}f_x - h_\ell f_{x+u}, \\ Z_Q(y) &= h_{m+v}f^{\,y} - h_m f^{\,y+v}. \end{aligned} \tag{10.3}$$

First-column and last-row expansion give the full parent polynomials

$$P_E = A_L\Phi(R,v) - h_m h_{\ell+u}\Phi(L+R,v) + h_\ell h_m\Phi(L+R+u,v), \tag{10.4}$$

$$\begin{aligned} P_M = {} & h_{\ell+u}h_{m+v}\Phi(L,R) - h_\ell h_{m+v}\Phi(L+u,R) \\ & - h_{\ell+u}h_m\Phi(L,R+v) + h_\ell h_m\Phi(L+u,R+v). \end{aligned} \tag{10.5}$$

Here A_L is a dense two-row Schur function. For $x > \ell$, the function $U_Q(x)$ is the connected five-row determinant with lists

$$X = (S+x+u, S+x, C_2, C_3, C_4), \qquad Y = (S+x-\ell, D_1, D_2, D_3, 0). \tag{10.6}$$

Its negative partition is $(1,1,1)$. The corresponding $Z_Q(y)$ has partition (3), and is the rotation of the U-diagram built on Q^* with singleton data m, v. Theorem 4.3 applies to these intermediates. Two U-diagrams with equal functions are literally equal, since rotation changes the negative partition; equality of a U- and a Z-diagram is rotation. Their lists then recover their parameters.

10.1. **End and middle placements do not mix.**

Lemma 10.2. *No end placement in* (10.1) *is Schur-equal to a middle placement in* (10.2).

Proof. Let $q = \min(u,v)$. The coefficient at the next index $N-q$, after corner removal, is

	E	M
$u < v$	$-h_m h_{\ell+u} f$	$-h_m h_{\ell+u} f$
$v < u$	$-h_\ell h_m f^{\,v}$	$-h_\ell h_{m+v} f$
$u = v$	$-h_m(h_{\ell+u}f + h_\ell f^{\,u})$	$-f(h_m h_{\ell+u} + h_\ell h_{m+u})$.

(10.7)

Every coefficient index is below $N-q$. Divisibility by f separates the last two end regimes from all middle regimes. Divisibility by $h_\ell h_m$ distinguishes those two end regimes from one another. After division by f, setting every positive-index generator equal to one distinguishes a strict middle marker from its tied marker.

Thus a cross-placement equality requires $u < v$ on both sides, after rotating the middle parent. The original singleton multiset $\{\ell, m\}$, together with $\{\ell+u, m\}$, determines which singleton was extended: the change in the sum of squares is $2\ell u + u^2$. Therefore u, ℓ, m, N, S agree.

Subtract

$$J = f(h_\ell h_m h_N - h_m h_{\ell+u} h_{N-u}).$$

At $K = \max(N-v, N-u-L)$, the next coefficients are

	$N-v > N-u-L$	$N-v < N-u-L$
E	$E_1 = -h_\ell h_m f^{\,v}$	$E_2 = A_L f$
M	$M_1 = -h_\ell h_{m+v} f$	$M_2 = h_m U_Q(L)$.

(10.8)

At equality of the indices use $E_0 = E_1 + E_2$, $M_0 = M_1 + M_2$. These coefficients are nonzero. The factors A_L and $U_Q(L)$ are irreducible of ordinary degrees two and five. In particular E_0 is divisible by neither f nor h_m. The nine comparisons reduce as follows:

	M_1	M_2	M_0
E_1	f-divisibility	irreducibility of U	first tie below
E_2	A_L is not a monomial	f-divisibility	f-divisibility
E_0	f-divisibility	h_m-divisibility	second tie below.

(10.9)

For $E_1 = M_0$, reduction modulo h_ℓ forces $m = \ell$. The common K fixes v, and the middle tie gives $L = v - u$. Opposite core orientations would require $h_{\ell+u} f_L = h_{\ell+v} f$, contrary to factorization. Equal orientations would contradict (9.6).

For $E_0 = M_0$, the tied indices give common v, L, with $v = L + u$. The remaining equation is

$$h_{\ell+u}(h_{m+L} f - h_m f_L) + h_\ell h_m (f_v - f_E^v) = 0,$$

where the subscript E specifies the end core orientation. Reduction modulo h_m forces $m = \ell+u$. Opposite orientations again give an impossible product equality; equal orientations give (9.6), up to sign. Both ties are excluded. □

10.2. **End placements.**

Lemma 10.3. *Two end placements have equal Schur functions only when their aligned diagrams are identical.*

Proof. The first marker in (10.7) distinguishes its three margin regimes.

If $u = v$, the marker recovers the singleton order: for unequal lengths, exactly h_m among the two singleton variables divides it. Subtraction recovers f^u and fixes the core orientation. The span fixes $L + R$. The last two terms of (10.4) are then common, leaving $A_L\Phi(R, v)$. Its connected factors of degrees two and five recover L, R.

Suppose $v < u$. The first marker fixes v and the core orientation, but leaves the singleton order. Put $T = L+R+u = N-S-v$, and subtract $h_\ell h_m \Phi(T, v)$. The negative of the coefficient at $K = \max(N - u, m + L + u)$ is

$$\begin{cases} h_m h_{\ell+u} f, & N - u > m + L + u, \\ h_\ell \Phi(R, v), & N - u < m + L + u, \\ h_m h_{\ell+u} f + h_\ell \Phi(R, v), & N - u = m + L + u. \end{cases} \tag{10.10}$$

The first type is divisible by f, the second has singleton content h_ℓ, and the tied type has neither singleton divisor when $\ell \neq m$. If $\ell = m$, a proposed equality of the second and tied types fixes $g_2 = T - K$, hence the same $\Phi(R, v)$, and leaves a nonzero extra term.

Within the first type, $u = N - K$ and the shifted singleton pair fix the order; the degree-two/degree-five factorization then fixes L, R. Within the second, the marker fixes ℓ, m, R and $L + u$. Canceling the common term involving $h_\ell h_{m+L+u}$ leaves equality of

$$h_{\ell+u}\big(h_{m+L}\Phi(R, v) - h_m \Phi(R + L, v)\big).$$

The bracket is connected of order six, so factorization fixes u, L. At the tie $u = N - K$, $g_2 = T - K$. If the singleton order changed, with $a_0 = S - D_1$, the row multiset would give

$$\{\ell + u, a_0 + m + g_2\} = \{m + u, a_0 + \ell + g_2\}.$$

For $\ell \neq m$, this forces $u = a_0 + g_2$, whereas the marker tie gives $u = S + v + g_2 > a_0 + g_2$. The order is fixed.

Suppose now $u < v$. Use the end row of (10.8). The types E_1, E_2, E_0 are respectively divisible by only h_ℓ, only f, or neither, among those two polynomials. For E_1, the marker fixes v and the core orientation, after which $A_L\Phi(R, v)$ fixes L, R. For E_0, the tie fixes $v = N - K$, $L = v - u$, and then R; its coefficient fixes orientation.

For E_2, the marker fixes $L = N - u - K$, while $R + v$ is known. If $v_- < v_+$, subtract the common first three last-column terms. At $K' = \max(N - v_-, S + v_+)$, the difference has coefficient

$$\mathbf{1}_{K'=N-v_-} h_\ell h_m f_-^{v_-} - \mathbf{1}_{K'=S+v_+} s_T. \tag{10.11}$$

Deleting the first core row and last column from the larger-margin parent gives T. Its subdiagonal indices are $\ell, m, C_2 - D_1, C_3 - D_2, C_4 - D_3 > 0$, so it is connected of order six. Its span $N - v_+$ and all product-coefficient indices are below K'. An irreducible degree-six function cannot equal a product with two singleton factors. Thus v, R are fixed.

It remains to compare Q and Q^* at fixed margins in this last case. Write $\mathcal{E}(w,v) = \Phi_Q(w,v) - \Phi_Q(v,w)$. Its leading coefficient is $-D_Q(w)$ at $S+v$ for $w<v$, or $D_Q(v)$ at $S+w$ for $w>v$. The end-parent difference is

$$A_L\mathcal{E}(R,v) - h_m h_{\ell+u}\mathcal{E}(L+R,v) + h_\ell h_m \mathcal{E}(L+R+u,v). \tag{10.12}$$

Put $w_0 = L+R+u$. If $v<w_0$, the last term has a private largest index. If $v>w_0$, the coefficient at $S+v$ has the private index $\kappa+w_0$. At $v=w_0$, that term vanishes and the top coefficient becomes

$$T_E = h_m h_{\ell+u} D_Q(L+R) - A_L D_Q(R).$$

Compare $t_0 = \kappa+L+R$ and $Z = m+L+u$. If $t_0>Z$, the first term has a private index. If $t_0<Z$, its h_Z-coefficient is $h_\ell D_Q(R)\neq 0$. At equality, $u=\kappa+R-m$ and

$$[h_Z]T_E = h_m h_{\ell+u} c_Q + h_\ell D_Q(R).$$

Its possible largest indices are $\ell+u$ and $m+u$. Unequal values give a private contribution; equal values give coefficient $2h_\ell c_Q \neq 0$. Thus a nonsymmetric core cannot reverse at fixed margins. □

10.3. **Middle placements.**

Lemma 10.4. *Two middle placements have equal Schur functions only when they are identical or whole-parent rotations.*

Proof. First fix $\ell, m, u, v, L+R$, and compare $L_- < L_+$, $R_- > R_+$, allowing either core orientation. At $K = \max(S+L_++u, S+R_-+v)$, expansion of (10.5) gives

$$[h_K](P_+ - P_-) = \mathbf{1}_{K=S+L_++u} h_\ell Z_{Q_+}(R_+) - \mathbf{1}_{K=S+R_-+v} h_m U_{Q_-}(L_-). \tag{10.13}$$

All coefficient indices are below K. A strict maximum is nonzero. At a tie, factorization forces $\ell=m$ and equality of the two connected five-row functions. Their single-row/single-column rigidity forces rotation, and (10.6) gives

$$u=v, \qquad R_+ = L_-, \qquad Q_+ = Q_-^*.$$

Since $L+R$ was fixed, this is whole-parent rotation.

When $u=v$, the first marker fixes that margin and $L+R$. Align the singleton order by rotating the parent; (10.13) then fixes L, R up to whole-parent rotation. Core orientation at fixed parameters is treated below.

For $u<v$, the middle row of (10.8) applies. The type M_1 is divisible by f; the type M_2 is divisible by h_m; M_0 is not divisible by h_m unless $\ell=m$. A proposed equality between a tied type and M_2, with $\ell=m$, fixes the same L, and the tie gives $v=L+u$. Cancellation leaves either an immediate nonzero term or

$$h_{\ell+u} D_Q(L) - h_\ell D_Q(L+u) = \pm h_{\ell+v} f,$$

excluded by (9.7). Thus the types agree.

Within M_1, the value $v=N-K$ is fixed, and (10.13) fixes L, R. Within M_2, the value $L = N-u-K$ and $U_Q(L)$ are recovered. The intermediate rigidity fixes the core orientation. Since $R+v$ is known, write

$$P_M = h_{m+v}\mathcal{A}_Q(L,R) - h_m \mathcal{A}_Q(L,R+v), \quad \mathcal{A}_Q(L,y) = h_{\ell+u}\Phi_Q(L,y) - h_\ell \Phi_Q(L+u,y).$$

The second term is common and $\mathcal{A}_Q(L,R)$ is connected of order six. Factorization of the first term fixes v, R. Within M_0, the tie fixes $v=N-K$, $L=v-u$, and R. Subtracting its known product term recovers $U_Q(L)$, hence orientation.

Finally compare the two core orientations at fixed parameters, assuming $Q\neq Q^*$. At $K = \max(S+L+u, S+R+v)$, the coefficient of the difference is

$$\begin{aligned}[h_K](P_Q - P_{Q^*}) = {} & \mathbf{1}_{K=S+L+u} h_\ell \big(Z_Q(R) - Z_{Q^*}(R)\big) \\ & + \mathbf{1}_{K=S+R+v} h_m \big(U_Q(L) - U_{Q^*}(L)\big).\end{aligned} \tag{10.14}$$

A strict maximum is nonzero by intermediate rigidity. At $L+u = R+v$, the coefficient simplifies to

$$T_M = h_m h_{\ell+u} D_Q(L) - h_\ell h_{m+v} D_Q(R). \tag{10.15}$$

If $L = R$, then $u = v$, and

$$T_M = (h_m h_{\ell+u} - h_\ell h_{m+u}) D_Q(L).$$

It vanishes only for $\ell = m$, when reversing the core is whole-parent rotation.

Assume $L > R$, so $v > u$, and put $p = \kappa + L$, $a = \ell + u$, $b = m + v$. Every strict maximum among p, a, b in (10.15) is private. If $a = b$, remove the common generator and use the private top of $D_Q(L)$. If $a = p > b$, the first product has a nonzero square coefficient; this also handles the triple tie. The only remaining comparison is $b = p > a$, for which

$$[h_p]T_M = h_m h_{\ell+u} c_Q - h_\ell D_Q(R).$$

The two tie equations imply $u = \kappa + R - m$. The possible largest indices in this coefficient are $\ell + u$ and $\kappa + R = m + u$. If $\ell \neq m$, one is private. If $\ell = m$, vanishing would require

$$D_Q(R) = h_{\kappa+R} c_Q,$$

contrary to Lemma 9.2. This excludes the final nonrotation possibility. □

Theorem 10.5. *A connected zero-free diagram whose maximal cofactor consists of a dense four-row component and two singletons is rotation-rigid against arbitrary connected mates.*

Proof. Lemma 10.1 exposes every mate in an end or middle form. Lemma 10.2 compares the two placements, and Lemmas 10.3 and 10.4 compare each placement internally, including both core orientations. The conclusion is therefore not restricted to mates sharing the source's negative partition. □

11. The sharp rigidity theorem

Proof of Theorem 1.2. An empty diagram contributes 1, whereas every nonempty connected component, including a single row or column, has a nonconstant irreducible Schur function [1]. Since s_D is nonconstant and irreducible, $s_H = s_D$ forces H to have exactly one nonempty component. Thus H is connected.

First work with zero-free diagrams at their minimum realizing row count. Put $m = n - 1$. By Lemma 3.4, the maximal cofactor is disconnected.

At $n = 7$, its component orders sum to six. Four or more components cost at least twelve negatives; among three-component orders only $1 + 1 + 4$ costs at most nine. The complete budget is

(11.1)

orders	cross cost	internal budget	rigidity input
$1 + 5$	5	≤ 4	five-row core: Lemmas 3.7 and 8.3
$2 + 4$	8	≤ 1	dense two-row and $\leq$ 1-negative four-row cores
$3 + 3$	9	0	two dense three-row cores
$1 + 1 + 4$	9	0	*Theorem* 10.5.

The second and third rows satisfy every hypothesis of Lemma 8.6: component and endpoint rigidity follow from Theorem 4.2 and lemma 3.5, and their endpoint-leading coefficients follow from Lemma 8.1. In particular the $3 + 3$ case uses only the endpoint-leading property, not Lemma 9.2. The first row uses arbitrary-mate rigidity of the five-row core and its endpoint extensions as well as its endpoint-leading property, not rigidity alone.

For $n \geq 8$, three components cost at least $2m - 3 \geq 11$. Two components both of size at least two cost at least $2(m - 2) \geq 10$. Thus only $1 + (n - 2)$ remains. Its cross cost is $n - 2$, and its internal budget is at most $11 - n$. For $n = 8, 9, 10, 11$, this gives respectively cores of orders $6, 7, 8, 9$ with at most $3, 2, 1, 0$ internal negatives. Lemmas 3.7, 8.1 and 8.3 apply. For $n \geq 12$, even the least cross cost exceeds nine. This covers every minimum-order case without a partition census.

Now induct on the actual row count above the minimum. By Lemma 3.4, the maximal cofactor is connected, retains $\rho(D)$, and has one fewer row. It and its endpoint extensions therefore satisfy the inductive rigidity assertion. Factorization forces the arbitrary mate's maximal cofactor to be connected and identifies it up to rotation. After alignment, Lemma 3.3 recovers the margins. A surviving swap equates opposite endpoint extensions. They cannot be translates; if they are rotations, undoing the extension makes the cofactor symmetric and the swap rotates the parent. This completes the zero-free induction.

For full boards, use the common row count and (3.1) on both source and mate. Their negative partitions, and hence the source's $n_{\min}$, are unchanged. The zero-free conclusion descends by injectivity and rotation covariance. Finally Proposition 12.1 supplies the sharpness assertion. □

Corollary 11.1. *A connected diagram with exactly ten negative indices and minimum realizing row count at least eight is rotation-rigid against arbitrary connected mates, with common betas allowed.*

Proof. At minimum row count eight, a disconnected seven-row cofactor has either orders $1+6$, with at most four internal negatives, or $2+5$, both dense. Three components cost at least eleven negatives. Apply Lemmas 3.7, 8.3 and 8.6. At minimum row counts nine through twelve only $1+(n-2)$ remains, with at most $12-n$ internal negatives, so the same singleton argument applies. Above twelve even the least cross cost exceeds ten. Connected bordering and homogenization finish exactly as in the proof of Theorem 1.2. □

Completion of Theorem 1.1(v). The partitions of six, up to conjugation, are

$$(6),\quad (5,1),\quad (4,2),\quad (4,1,1),\quad (3,3),\quad (3,2,1).$$

Their minimum realizing row counts are respectively $8,7,6,6,6,5$, by Theorem 2.5. The first two pairs are rigid by Theorem 1.2; the next three by Theorem 8.7. The staircase is classified by Theorem 6.2 and proposition 7.13. As in the proof of parts (i)–(iii), a Schur-equal skew diagram is connected by irreducibility. Thus all comparisons include arbitrary skew mates. □

12. Sharpness and failures beyond the hypotheses

12.1. **Seven-row ten-negative families.**

Proposition 12.1. *Let a, b, d be integers with $a \geq 1$, $d \geq 0$, and $b > a + d$. The profiles* (12.2) *and* (12.3) *define two families of four distinct connected Schur-equal diagrams F_i, F_i^*, H_i, H_i^*, $i = A, B$, of respective areas $3a + 11b + 7d$ and $4a + 11b + 7d$. In either family their negative partitions are, respectively,*

$$(3,3,2,1,1),\quad (5,3,2),\quad (4,2,2,1,1),\quad (5,3,1,1). \tag{12.1}$$

All have ten negatives and minimum realizing row count seven. Both families have proper legal marked-composition presentations.

Proof. Define family A by

$$\begin{aligned} F_A = (&a+d, b+d, a+b+d, 2b+d, 2b+d, a+2b+d, 3b+d;\\ &d, a+d, b+d, b+d, a+b+d, 2b+d),\\ H_A = (&a+d, b+d, a+b+d, 2b+d, a+2b+d, 3b+d, 2b+d;\\ &d, a+d, b+d, a+b+d, 2b+d, b+d), \end{aligned} \tag{12.2}$$

and family B by

$$\begin{aligned} F_B = (&a+d, b+d, a+b+d, 2b+d, a+2b+d, a+2b+d, 3b+d;\\ &d, a+d, b+d, a+b+d, a+b+d, 2b+d),\\ H_B = (&a+d, b+d, a+b+d, 2b+d, a+2b+d, 3b+d, a+2b+d;\\ &d, a+d, b+d, a+b+d, 2b+d, a+b+d). \end{aligned} \tag{12.3}$$

The profile inequalities (2.1) follow from the parameter bounds. Both have common column list $Y = (6b, 5b, 4b, 3b, 2b, b, 0)$. Their row lists are

$$\begin{aligned} X_{F_A} &= (a+6b, 6b, a+5b, 5b, 4b, a+3b, 3b) + d, \\ X_{H_A} &= (a+6b, 6b, a+5b, 5b, a+4b, 4b, 2b) + d, \end{aligned} \tag{12.4}$$

and

$$\begin{aligned} X_{F_B} &= (a+6b, 6b, a+5b, 5b, a+4b, a+3b, 3b) + d, \\ X_{H_B} &= (a+6b, 6b, a+5b, 5b, a+4b, 4b, a+2b) + d. \end{aligned} \tag{12.5}$$

Here adding d means adding it to every entry. Since $0 \le d < a+d < b$, counting columns strictly above each row beta gives (12.1); rotation gives the conjugate partitions. Theorem 2.5 gives minimum seven, and summing the rows gives the areas. In each family the last-row lengths differ and both exceed the common first-row length, while the displayed profiles are not identical. Thus the four members are distinct. They are zero-free exactly when $d > 0$.

For the marked compositions of McNamara and van Willigenburg [5], take

$$\begin{aligned} D_A &= (3,4,4;2,3), \qquad D_B = (2,3,3,3;1,2,2), \\ E &= (a+d, b+d; d), \qquad W = (a+d), \end{aligned} \tag{12.6}$$

in attachment case (c). Put $L = a+d$, $M = b+d$, $\beta = b-1$, and use integer cell coordinates

$$\begin{aligned} E &= \{(0,j) : \beta \le j < \beta + L\} \cup \{(1,j) : 0 \le j < M\}, \\ W_{\mathrm{sw}} &= \{(1,j) : 0 \le j < L\}, \qquad W_{\mathrm{ne}} = \{(0,j) : \beta \le j < \beta + L\}. \end{aligned}$$

The inequality $\beta \ge L$ makes each occurrence maximal on its content diagonals. Their clearance is $b - a - d + 1 \ge 2$. Deleting the northeast occurrence leaves a row; deleting the southwest occurrence leaves two intervals overlapping in $d+1$ cells. The middle part is the single row $\{(1,j) : L \le j < M\}$, whose hatted part is empty. These verify Hypotheses I–IV, respectively. The southwest attachment is horizontal and the northeast attachment vertical. Both outer diagrams are connected and proper.

Put $\tau = (-1, b-1)$, $\eta = (1,1)$, and

$$\mathcal{U} = \bigcup_{j=0}^{5} (E + j\tau) \ \cup \ \bigcup_{j=1}^{4} (E + \eta + j\tau).$$

For family A the northwest-ribbon rule gives

$$\begin{aligned} D_A \circ_W E &= \mathcal{U} \cup (E + 2\eta + 3\tau) = H_A, \\ D_A^* \circ_W E &= \mathcal{U} \cup (E + 2\eta + 2\tau) = F_A. \end{aligned} \tag{12.7}$$

Its supplemental interfaces are already covered by these copies. The normalized left endpoints of the seven rows are $(6-r)(b-1)$, $r = 0, \dots, 6$; the last copy extends rows $r = 4, 5$, or $r = 5, 6$, to lengths $a + 2b + d$ and $3b + d$. For family B the supplemental interface gives

$$\begin{aligned} D_B \circ_W E &= \mathcal{U} \cup (E + 2\eta + 2\tau) \cup (W_{\mathrm{ne}} + 2\eta + 3\tau) = F_B, \\ D_B^* \circ_W E &= \mathcal{U} \cup (E + 2\eta + 3\tau) \cup (W_{\mathrm{ne}} + 2\eta + \tau) = H_B. \end{aligned} \tag{12.8}$$

Reading the intervals gives the displayed profiles. Outer reversal and whole-parent rotation give all four constructed members.

Their equalities also have a direct determinant proof. Localize at h_d, and put

$$U(t) = \sum_{j=0}^{6} h_{jb+d} t^j, \qquad V(t) = \sum_{j=0}^{6} h_{a+jb+d} t^j, \qquad \frac{V(t)}{U(t)} \equiv \sum_{j=0}^{6} c_j t^j \pmod{t^7}.$$

Toeplitz elimination expresses each original determinant as h_d^7 times a minor of $T(c)_{ij} = c_{i-j}$, where $c_k = 0$ for $k < 0$. For family A the two index pairs are

$$(\{3,5,6\}, \{0,1,2\}), \qquad (\{4,5,6\}, \{0,1,3\}). \tag{12.9}$$

For family B the equality is

$$\det\begin{pmatrix} c_3 & c_2 & c_1 & 0 \\ c_4 & c_3 & c_2 & c_0 \\ c_5 & c_4 & c_3 & c_1 \\ c_6 & c_5 & c_4 & c_2 \end{pmatrix} = \det\begin{pmatrix} c_2 & c_1 & c_0 & 0 \\ c_4 & c_3 & c_2 & c_1 \\ c_5 & c_4 & c_3 & c_2 \\ c_6 & c_5 & c_4 & c_3 \end{pmatrix}, \tag{12.10}$$

with index pairs $(\{3,4,5,6\},\{0,1,2,4\})$ and $(\{2,4,5,6\},\{0,1,2,3\})$. In each case reflection $i \mapsto 6-i$, followed by transposition, identifies the two minors. This is the classical Toeplitz-minor reversal of [3]. The determinant ring embeds in the localization, so the equalities hold in the original ring. At $d=0$, the localized element is already $h_0 = 1$. □

12.2. **Six-row quartets.**

Proposition 12.2. *For every $1 \le q \le b < g$, let*

$$X = (b+5g, b+4g, b+3g, b+2g, b+g, b) \tag{12.11}$$

and take the following column lists:

ν	$Y(F_\nu)$	$Y(H_\nu)$
7	$(3g+q, 2g, g+q, g, q, 0)$	$(3g, 2g+q, g+q, g, q, 0)$
8	$(4g+q, 2g+q, g+q, g, q, 0)$	$(3g+q, 2g+q, 2g, g+q, q, 0)$
9	$(4g+q, 3g+q, g+q, g, q, 0)$	$(3g+q, 3g, 2g+q, g+q, q, 0)$.

(12.12)

For each row of the table, $F_\nu, F_\nu^*, H_\nu, H_\nu^*$ are four distinct connected Schur-equal diagrams. Their minimum realizing row count is six. Their source partitions are

ν	$\rho(F_\nu)$	$\rho(H_\nu)$
7	$(4,2,1)$	$(4,2,1)$
8	$(4,2,1,1)$	$(4,3,1)$
9	$(4,2,2,1)$	$(4,3,2)$.

(12.13)

No upper bound on these connected classes is asserted.

Proof. Put $s = b-q$, $A = s+g$, $B = s+2g$. The profiles are

F_7	$(B, B+q, B, A+q, A, b; A, A+q, A, b, s)$
H_7	$(B+q, B, B, A+q, A, b; A+q, A, A, b, s)$
F_8	$(A, B, B, A+q, A, b; s, A, A, b, s)$
H_8	$(B, B, A+q, A, A, b; A, A, b, s, s)$
F_9	$(A, A, B, A+q, A, b; s, s, A, b, s)$
H_9	$(B, A+q, A, A, A, b; A, b, s, s, s)$.

(12.14)

They satisfy (2.1), including $s=0$. Comparing (12.11) and (12.12) gives (12.13); Theorem 2.5 gives minimum six. The first row distinguishes each F_ν from H_ν. Every displayed last row has length b, and every displayed first row is longer, so neither is a rotation of either displayed source. This gives fourfold distinctness independently of the determinant equality.

Reverse the six matrix rows and index them by $k = 0, \dots, 5$. Define

$$U_j = (h_{b+(k-j)g})_{k=0}^{5}, \qquad V_j = (h_{b+(k-j)g-q})_{k=0}^{5}.$$

For the down-shift L on six-vectors, $LU_j = U_{j+1}$ and $LV_j = V_{j+1}$, because $b < g$. Use the shift identity from the proof of (5.1):

$$\sum_{i=1}^{6} \det(z_1, \dots, Lz_i, \dots, z_6) = 0. \tag{12.15}$$

For the seed word $(V_2, U_2, V_1, U_1, V_0, U_0)$, four summands have repeated columns, and the other two give

$$\det(V_3, U_2, V_1, U_1, V_0, U_0) = \det(U_3, V_2, V_1, U_1, V_0, U_0).$$

For the seed word $(V_3, V_2, V_1, U_1, V_0, U_0)$, the same four vanishing positions give

$$\det(V_4, V_2, V_1, U_1, V_0, U_0) = \det(V_3, V_2, U_2, V_1, V_0, U_0).$$

These are the seven- and eight-negative equalities.

The nine-negative equality uses a different elimination. Set $u_k = h_{b+kg}$, $v_k = h_{b+kg-q}$, and let $T_v = (v_{i-j})_{0\le i,j\le 5}$, with $v_j = 0$ for $j < 0$. In the localization at $v_0 = h_{b-q}$, define

$$\sum_{k=0}^{5} c_k z^k \equiv \frac{\sum_{k=0}^{5} u_k z^k}{\sum_{k=0}^{5} v_k z^k} \pmod{z^6}, \qquad C_j = \sum_{k=j}^{5} c_{k-j} e_k.$$

Then $T_v^{-1} V_j = e_j$, $T_v^{-1} U_j = C_j$. The two transformed determinants are

$$\det(e_4, e_3, e_1, C_1, e_0, C_0) = c_2 c_4 - c_1 c_5 = \det(e_3, C_3, e_2, e_1, e_0, C_0). \tag{12.16}$$

Restoring T_v proves equality. The original polynomials embed in the localization, so the identity holds before localization as well. In particular $b = q$ is included, with $v_0 = h_0 = 1$. The reduced two-by-two Toeplitz-minor interpretation belongs to the correspondence of [3]; it is not one further application of (12.15). □

12.3. **A non-dense placement resonance.**

Proposition 12.3. *Assume $1 \le q < b < g$, set $s = b - q$, and let*

$$Q = (b, s+g, s+g; s, s).$$

This connected three-row component is one-negative and rotation-rigid, but satisfies

$$h_b s_{Q^{\uparrow(g-q)}} - h_{s+g} s_Q = h_s \big(s_{Q^{\uparrow g}} - s_{Q^{\downarrow g}} \big). \tag{12.17}$$

At $t = g$, its endpoint difference has a nonzero lower remainder below its moving leading term. Its maximal cofactor is h_s^2, a product of two singletons. This resonance permits the distinct middle and end placements in the eight-negative quartet of Proposition 12.2.

Proof. The matrix of Q is

$$\begin{pmatrix} h_b & h_{b+g} & h_{b+2g} \\ h_s & h_{s+g} & h_{s+2g} \\ 0 & h_s & h_{s+g} \end{pmatrix}.$$

It has exactly one negative index, so Theorem 4.2 gives arbitrary-mate rigidity. Direct expansion gives

$$\begin{aligned} f = s_Q &= h_b h_{s+g}^2 - h_b h_s h_{s+2g} - h_{b+g} h_s h_{s+g} + h_{b+2g} h_s^2, \\ s_{Q^{\uparrow(g-q)}} &= h_{s+g}^3 - 2 h_s h_{s+g} h_{s+2g} + h_s^2 h_{s+3g}, \\ D_Q(g) &= h_{b+g} h_{s+g}^2 - h_{b+2g} h_s h_{s+g} - h_b h_{s+g} h_{s+2g} + h_b h_s h_{s+3g}. \end{aligned}$$

These prove (12.17). Here $\kappa = s + 2g$, and

$$[h_{\kappa+g}] D_Q(g) = h_b h_s \neq 0.$$

After removing this term, the largest remaining index is $b + 2g$, with coefficient $-h_s h_{s+g} \neq 0$. Thus even nondivisibility by the moving top generator survives the resonance. On the other hand,

$$[h_{b+2g}] f = h_s^2,$$

so ordinary degree two must not be substituted for irreducible nonlinearity of the maximal cofactor.

In (12.12), the maximal cofactors of F_8, H_8 have component orders (s, K, s) and (K, s, s), where

$$K = ((b+2g, b+g, b); (g+q, q, 0)), \qquad K^* = Q.$$

Both corner coefficients are $-h_s^2 s_K$. The remaining tied comparison is exactly (12.17), with $u = q$, $x = g - q > s$, and $v = g = x + u$. All these parameters are strictly positive. This is an interior end-to-middle equality with the same rigid nonlinear component, not an ambiguity in recovering its function. □

For seven negatives the four-row maximal-cofactor components themselves can be nonrotation mates: their profiles are

$$(A, A+q, A, b; s, b, s), \qquad (A+q, A, A, b; b, s, s),$$

with $s = b - q > 0$, $A = s + g$, and negative partition $(2, 1)$. They belong to the four-row exceptional family (4.1). For nine negatives the maximal cofactors instead have four components, with orders (s, s, R, s) and (R, s, s, s), where $R = (A, b; s)$. Thus the nine-negative construction is not another three-component resonance.

Remark 12.4. The staircase $(4, 3, 2, 1)$ has ten negatives but minimum realizing row count six. A classical ribbon example is

$$\begin{aligned} F &= (a, b, a, a+b, a+b, b; 0, 0, 0, 0, 0), \\ H &= (a, a+b, a+b, b, a, b; 0, 0, 0, 0, 0), \end{aligned} \qquad a, b > 0, \quad a \neq b.$$

For $\gamma = (a, b)$, these are $\gamma \cdot \gamma^{\odot 3}$ and $\gamma^{\odot 3} \cdot \gamma$. The ribbon product identity, in the factor-reversal theory of Billera, Thomas and van Willigenburg [2], gives

$$s_F = r_\gamma r_{\gamma^{\odot 3}} - r_{\gamma^{\odot 4}} = s_H.$$

Their first and last row lengths also distinguish the rotation orbits. This ribbon construction is different from the seven-row sharpness families. It supplies no generation theorem for all sources with the staircase pattern, in particular for nonribbon sources with that pattern.

13. Ten negatives at minimum row count seven

13.1. **Five-negative cofactor components.**

Lemma 13.1. *Let $Q = (C, D) \neq Q^*$ be connected and zero-free with five rows and five negative indices, normalized by $C_1 = S, D_5 = 0$. There are an integer $0 < k_Q < S$ and a nonzero polynomial c_Q, independent of $t > 0$, such that*

$$\max \operatorname{index} D_Q(t) = k_Q + t, \qquad [h_{k_Q+t}] D_Q(t) = c_Q. \tag{13.1}$$

Every index in c_Q is at most k_Q.

Proof. Put $\kappa = \max(C_2, S - D_4)$. In (3.8), all fixed cofactor indices are at most κ. An untied top has one nonzero minor as coefficient. At a tie put $q = S - C_2 = D_4$. The coefficient is, up to sign, $D_R(q)$, where R is the maximal cofactor of Q.

The four positive rows of R contain five negatives. By Lemmas 2.3 and 3.2, its components are two singletons and a dense two-row diagram. If the nonlinear component is at an end, factorization separates its extended function from its unextended function, so $D_R(q) \neq 0$. If it is in the middle,

$$D_R(q) = (h_{\ell+q} h_m - h_\ell h_{m+q}) s_T,$$

which vanishes exactly when $\ell = m$. In every nonvanishing case, (13.1) holds with $k_Q = \kappa$.

In the remaining case the profile has the form

$$Q = (b+q, x, y, z, b+q; b, c, d, b), \qquad x = b + c + u, \quad z = b + d + w, \quad y = c + d - \sigma,$$

where $b, q, c, d, u, w, \sigma > 0$ and $\sigma < \min(c, d)$. The two negative separating entries give $u, w > 0$; the dense two-row middle component gives the inequalities on σ. Put $X = x+y-c$, $Y = y+z-d$, and

$$\mathcal{A}_{r,s}(t) = -\det \begin{pmatrix} h_b & h_{b+q} & h_{b+q+t} \\ h_r & h_{r+q} & h_{r+q+t} \\ h_s & h_{s+q} & h_{s+q+t} \end{pmatrix}.$$

Expansion along the first row and last column gives

$$D_Q(t) = h_b \big(h_y \mathcal{A}_{x,z}(t) - h_c \mathcal{A}_{X,z}(t) - h_d \mathcal{A}_{x,Y}(t) + h_\sigma \mathcal{A}_{X,Y}(t) \big). \tag{13.2}$$

Let $K_j = h_{b+q}h_j - h_b h_{j+q}$. The coefficient at $h_{\max(X,Y)+q+t}$ is

$$h_b \begin{cases} h_c K_z - h_\sigma K_Y, & X > Y, \\ -h_d K_x + h_\sigma K_X, & Y > X, \\ h_c K_z - h_d K_x, & X = Y. \end{cases}$$

For $X > Y$, its h_{Y+q}-coefficient is $h_b^2 h_\sigma \neq 0$; the reversed case is analogous. At $X = Y$, nonsymmetry forces $x \neq z$: otherwise $c = d$ and $u = w$. The larger of h_{x+q}, h_{z+q} then has a nonzero coefficient. Every fixed index in (13.2) is at most $k_Q = \max(X, Y)+q$, and this positive offset is below S. The same coefficient works for every $t > 0$. □

The offset in Lemma 13.1 need not be the ordinary boundary value $\max(C_2, S - D_4)$.

Lemma 13.2. *Let Q be a connected zero-free five-row five-negative diagram whose complete connected class is $\{Q, Q^*\}$. Every connected zero-free parent with maximal cofactor consisting of a singleton and Q is rotation-rigid.*

Proof. Factorization recovers the singleton and Q up to rotation. The marker (3.10) and the separator (3.11) recover the margins as in the proof of Lemma 3.7. Once the core has been recovered, fixing its orientation when $v \leq u$ requires only $D_Q(t) = 0$ exactly when $Q = Q^*$, supplied by Lemma 5.5; arbitrary-mate rigidity of the endpoint extensions is unnecessary.

For $u < v$, after the margins are fixed, only core reversal remains. Put $a = s + g$ and $\mathcal{E}(w, v) = \Phi_Q(w, v) - \Phi_Q(v, w)$. The difference is

$$h_{s+u}\mathcal{E}(a, v) - h_s\mathcal{E}(a + u, v).$$

A zero summand leaves the other nonzero. For $v < a + u$ the second nonzero summand has a private largest index. For $v > a + u$, the coefficient at $S + v$, up to sign, is

$$h_{s+u}D_Q(a) - h_s D_Q(a + u).$$

Its second term has the private index $k_Q + a + u$, larger than both $k_Q + a$ and $s + u$, by Lemma 13.1 and $a > s$. Thus a nonsymmetric core cannot reverse. For a symmetric core the reversal is literal. □

13.2. **A one-negative four-row component.**

Lemma 13.3. *Let $Q \neq Q^*$ be connected and zero-free with four rows and one negative index. Normalize its lists as*

$$C_1 = S > C_2 > C_3 > D_1 > C_4 > D_2 > D_3 > D_4 = 0. \tag{13.3}$$

Put $\kappa = \max(C_2, S - D_3)$, and let $c_Q \neq 0$ be the fixed leading coefficient supplied by Lemma 8.1. Then

$$D_Q(t) \neq h_{\kappa+t}c_Q \qquad (t > 0). \tag{13.4}$$

Proof. Every diagonal index is strictly below κ. If the first and last row lengths differ, the minimum-square monomials of the two endpoint extensions have different square weights. Their difference is twice t times that row-length difference. The smaller minimum cannot occur in the other polynomial, and all its indices are below $\kappa + t$. It supplies a nonzero lower term.

Suppose the end rows agree, so $S = C_4 + D_1$, and put $r = D_1 - C_4 > 0$. If $r = 1$ or 2, specialize to three variables. The rank-three formula in the proof of Lemma 9.2 gives zero also at indices $-1, -2$. Thus the one missing entry already has its correct value in that matrix, and both four-row endpoint determinants vanish. At a strict boundary, c_Q is a nonzero three-row Schur function and stays nonzero. At a tied boundary it is, up to sign, $D_T(D_3)$, where T is the connected three-row maximal cofactor. This difference is nonzero by Lemma 3.5: otherwise the opposite endpoint extensions would be rotations and T, hence its equal-margin border Q, would be symmetric. A skew Schur function with at most three rows has Schur expansion supported on partitions of length at most three, and specialization to three variables is injective on their span. Consequently c_Q remains nonzero.

For $r \geq 3$, specialize to the alphabet $(1,1)$, so $h_j \mapsto j+1$ for $j \geq 0$. The virtual matrix $(C_i - D_j + 1)$ has rank at most two. Correcting its one negative-index entry from $1-r$ to zero raises rank by at most one, so both four-row endpoint determinants vanish. A strict leading coefficient is a three-row one-negative determinant; its value is, up to sign, $r-1$ times two positive row and column beta gaps, and is nonzero. At a tie, with $T = (C_2, C_3, C_4)/(D_1, D_2, D_3)$, the same correction gives

$$D_T(t)(1,1) = (r-1)t\big((D_2 - D_3) - (C_2 - C_3)\big). \tag{13.5}$$

If the inner gaps agreed, that relation, $S - C_2 = D_3$, and $S = C_4 + D_1$ would give all three reflection equations for $Q = Q^*$. Hence c_Q is nonzero on this specialization too. In either case a vanishing specialized endpoint difference cannot equal its nonzero specialized leading term, proving (13.4). □

Lemma 13.4. *For a connected zero-free one-negative four-row Q, both* (9.6) *and* (9.7) *hold with their stated hypotheses $x > \ell > 0$, $u > 0$, $v = x + u$.*

Proof. Write $f = s_Q$, $f_x = s_{Q^{\uparrow x}}$, and $f^{\,y} = s_{Q^{\downarrow y}}$. For symmetric Q, the first inequality follows from factorization of distinct connected endpoint functions against coordinate factors; the second is immediate. For nonsymmetric Q, normalize its lists as (13.3) and use κ, c_Q from Lemma 8.1. The cofactor

$$B_Q = \det(h_{C_i - D_j})_{2 \leq i \leq 4,\ 1 \leq j \leq 3}, \qquad [h_S]f = -B_Q,$$

is connected and one-negative: its subdiagonal indices are $C_3 - D_1 > 0$ and $C_4 - D_2 > 0$. It is therefore irreducible.

For (9.6), the possible largest indices are $S+x, \ell+v, \kappa+v$. If $\ell < \kappa$, a tie of the first and third has coefficient $-h_{\ell+u}B_Q - h_\ell c_Q$, nonzero modulo h_ℓ. If $\ell > \kappa$, a tie of the first and second has coefficient $-h_S B_Q - f$, nonzero modulo h_S. For $\ell = \kappa$, the remaining coefficients are $-f - h_\ell c_Q$ or $-h_S B_Q - f - h_\ell c_Q$. The first is nonzero modulo h_ℓ; the second retains the coefficient-one diagonal monomial of f, all of whose indices are below κ. Every strict maximum is nonzero.

The left side of (9.7) has largest index $\kappa + v$, with coefficient $-h_\ell c_Q$. For $\ell < \kappa$, the right side can have that largest index only at $S = \kappa + v$, where its coefficient is not divisible by h_ℓ. For $\ell > \kappa$, it has a larger index. For $\ell = \kappa$, it has either a larger index, a coefficient f not divisible by h_ℓ, or a nonzero square of its top generator. None agrees with the left side. □

Proposition 13.5. *A connected zero-free diagram whose maximal cofactor consists of a one-negative four-row component and two singletons is rotation-rigid against arbitrary connected mates.*

Proof. Factorization of the maximal coefficient recovers the two singleton functions and the connected four-row function. By Theorem 4.2, the latter determines Q up to rotation. Normalize it as in (13.3). Every mate is therefore an end or middle placement in (10.1) or (10.2), with this one-negative Q. Both singleton orders are included, with $L = \ell + g_1$, $R = m + g_2$ and all six parameters positive. The expansions (10.4) and (10.5) remain identities: the changed entry stays inside their core determinants.

For $x > \ell$, the intermediate $U_Q(x)$ in (10.6) now has precisely the negative sites

$$(3,1),\ (4,1),\ (5,1),\ (5,2).$$

Thus $\rho(U) = (2,1,1)$, while $\rho(Z) = (3,1)$. Proposition 4.8 gives arbitrary-mate rigidity of both. Equal U-type functions are literally equal, since rotation changes their negative partition; equality of a U-type and a Z-type function forces rotation. Their lists recover the parameters as before.

The first and third markers are still (10.7) and (10.8). The irreducibility and coordinate-divisibility exclusions in (10.9) apply unchanged. Its two remaining ties are excluded by Lemma 13.4, so end and middle placements cannot mix.

Within end placements, (10.10) recovers the singleton order and the margins in the bottom-dominant regime. The moving-bottom complement in (10.11) remains connected: its subdiagonal entries are $\ell, m, C_2 - D_1, C_3 - D_2, C_4 - D_3 > 0$. Irreducibility separates it from a product

with singleton factors. The fixed-core-orientation comparison (10.12) uses only the nonzero fixed endpoint leader, supplied here by Lemma 8.1. Hence aligned end placements agree only literally.

Within middle placements, the tied movement coefficient in (10.13) is settled by the hook rigidity of U and Z. The comparisons of marker types use Lemma 13.4. The remaining fixed-orientation coefficient is (10.15). Its strict maxima and square coefficients are nonzero by the endpoint leader. The final tie, with $\ell = m$, would require $D_Q(R) = h_{\kappa+Rc_Q}$, contrary to Lemma 13.3. Middle placements therefore agree only literally or under whole-parent rotation. □

13.3. **Singletons over exceptional five-row cores.** In this subsection let F_i, H_i, $i = 1, 2$, be (5.1) with $b = p, g = G$, where $0 < q < p < G$, and put $S = p + 4G$. Zero-freeness is equivalent to $p > q$: at $p = q$, the last row beta equals the penultimate column beta. The four choices F_i, H_i, F_i^*, H_i^* include both opening types and both orientations. Set

$$a_j = h_{p+jG}, \qquad b_j = h_{p+jG-q}, \qquad a_j(t) = h_{p+jG+t}, \quad b_j(t) = h_{p+jG-q+t}.$$

Negative subscripts denote zero. The generators a_j, b_j with $j \geq 0$ are jointly independent. Define

$$\begin{aligned} D_1 &= a_0^2 b_2 - a_0 a_1 b_1 - a_0 a_2 b_0 + a_1^2 b_0, \\ D_2 &= a_0 b_0 b_2 - a_0 b_1^2 + a_1 b_0 b_1 - a_2 b_0^2, \\ W &= a_0 b_1 - a_1 b_0, \qquad K_1 = a_0^2, \quad K_2 = a_0 b_0. \end{aligned} \tag{13.6}$$

For each i, write

$$U_i(t) = s_{F_i^{\uparrow t}} - s_{H_i^{\uparrow t}}, \qquad V_i(t) = s_{F_i^{\downarrow t}} - s_{H_i^{\downarrow t}}, \qquad C_i(t) = U_i(t) + V_i(t).$$

Lemma 13.6. *The nonzero leading coefficients of these differences are*

difference	*largest index*	*coefficient*
$U_i(t),\ t > 0$	$p + 3G + t$	$-b_0 D_i$
$V_i(t),\ t > 0$	$p + 3G + t$	$b_0 D_i$
$C_i(t),\ t > q$	$p + 3G - q + t$	$a_0 D_i$
$s_{H_i^{\uparrow t}} - s_{F_i^{\downarrow t}},\ t > 0$	$p + 4G - q + t$	$-K_i W$
$s_{F_i^{\uparrow t}} - s_{H_i^{\downarrow t}},\ t > 0$	$p + 4G - q + t$	$-K_i W$.

(13.7)

For U_i, V_i and the two mixed differences, every fixed index is at most the leading offset. These offsets are positive and less than S.

Proof. For rows $r = 4, 3, 2, 1, 0$, the four Jacobi–Trudi matrices have row vectors

F_1	$a_{r-3}, b_{r-1}, a_{r-1}, b_r, a_r$
H_1	$b_{r-2}, a_{r-2}, a_{r-1}, b_r, a_r$
F_2	$b_{r-3}, b_{r-1}, a_{r-1}, b_r, a_r$
H_2	$b_{r-2}, a_{r-2}, b_{r-1}, b_r, a_r$.

(13.8)

Shift the first row for an up extension and the last column for a down extension. Cofactor expansion gives (13.7). For example, the $a_3(t)$-cofactor difference in U_1 is $-a_0 D_2 - W^2 = -b_0 D_1$, since $b_0 D_1 - a_0 D_2 = W^2$; in U_2 it is $-b_0 D_2$. The mixed $b_4(t)$-cofactor is the triangular minor $-K_i W$. The $a_3(t)$-terms of U_i, V_i cancel in their sum, leaving $b_3(t)$ with coefficient $a_0 D_i$.

In U_i, V_i, every fixed index is at most $p + 3G$. For $t > q$, the index of $b_3(t)$ is above this bound. In the mixed comparisons, the common corner term cancels; every fixed index is at most $p + 4G - q$, and $b_4(t)$ is the moving leader. Finally, the a_2-coefficients of D_1, D_2 are $-a_0 b_0, -b_0^2$, and the a_1-coefficient of W is $-b_0$. These are nonzero. The stated bounds exclude cancellation of any leader with a fixed index. □

Proposition 13.7. *Let Q be one of F_i, H_i, F_i^*, H_i^*, and let P_Q be its singleton context* (3.6)*, with $s, u, v, g > 0$. Its complete connected class is its rotation orbit, except when $Q \in \{F_i^*, H_i^*\}$ and*

$$(s, u, v, g) = (p - q, q, G, G - p). \tag{13.9}$$

Then the class consists of the two contexts over F_i^*, H_i^* and their rotations. These are the two seven-row families of Proposition 12.1, with $(a, b, d) = (q, G, p - q)$: family B for $i = 1$, and family A for $i = 2$.

Proof. Write $N = S + s + g + u + v$ for the parent span. Factorization recovers the singleton length s and the five-row core function. Theorem 5.1 leaves precisely the four core choices above. Rotate any mate to put the singleton first. The marker (3.10) distinguishes $u < v$, $v < u$, and $u = v$. If $v \le u$, its bottom endpoint function distinguishes the four choices, by Lemmas 5.5 and 13.6. The remaining margin is recovered by the factorization argument in the proof of Lemma 3.7, after this literal core recovery. Such a parent is rigid.

For $u < v$, the marker fixes u. Subtracting

$$s_Q(h_s h_N - h_{s+u} h_{N-u})$$

leaves the separator (3.11). Its derivation uses only the common core function and span, not a prescribed representative. The product and connected complementary determinant remain nonzero, and the tied coefficient is their same-sign Schur-positive sum. Thus v, then g, is fixed even across different core choices. Every remaining comparison has identical s, u, v, g.

Put $A = s + g > s$. For distinct core choices Q, R, write

$$\Delta^+(t) = s_{Q^{\uparrow t}} - s_{R^{\uparrow t}}, \qquad \Delta^-(t) = s_{Q^{\downarrow t}} - s_{R^{\downarrow t}}.$$

The up differences have positive fixed leading offsets by Lemmas 13.1 and 13.6. In $\Phi_Q(x, y) - \Phi_R(x, y)$, the corner term cancels. Off the tie $x = y$, the largest index is $S + \max(x, y)$, with the corresponding nonzero endpoint difference as coefficient. If $v < A + u$, the second border difference in (3.7) has the private largest index $S + A + u$. If $v > A + u$, the coefficient at $S + v$ is

$$h_{s+u}\Delta^+(A) - h_s\Delta^+(A + u),$$

which has a private second moving index since $A > s$. Equality therefore requires

$$v = A + u. \tag{13.10}$$

At this tie, the coefficient of h_{S+v}^2 cancels: it is the common maximal cofactor of the equal cores. The remaining coefficient is

$$E = h_{s+u}\Delta^+(A) - h_s\big(\Delta^+(v) + \Delta^-(v)\big). \tag{13.11}$$

For opposite orientations of the same core the parenthesized sum is zero, and $E \neq 0$. Four cross-orbit pairings remain.

Cancel the common rows in their row multisets. The remaining two or three entries give

(13.12)

pair	$i = 1$	$i = 2$
F_i, H_i	$A = G$	$A = G$
F_i^*, H_i^*	$v = G$	$v = G$
F_i, H_i^*	$(A, v) = (G, G)$ or $(3G - q, 2G)$	(G, G) or $(3G - q, 2G - q)$
H_i, F_i^*	$(A, v) = (G, G)$ or $(2G, 3G - q)$	(G, G) or $(2G - q, 3G - q)$.

For example, the first mixed row in family 1 is

$$\{A + G, 3G - q, v\} = \{A, 2G, 2G - q + v\}.$$

The entry $3G - q$ matches either A or $2G - q + v$, giving exactly the two listed possibilities. Family 2 replaces $A + G$ by $A + G - q$ and $2G$ by $2G - q$. Interchanging the exterior positions gives the last row. The first two rows follow by comparing the products of the two remaining entries. Because $u = v - A > 0$, the third row and every (G, G) alternative are impossible.

For F_i, H_i, one has $A = G$, $v = G + u$, and $s < G$. The candidate largest indices in E are $S, s + u, S - q + u$. The last exceeds $s + u$, so cancellation forces $u = q$. At that value its h_S-coefficient is

$$-D_i(h_{s+q}b_0 + h_s a_0) \neq 0.$$

For F_i^*, H_i^*, one has $v = G$, $A = G - u$, and $s + u < G$. The leaders $S - u, S - q$ force $u = q$, and their coefficient is

$$D_i(h_{s+q}b_0 - h_s a_0).$$

Equality of these coordinate monomials forces $s = p - q$, hence (13.9).

For H_i, F_i^* in family 1, the row conditions give $(A, v, u) = (2G, 3G - q, G - q)$. The first term of E has leader $p + 6G - q$, above the other leader $p + 6G - 2q$. For family 2, they give $(A, v, u) = (2G - q, 3G - q, G)$. The common leading coefficient, apart from the nonzero factor a_0, is

$$h_s D_2 - h_{s+G}b_0 W.$$

For $s < p + G$ its private largest index is $p + 2G$; for $s > p + G$ it is $s + G$. At $s = p + G$, the $a_2 = h_{p+2G}$-coefficient is $-a_0 b_0 b_1 \neq 0$. This includes the admissible tie $p + q < G$.

All six unordered core pairs have been compared. The remaining locus is sufficient by the identities in Proposition 12.1. Its four members are distinct, so this is the complete class. □

13.4. **The seven-row base and its borders.**

Proposition 13.8. *Every connected zero-free seven-row diagram with $\nu = 10$ and $n_{\min} = 7$ is rotation-rigid, except for the two complete four-member families in Proposition 12.1, with $d > 0$.*

Proof. The maximal cofactor has six positive rows and is disconnected by Lemma 3.4. The complete budget is

(13.13)

component orders	cross cost	internal negatives
$1 + 5$	5	5
$2 + 4$	8	2
$3 + 3$	9	1
$1 + 1 + 4$	9	1.

Other three-component orders cost at least eleven, and four components cost at least twelve. Two-row connected components are dense, and a three-row component has at most one negative.

For $2 + 4$, apply Lemma 8.6 using Theorem 4.2 and lemmas 3.5, 8.1 and 8.2. For $3 + 3$, the same lifting lemma uses one dense and one one-negative component, with their endpoint leaders from Lemma 8.1. The $1 + 1 + 4$ case is Proposition 13.5. For $1 + 5$, Theorem 5.1 says that the five-row core is rigid or belongs to one of its two exceptional classes. Apply Lemma 13.2 in the first case and Proposition 13.7 in the second. Each argument compares with arbitrary connected mates. □

The five distinct gap values in either seven-row family are

$$d < a + d < b + d < a + b + d < 2b + d.$$

Family A repeats the third value and family B the fourth. Their gap multisets therefore determine the family and d, a, b. The four ordered gap words within each class are pairwise different.

Lemma 13.9. *Let R be connected and zero-free with seven rows, ten negatives, and minimum realizing row count seven. For $t > 0$, $s_{R^{\uparrow t}} = s_{R^{\downarrow t}}$ if and only if $R = R^*$. Every positive endpoint extension of a member of either exceptional seven-row quartet is rotation-rigid. Endpoint extensions from its two different rotation orbits cannot have equal functions.*

Proof. Both extensions stay in the domain of Proposition 13.8 and retain the same ordered gap word. The four gap words in an exceptional class are distinct, so two distinct opposite endpoint extensions cannot form an exceptional pair. The base classification makes them rotations, and undoing the common top extension gives $R = R^*$. The converse follows from rotation.

A positive endpoint extension of an exceptional member has the same gap multiset and larger area. The gap multiset fixes its family's parameters, so the extension cannot be exceptional; it

is rigid by the base theorem. Cross-orbit extensions cannot be identical or rotations, because the original gap words are different even after reversal. Equality of their functions would force a common extension size by weighted degree, and then contradict this rigidity. □

Lemma 13.10. *Every positive border over any member of either seven-row family in Proposition 12.1 is rotation-rigid against arbitrary connected mates. Common betas are allowed.*

Proof. First take $d > 0$. Factorization and Proposition 13.8 recover the connected seven-row child up to its four-member class. Lemma 3.3 recovers the unordered margins. Unequal margins would equate cross-orbit endpoint functions, excluded by Lemma 13.9. Within one child orbit, the same lemma excludes every margin swap other than whole-parent rotation.

At equal margins (t, t), a core rotation only rotates the parent. For opposite child orbits, canceling common parent rows leaves

$$\begin{array}{c|c} A & \{3b+d, 3b+d+t\} = \{4b+d, 2b+d+t\}, \\ B & \{a+3b+d, 3b+d+t\} = \{4b+d, a+2b+d+t\}. \end{array} \tag{13.14}$$

Each forces $t = b$.

Put $U_j = h_{jb+d}$, $V_j = h_{a+jb+d}$, with negative subscripts interpreted as zero. With columns indexed $7, 6, \ldots, 0$, a symbol U_r below means the row $(U_{r-j})_j$, and similarly for V_r. The four bordered matrices have rows

$$\begin{array}{c|l} A, F & V_8, V_7, U_7, V_6, U_6, U_5, V_4, U_4 \\ A, H & V_8, V_7, U_7, V_6, U_6, V_5, U_5, U_3 \\ B, F & V_8, V_7, U_7, V_6, U_6, V_5, V_4, U_4 \\ B, H & V_8, V_7, U_7, V_6, U_6, V_5, U_5, V_3. \end{array} \tag{13.15}$$

Their differences satisfy

$$\begin{aligned} [U_1^4 V_0 V_3 V_5 V_7](s_{\mathfrak{B}_{b,b} F_A} - s_{\mathfrak{B}_{b,b} H_A}) &= -1, \\ [U_1^3 V_0 V_1 V_3 V_5 V_7](s_{\mathfrak{B}_{b,b} F_B} - s_{\mathfrak{B}_{b,b} H_B}) &= -1. \end{aligned} \tag{13.16}$$

For A the source contributes nothing, and the mate contributes the positive column word $(7, 1, 2, 5, 3, 8, 4, 6)$. For B both share the positive word $(1, 8, 2, 7, 3, 6, 4, 5)$, and only the mate has the preceding additional word. The forced U_1-sites and the distinct V-indices exhaust the possibilities. The alphabets are independent since $0 < a < b$, so these coefficients exclude the tie.

For $d = 0$, homogenization sends the family parameters to $(2a, 2b, 2d+1)$ and the two border margins to twice themselves. The zero-free result descends by Lemma 3.1. □

Proof of Theorem 1.3. The zero-free seven-row base is Proposition 13.8. At eight rows the maximal cofactor is connected, has ten negatives, and has minimum realizing row count seven by Lemma 3.4. If that child is rigid, align it and apply Lemma 3.3 with the opposite-endpoint isolation in Lemma 13.9. If it is exceptional, use Lemma 13.10. Every eight-row source is therefore rigid.

Above eight rows, the child and its positive endpoint extensions have at least eight rows and retain the same negative partition. The inductive rigidity assertion and Lemma 3.3 give rigidity at the next height. This argument does not require endpoint extensions of an arbitrary rigid seven-row source to be rigid.

Apply Lemma 3.1 to a full-board equality at its common row count. At height at least eight, rigidity descends immediately. At height seven, a nonrotation image belongs to family A or B. Rotate to a displayed orientation, whose first three gaps are $d', a'+d', b'+d'$. All image gaps are odd, so d' is odd and a', b' are even. The inverse parameters

$$a = a'/2, \qquad b = b'/2, \qquad d = (d'-1)/2$$

satisfy $a \geq 1, d \geq 0, b > a + d$. Reversing the homogenization gives exactly the displayed full-board family and all four of its members. Injectivity and rotation covariance exclude additional mates. The boundary $q = p$ in the five-row core corresponds to $d = 0$; $G = p+1$ gives separating gap one. Both are included. □

Together with Corollary 11.1, this classifies all connected sources with $\nu = 10$ and $n_{\min} \geq 7$. The minimum-six stratum is not included.

14. Concluding remarks

The classification through six negatives has only the seven displayed exceptional schemes, all confined to four or five rows. When a negative pattern requires at least seven rows, nine negatives force rigidity. At ten negatives under the same minimum-row condition, the two seven-row families in Theorem 1.3 are the complete exceptions; every such source with at least eight rows is rigid. Minimum realizing row count, unlike the negative partition itself, is constant on a connected Schur class. It must not be replaced by the actual row count.

A census of connected skew diagrams with eight through thirteen cells gives eighteen Schur classes containing more than one rotation orbit. It enumerates full profiles, identifies diagrams up to translation and rotation, and counts each Schur class once rather than counting pairs of diagrams. Nine classes lie in this paper's classification domain; three further classes are classical ribbon classes [2]. The remaining six are nonribbon classes outside this domain, for which no geometric account is asserted here. This is a bounded frontier, not a proportion of the general problem solved and not a claim that those six classes admit no legal presentations.

The six-row quartets in Proposition 12.2 exhibit three remaining cases: a nonlinear component can itself have nonrotation mates, a rigid non-dense component can change placement, and four components require a further assembly comparison. Their displayed members establish no upper class bound. (12.6) supplies legal reversal presentations for the two ten-negative families.

Problem 14.1. What are the complete connected classes containing the seven-, eight-, and nine-negative six-row quartets in Proposition 12.2? Four constructed members give a lower inclusion, not an upper bound.

Problem 14.2. Which non-dense components admit a mixed endpoint resonance producing different cofactor placements? A criterion must distinguish an irreducible nonlinear cofactor from a product of singletons of the same ordinary degree, as Proposition 12.3 requires.

Appendix A. The four-row calculation

Proof of Proposition 4.1. We may first work zero-free by Lemma 3.1. The maximal cofactor has three positive rows. Its factorization distinguishes the three possibilities: one connected three-row component, a singleton and a connected two-row component, or three singletons. The arbitrary mate has the same order and zero-freeness by Lemma 2.1, and the same component inventory by irreducibility.

The connected case is Lemmas 3.3 and 3.5. In the singleton-plus-two-row case, rotate to put the singleton first. The two-row component is dense; Corollary 3.8 applies. Only its dense clause is used here, whose proof depends on nonnegative-index rigidity, not on Theorem 4.2 or the present four-row classification.

Three singletons mean exactly $r_2 > z_1 + z_2$, $r_3 > z_2 + z_3$, the upper Hessenberg case. Use the eight terms of (6.2) and put $c = \min(z_1, z_2, z_3)$. The generator h_c occurs only at minimum subdiagonal sites. Setting it to zero splits the board into connected blocks of at most three rows. Their irreducible polynomials recover the unordered blocks up to rotation. Their row counts distinguish the $1 + 3$ and $2 + 2$ splits.

One minimum at an end. Orient the profile as $(p, q, r, s; c, a, b)$, with $a, b > c$, and put $t = p - c > 0$. Only reversal of the three-row block remains. The coefficient of $-h_c$ is

$$h_{q+t}h_r h_s - h_{q+r+t-a}h_a h_s - h_{q+t}h_{r+s-b}h_b + h_N h_a h_b.$$

If $a < b$, its private h_a-coefficient, compared with the reversed block, forces $s = q + r - a$. After substitution the remaining difference is

$$h_r(h_{q+t}h_s - h_{s+t}h_q) - h_b(h_{q+t}h_{r+s-b} - h_{r+s+t-b}h_q).$$

Its largest label $r + s + t - b$ is unique and has nonzero coefficient; $r > a + b$ verifies the strict comparisons. The case $a > b$ reverses this argument. If $a = b, q \neq s$, the largest remaining

label is $r+t+\max(q,s)-a$, again with nonzero coefficient. For $q=s$ the block was already symmetric.

One minimum in the middle. The profile is $(p,q,r,s;a,c,b)$, with $a,b>c$. Fix the order of the two recovered two-row blocks by global rotation. The coefficient of $-h_c$ is

$$h_{q+r-c}h_ph_s - h_{p+q+r-a-c}h_ah_s - h_{q+r+s-b-c}h_bh_p + h_Nh_ah_b.$$

Flipping only the left block changes it, up to sign, by

$$h_s(h_ph_{q+u} - h_qh_{p+u}) - h_b(h_ph_{q+v} - h_qh_{p+v}),$$

where $u=r-c>0$, $v=r+s-b-c>u$. If $p\neq q$, the largest label $\max(p,q)+v$ survives. The right-block argument is its rotation. For $a<b$, the private h_a-coefficient fixes the right-block orientation; $b<a$ is symmetric. For $a=b$, comparing both-block flips and taking the part of the h_c-coefficient free of h_a gives

$$h_{q+r-c}h_ph_s = h_{p+s-c}h_qh_r.$$

The merged labels exceed their summands. Their equality forces $q+r=p+s$, then $\{p,s\}=\{q,r\}$. Either both blocks are symmetric, or $p=r, q=s$, giving global rotation.

Three minima. Subtract c from every row and gap. The index substitution on the homogeneous degree-four determinant is injective when the unit variable is retained, and the result is a four-row ribbon. Such a ribbon has no nonrotation mate by [2]: two nonsymmetric composition factors would require at least five rows, using $\ell(\alpha\circ\beta)=\ell(\alpha)+|\alpha|(\ell(\beta)-1)$, since each has at least three cells and two rows. Restoring c is injective and commutes with rotation.

Two minima. Let the gaps be $\{c,c,d\}$, $d>c$. The cut recovers singleton rows $p,q>c$ and a block with rows $a,b>d$ and gap d. Modulo global rotation all arrangements have one of the forms

$$L=(p,q,a,b;c,c,d), \qquad M=(p,a,b,q;c,d,c),$$

allowing both singleton orders and both block orientations, subject to the Hessenberg inequalities. Put $t=a+b-d$, $N=p+q+t-2c$. The coefficients of h_c^2 and h_c are

$$\begin{aligned} Q_L &= h_{p+q+a-2c}h_b - h_Nh_d, & Q_M &= h_{p+a-c}h_{q+b-c} - h_Nh_d,\\ \ell_L &= -h_{p+q-c}h_ah_b + h_{p+q-c}h_th_d - h_ph_{q+a-c}h_b + h_ph_{q+t-c}h_d,\\ \ell_M &= -h_{p+a-c}h_bh_q - h_ph_ah_{q+b-c} + h_{p+t-c}h_dh_q + h_ph_{q+t-c}h_d. \end{aligned}$$

All remaining indices exceed c, and the constant terms agree. For L against L, quadratic equality fixes the block orientation; swapping unequal singletons then has largest label $\max(p,q)+t-c$ with coefficient $\pm h_{\min(p,q)}h_d\neq 0$. For M against M, the quadratic terms agree only when $p=q$ or $a=b$, giving rotation or identity.

For $L(p,q,a,b)$ against $M(p,a,b,q)$, quadratic equality forces $b=p+a-c$. In the linear difference the largest possible labels are $L_1=q+b-c$ and $L_2=p+t-c$. Their coefficients are h_ph_a, $-h_qh_d$, or at the tie $h_ph_a-h_qh_d$. Only the tie can vanish, and it forces $p=d, q=a$, because $a>d$. Then $b=a+d-c$, and the source inequality is $a>c+d$. Renaming d as w and a as b gives (4.1). Substituting these equalities in the eight terms proves full polynomial equality. For the other alignment $M(p,b,a,q)$, quadratic equality instead forces $b=q+a-c$; the linear difference has unique largest label $p+t-c$, with coefficient $-h_qh_d\neq 0$. These comparisons exhaust all arrangements, including coincident rows.

The zero-free exceptional parameters satisfy $1\le c<w$, $b>c+w$. For a Γ-image its bare profile parameters c,w,b are odd. Replacing them by $(c-1)/2,(w-1)/2,(b-1)/2$ gives exactly $0\le c<w$, $b>c+w$; conversely substitution in the interval formula gives the identity at $c=0$ as well. The different overlap words distinguish the two rotation orbits, and the displayed repeated-row and gap data recover their parameters. The three strictly negative indices follow from $2c-b<0$, $c+w-b<0$, and the still smaller bottom-left index. No other index is negative. This proves the complete full-board statement. □

Appendix B. First-order determination at a contact

The q-components and the augmentation ε are those of Section 6. The result below determines a component function from its first jet; it does not recover that jet from a parent polynomial.

Theorem B.1 (First-jet rigidity)**.** *Let A, B be q-components with at most four rows. Then*

$$s_A\big|_{h_q=0} = s_B\big|_{h_q=0}, \qquad [h_q]s_A = [h_q]s_B \quad \Longrightarrow \quad s_A = s_B.$$

Equivalently, equal restrictions of distinct full functions satisfy $\nu_{h_q}(s_A - s_B) = 1$, where ν_{h_q} is the order of divisibility by the independent generator h_q.

Proof. Put $S_j(A) = [h_q^j]s_A$. The inherited boundary preserves the private full-interval term in S_0, whose coefficient is $(-1)^{n-1}\prod_i h_{g_i}$. Thus $S_0 \neq 0$ recovers the row count, full interval, and gap multiset. Equal restrictions have the same row count, so we work at each order separately.

For four rows use $(a, b, c, d; r, s, t)$ and the indices of (6.5). The boundary gives $d, w, y, T > q$. Since $u > \max(a, b)$, $v > \max(b, c)$, and $x > \max(u, v)$, the following list of contact sets is exhaustive. A word such as ac means that exactly a, c among a, b, c, u, v, x equal q; comma-separated words denote alternatives, not simultaneous contacts. Equalities among the other indices are allowed. The independent generators are collected before applying ε. In particular, the boundary excludes an h_q^4-term; the only possible unknown coefficients after the first jet are S_2, S_3.

indices equal to q	$(\varepsilon(S_0), \varepsilon(S_1))$	S_2	S_3
$\varnothing, a, b, c, u$	$(0, 0)$	0	0
v, uv	$(1, -1)$	0	0
x	$(-1, 1)$	0	0
ab	$(0, 0)$	$R_t(c, d)$	0
ac	$(1, -2)$	$h_b h_d$	0
bc	$(1, -2)$	$h_a h_d$	0
av	$(0, 1)$	$-h_s h_d$	0
uc	$(-1, 2)$	$-h_r h_d$	0
abc	$(1, -1)$	$-h_t h_{q+d-t}$	h_d.

All higher coefficients vanish. The table is obtained by substitution in the displayed eight terms, retaining every numerical coincidence. It suffices to prove that the first jet determines the listed higher coefficients whenever one of the last six cases occurs.

For av, the scalar jet identifies the case. Its first coefficient is

$$h_b h_c h_d - h_t h_b h_{c+d-t} + h_s h_t h_{q+d-t}.$$

The last index is private and largest, with coefficient $h_s h_t$. Removing that term leaves $h_b R_t(c, d)$. Its monomial content recovers b; its primitive binomial recovers $t, \{c, d\}$. Since $c < q < d$, the pair is oriented, and $s = b + c - q$. Thus S_2 is recovered.

For uc, again the scalar jet identifies the case, and

$$S_1 = h_a h_b h_d + h_r h_t h_{q+d-t}.$$

The last index is private. After removing its term, the only index above q in the remaining monomial is d. The last index then recovers t, and its coefficient recovers r. Hence $S_2 = -h_r h_d$ is fixed.

For bc,

$$S_1 = -h_r h_{a+q-r} h_d - h_t h_a h_{q+d-t}.$$

Every coefficient is negative. This distinguishes it from ac, whose first coefficient has a private positive largest term, as below. For bc, $a \neq q$, so Lemma 6.5 applies directly to S_0. It recovers a, r after a valid orientation, or both endpoint pairs at a tie. The middle rows are q, q, so the recovered full interval and gap multiset give $d = T - a - 2q + r + s + t$. Hence $S_2 = h_a h_d$ is determined. No residual-classification theorem is used in this first-jet argument.

For ac, the first coefficient is

$$\begin{aligned} S_1 = & -h_r h_{q+b-r} h_d - h_s h_{q+b-s} h_d - h_t h_b h_{q+d-t} \\ & + h_s h_t h_{b+q+d-s-t}. \end{aligned}$$

Its last index y is strictly largest and has positive coefficient. The known difference $T-y = q-r$ recovers r. Put $U = q+b-r$. The restriction is

$$S_0 = -h_r P_{(U,q,d;s,t)}, \qquad U > q+s, \quad d > q+t.$$

This middle-row three-row restriction determines the contracted profile up to rotation. To see this directly, its largest coefficient recovers $\{s,t\}$. If the gaps differ, its two negative monomials are identified by their unique gap factors below q; swapping both high-index pairs would require a positive shift in one equation and a negative shift in the other. If $s=t$, their least high index and common total recover $\{U,d\}$, including coincident monomials.

The only possible distinct endpoint lift is therefore, with $\delta = q-r > 0$,

$$(q, U-\delta, q, W; r, s, t) \quad \text{against} \quad (q, W-\delta, q, U; r, t, s).$$

Their contracted full functions are rotations. The first-coefficient comparison is consequently between the two tail restrictions. After the common full-interval term is removed, their only possible largest indices are $q+W-t$ and $q+U-s$, with coefficient contributions $-h_t h_{U-\delta}$ and $h_s h_{W-\delta}$. The barred specializations do not vanish: those two rows are not q in the ac case. At a tie, a crossed matching is impossible because $U-\delta > s$ and $W-\delta > t$. Hence equality forces $U=W, s=t$, and the lifts coincide. This recovers S_2.

For abc,

$$S_1 = -h_r h_{2q-r} h_d - h_s h_{2q-s} h_d + h_s h_t h_{2q+d-s-t}.$$

Its last term is private and largest. The competing lower-degree v-case has a single negative first-coefficient monomial, so cannot match. The uv-case has

$$S_1 = -h_r h_c h_d - h_s h_a h_d + h_r h_t h_{c+d-t}.$$

If its private positive term matched, subtract it on both sides. Each remaining abc monomial has exactly one index below q, whereas each remaining uv monomial has two. They cannot coincide. For two abc profiles, the gap multiset and T give d. The largest first-coefficient term recovers $\{s,t\}$, hence r. Removing it and the known r-term leaves $-h_s h_{2q-s} h_d$, whose low index recovers s, then t. Both S_2 and S_3 are determined.

The remaining case is ab. Its first coefficient is

$$S_1 = -h_s R_t(q+c-s, d), \qquad c \neq q.$$

The lower-degree competitors with scalar jet $(0,0)$ have first coefficients

$$0, \quad s_{(b,c,d;s,t)}, \quad h_a R_t(c,d), \quad h_d R_r(a,b), \quad -h_r R_t(c,d),$$

for contact sets $\varnothing, a, b, c, u$, respectively. Under $h_j = \epsilon^{j^2}$, a full Schur polynomial has positive initial coefficient by Lemma 2.1; multiplication by a monomial preserves that sign. Thus only the negative u-case can compete.

Write this candidate as $(a', b', C, D; s, r, t)$. Equality of first coefficients and the recovered gap multiset force

$$a' + b' - s = q, \qquad \{C, D\} = \{q+c-s, d\}, \qquad C, D > q+t.$$

The source restriction is $-h_r s_{(2q-r,c,d;s,t)}$. Divisibility of the candidate restriction by h_r, after cutting its middle seam, forces $a' = r$: $b' > r$ and $C, D > q > r$, and the remaining two-row factor does not vanish. Consequently

$$r > s, \qquad q > 2r, \qquad b' = B_0 = q+s-r.$$

Set

$$\begin{aligned} v_0 &= B_0 + C - r, \quad x_0 = q + C - r, \quad w_0 = C + D - t, \\ y_0 &= B_0 + C + D - r - t, \quad T_0 = B_0 + C + D - s - t. \end{aligned}$$

The candidate restriction divided by h_r is

$$h_{B_0}h_Ch_D - h_rh_{v_0}h_D - h_th_{B_0}h_{w_0} + h_sh_{x_0}h_D + h_rh_th_{y_0} - h_sh_th_{T_0}.$$

Here

$$T_0 > y_0 > w_0 > \max(x_0, D), \qquad x_0 > v_0 > C > B_0 > r > s.$$

The three largest-index terms cannot cancel. After canceling the common h_D from the remaining potential pairings, their least indices and $x_0 > v_0$ exclude every opposite-sign coincidence. The norm is six. The source restriction divided by h_r is a full three-row Schur function of norm four: its full interval is private, and its diagonal cannot equal an adjacent-merge term because the merged index exceeds both rows. Same-sign collisions preserve this norm. The candidate is therefore impossible.

For two ab profiles, the first coefficient recovers $s, t, \{c+q-s, d\}$, and the gap multiset recovers r. The source restriction divided by $-h_r$ is a full three-row polynomial, whose minimum-square monomial recovers $\{2q-r, c, d\}$. Remove one occurrence of $2q-r$. The two multisets $\{c, d\}$ and $\{c+q-s, d\}$, with $q-s>0$, recover c, d in order: subtract their sums of squares. Hence $S_2 = R_t(c, d)$ is fixed.

This exhausts the four-row table. With three rows, the boundary gives $c, b+c-s, a+b+c-r-s > q$. Degree at least two in h_q occurs only when $a=b=q$, with

$$S_2 = h_c, \qquad S_1 = -h_sh_{q+c-s}, \qquad (\varepsilon(S_0), \varepsilon(S_1)) = (0, -1).$$

The other scalar jets are $(0,0), (-1,1), (1,-1)$. Thus this case is identified and its first coefficient recovers s, c. Two-row polynomials have degree at most one, and a singleton is independent of h_q. The theorem follows. □

Remark B.2. The inherited inequality $d > q+t$ preserves the full-interval term and confines the possible q-indices to the contact table. On the norm-four locus $c=q, s=t, d=r, a=q+r-s$, one has $r > q+s$, $b > r+s$, so only c equals q. For example, at $A = (8,8,3,6;6,1,1)$ and $q=3$,

$$[h_3]s_A = h_6h_8^2 - h_6^2h_{10}, \qquad [h_3^j]s_A = 0 \quad (j \geq 2).$$

The cancellation in the restriction does not conceal a higher h_q-coefficient. The theorem compares the original component functions, not merely their primitive associates, and it does not identify their first jets from a parent polynomial. Indeed, for $e > q$,

$$[h_q]s_{A\cdot_q(e)} = h_e[h_q]s_A - s_{A\odot_q(e)}\big|_{h_q=0}.$$

The contraction term remains part of global compatibility. The first-jet theorem is an additional component-reconstruction result; the classification in Section 7 uses the separate compatibility lemmas.

In congruence notation, the theorem reads

$$s_A \equiv s_B \pmod{h_q^2} \iff s_A = s_B \qquad \text{for } q\text{-components with at most four rows.} \tag{B.1}$$

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